RECOGNITION OF RIBBON-OUTER SKEW SCHUR COMPOSITIONS

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Abstract. We prove a recognition theorem for skew diagram compositions satisfying Hypotheses I–IV of McNamara and van Willigenburg. The inner diagram carries two translated end copies of an overlap, or boundary edges for an empty overlap, and their attachments; the period is their content displacement. A composition is proper when both factors are connected and have at least two cells. Minimality at a fixed parent first minimizes the overlap's diagonal span, then the inner diagram's. For Schur-equal minimal compositions of diagrams with at least 24 cells, assume the first has a ribbon outer and no smaller period. We recover its marked inner and outer Schur function on the other diagram, up to rotation of the full marking data and permitted attachment changes with their outer involution. The original Schur class contains no proper row- or column-inner composition. At every smaller cell count, we assume that proper-composition existence transfers under Schur equality, every class with more than one rotation orbit contains a proper composition, and Schur-equal minimal compositions have the same marked inner and outer Schur function modulo these identifications. A finite Jacobi–Trudi identity and two boundary records construct the legal factors before minimality is applied.

1. Introduction

The skew diagram composition of McNamara and van Willigenburg [2] assembles translated copies of a connected inner diagram E according to a connected outer diagram D. The end data specify two copies of an overlap W, their displacement, and their horizontal or vertical attachments. For nonempty W, the copies contain the southwest and northeast cells of E; for empty W, the marks are boundary edges. We retain E and these data as a *marked inner* ω and write $D \circ_\omega E$, following [9]. Different assemblies can have the same skew Schur function [4]. The recognition problem asks whether that function determines a prescribed marked inner, rather than only its Schur function. A *Schur class* consists of connected diagrams with the same skew Schur function.

Under Hypotheses I–IV of McNamara and van Willigenburg, composition and factor reversal preserve skew Schur equality, by Olson-Harris's theorems [3]. The earlier results of McNamara and van Willigenburg imposed the additional attachment restriction of Hypothesis V [2]. A marking is *legal* if it satisfies I–IV.

Following [9], a *chart* of F is a chosen legal composition presentation $F = D \circ_\omega E$, including its full marking data. It is *proper* when both connected factors have at least two cells. The converse problem requires such charts on the diagrams being compared. A factor of a determinant does not supply a chart. A *line-inner* chart has an inner that is a single row or column. In a *ribbon-outer* chart, D is a ribbon; the whole diagram F need not be one.

The content of a cell (i, j) is $j - i$. If translation by $v = (v_1, v_2)$ sends the lower end mark to the upper one, the *period* is their content displacement $p = v_2 - v_1$ [9]. Write N_A for the number of occupied content diagonals of a diagram A, and $w = N_W$ for the *overlap width*, with $w = 0$ for the empty overlap. Then

$$N_E = p + w, \qquad N_F = pN_D + w, \qquad 0 \le w < p. \tag{1.1}$$

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A chart is *source-minimal* when its score (w, N_E) is lexicographically least among all proper charts of that same parent. This is the minimal-factorization convention of McNamara and van Willigenburg [2]. The qualifier *source* specifies that the parent F is held fixed, rather than varied over its Schur class [9].

The attachments are *ordinary* when both are horizontal or both vertical, and *mixed* when they differ. A permitted *gauge change* switches between two legal ordinary markings of the same inner, or between the corresponding empty-edge readings, together with the fundamental involution on the outer Schur function.

The *prescribed full marked fiber* $\mathcal{F}(\omega, s_D)$ is the set of diagrams admitting proper compositions through the fixed marked inner ω, or its transported rotation, with outer Schur function s_D [9]. Transported rotation means a half-turn of the inner and both end marks, interchanging the lower and upper ends and transporting their attachments. The fiber includes the permitted gauge changes with their action on the outer function; the markings are not identified while that function is held fixed. Precise conventions are given in Section 2.

A diagram is *rooted* if it has a proper chart. The *key* of a source-minimal chart is its full marked inner and outer Schur function, modulo the rotations and gauges just described [9]. Area means the number of occupied cells. For the original cell count n, we use the joint smaller-area hypothesis $\mathsf{J}_{<n}$ of [10]. At each Schur class of area strictly less than n, we assume that one rooted member implies all members are rooted; that a class with more than one rotation orbit contains a rooted member; and that rooted members have the same source-minimal key. No assertion at area n is included.

Theorem 1.1 (Ribbon-outer recognition)**.** *Let $n \geq 24$, and assume $\mathsf{J}_{<n}$. Let*

$$F = D \circ_\omega E, \qquad H = C \circ_\eta U, \qquad s_F = s_H, \qquad |F| = n \tag{1.2}$$

be actual proper source-minimal charts satisfying Hypotheses I–IV of McNamara and van Willigenburg. Let their periods be $p \geq q$, respectively. Suppose D is a ribbon and no member of their whole Schur class has a proper line-inner chart. Then $p = q$ and

$$H \in \mathcal{F}(\omega, s_D). \tag{1.3}$$

The fiber is prescribed by the original displayed chart of F. Its only identifications are transported full-mark rotation and the genuine ordinary or empty-overlap gauge, with its action on the outer function.

The period equality is the ribbon-outer period-rigidity theorem of [10], under exactly these original chart and strictly smaller-area hypotheses. Once the periods agree, the assertion proved here constructs a proper chart of H through the prescribed marked inner of F, with outer function s_D, up to the full-mark rotations and gauges above. Equal widths follow from (1.1); an extreme Jacobi–Trudi cofactor comparison makes C a ribbon as well [7]. The result does not assert uniqueness of all factorizations, choose a new fiber after the comparison, or supply rootedness transfer or root existence at the current area.

Two distinctions govern the proof. First, factor-function recovery is not marked-factor recovery. The ordinary and mixed ribbon inverses at a supplied period are the factor-recovery theorems of [8]. They use Hall adjoints and a weighted derivation, respectively. If the recovered inner function has several realizations, its full mark has still to be placed on the second diagram. Where the proof also supplies the function of a first-row deletion, reconstruction from these two boundary observations identifies the connected diagram up to translation [5, Reconstruction from two boundary observations]. Second, same-parent gluing compares charts that already exist on one diagram [9]. It cannot supply the missing chart on another Schur-equal parent.

Corner descent compares the extreme Jacobi–Trudi cofactors. The interior of a diagram consists of the northwest cells of its 2×2 blocks. The corner transport theorem [7] records the compositions arising from connected components of the outer interior inside the whole composition's cofactor. A ribbon has no 2×2 block, so this descent provides no outer-interior child composition.

Instead, we compare compositions whose outer diagram is a horizontal or vertical domino. For a nonline inner, outer-independent minimality [7] preserves a source-minimal chart when its

connected proper outer is changed. Whenever these test diagrams have fewer cells, $\mathsf{J}_{<n}$ settles the comparison; the remaining original outers have two cells. If the recovered inners E and U are not related by rotation, they themselves have fewer than n cells. The hypothesis $\mathsf{J}_{<n}$ then gives compositions of E and U through the same full marked inner. These are the *finer charts*: they factor the inner diagrams themselves. The remaining algebra compares the Jacobi–Trudi determinants of the finer outer diagrams, retaining the complete inverse matrices in each chart's own rational coordinates.

On the mixed branch, the composition equality supplies a finite identity for rational coefficient functions describing permitted variations of these determinants. At the actual private gate, the aliases and fine weight make a nonzero response a positive integer power of the returned clock with constant rational amplitude. Whole-field stationary descent [6, Whole moving potential] excludes such a response on distinct equal-function diagrams. Its use here requires the actual coarse span, raw independence in each rebuilt field, the full moving homomorphism, and every coefficient through the terminal degree. These are supplied by the physical comparison, not assumed as an algebraic replacement for it. The resulting vanishing coefficient is not yet a geometric factorization.

Substituting this vanishing coefficient into the two-copy identity produces connected diagrams with equal whole Schur functions and strictly fewer cells. Only after these smaller diagrams have been constructed is $\mathsf{J}_{<n}$ applied to them. Their two boundary words, the binary step records of the northwest and southeast boundaries, supply the native hypotheses of the decoder and residue folds in [9]. Those constructions retain I–IV, both full end marks and their attachments. The resulting period divides that of the finer charts. The aligned composition theorem [9] then places a new chart on the same diagram at which the relevant source minimum was chosen. Only this realized chart's score is compared with the minimum. The ordinary and mixed attachment cases require different boundary arguments.

Section 2 fixes the chart and descent conventions. Section 3 gives factor readout, the literal-inner finish, the reduction to two-cell outers, and then the nonliteral common finer source. Section 4 derives the actual finite equations and applies coframe separation [6]. Section 5 specifies the imported geometric interfaces; Section 6 constructs their actual inputs and outputs, and Section 7 completes recognition. The appendices prove the physical determinant identities, exceptional cell constructions, and complete finite comparisons used by these reductions.

2. Marked charts and strict descent

2.1. **Diagrams and the two boundary words.** Diagrams are finite skew Young diagrams in English notation, identified up to translation. Connected means nonempty and edge-connected. The area $|S|$ is the number of cells; N_S is the number of occupied content diagonals. The content of (i,j) is $j-i$, and $e=(1,1)$. For connected diagrams, $N_S = r(S)+c(S)-1$. We write S^* for rotation through 180 degrees and $\mathrm{AT}(i,j)=(-j,-i)$. The involution ι on symmetric functions satisfies $\iota(h_j)=e_j$ and $\iota(s_S)=s_{\mathrm{AT}(S)}$. A line is a row or column; a ribbon is a connected diagram without a 2×2 block.

We work in $\Lambda=\mathbb{Q}[h_1,h_2,\ldots]$, with $h_0=1$ and $h_j=0$ for $j<0$. A southwest-normalized connected diagram of span N has full beta lists

$$X=(N,x_1,\ldots,x_{r-1}),\qquad Y=(y_0,\ldots,y_{r-2},0),\qquad s_S=\det(h_{x-y})_{x\in X,\,y\in Y}, \tag{2.1}$$

with both lists strictly decreasing. Equal labels in the two lists are retained. They correspond to actual unit entries, not to deleted cells.

If $[\ell_k,h_k]$ is the row interval on content k, define

$$A_k=\ell_{k-1}-\ell_k=\mathbf{1}_{k\in Y},\qquad B_k=h_{k-1}-h_k=\mathbf{1}_{k\in X},\qquad 1\le k<N. \tag{2.2}$$

Both words are binary. Their height is

$$H_S(k)=h_k-\ell_k+1=\#\{y\in Y: y\le k\}-\#\{x\in X: x\le k\},\qquad H_S(k)-H_S(k-1)=A_k-B_k. \tag{2.3}$$

There are no artificial steps at 0 or N. The height word alone forgets the distinction between the pairs $(0,0)$ and $(1,1)$.

Lemma 2.1. *A finite cell set is a connected skew diagram if its contents form an interval, each diagonal is a row interval, its two endpoint sequences have steps zero or minus one, and its extreme diagonals are singletons. Conversely, every connected skew diagram has these properties. Thus positive heights with singleton ends and the two binary boundary words determine a connected skew diagram up to translation.*

Proof. Starting from the southwest singleton, the two words successively give the endpoints of every diagonal interval. The step conditions are exactly the two skew-boundary conditions. Every cell outside the last diagonal has an eastern or northern neighbor on the next diagonal: the next interval contains either its row or the preceding row. Hence the set is edge-connected. The converse follows by following the two boundaries of a connected skew diagram. □

The area, row count, column count, span, and row-overlap multisets are Schur invariants [4]. Connected skew Schur functions are irreducible [1]. We use primitivity in the complete generators to fix scalar normalizations. The interior $S^- = \{x \in S : x + e \in S\}$ satisfies

$$(-1)^{r(S)-1}[h_{N_S}]s_S = s_{S^-}. \tag{2.4}$$

The right side is the product of the Schur functions of its connected components. This is the extreme cofactor in (2.1).

2.2. **Full legal marks.** A marked inner ω consists of a connected diagram E, two congruent connected end diagrams W_- and $W_+ = W_- + v$, and their attachment case. The lower copy contains the southwest cell of E, the upper copy its northeast cell. Put

$$O = E \setminus (W_- \cup W_+), \qquad w = N_W, \qquad p = c(v) = N_E - w.$$

The four cases, listed as lower and upper attachments, are

case	(a)	(b)	(c)	(d)
attachments	(H,H)	(V,V)	(H,V)	(V,H).

(2.5)

Cases (a) and (b) are ordinary; cases (c) and (d) are mixed. A horizontal lower attachment uses the western neighbor of the southwest cell of O, and a vertical lower attachment its southern neighbor. The upper attachments use the eastern or northern neighbor of the northeast cell of O.

We use the following complete form of Hypotheses I–IV [2, 9].

(I) No strictly larger skew diagram containing W_- on the same content diagonals has translates at both ends of E.
(II) There is an occupied diagonal between the two copies meeting neither: $0 \le w < p$.
(III) Removing either end copy leaves a connected skew diagram.
(IV) In $\boldsymbol{E} = \bigcup_{k\in\mathbb{Z}}(E + kv)$, no hatted-middle copy is adjacent to a hatted-mark copy, where

$$\widehat{O}_1 = \{x \in O_1 : x + e \in O_1\}, \qquad \widehat{W}_1 = \{x \in \boldsymbol{E} : x + e \in W_1\} \cup \{x \in W_1 : x + e \in \boldsymbol{E}\}. \tag{2.6}$$

Both sets in the union and every integer translate are included. Adjacency means sharing an edge.

At width zero the mark is read horizontally: the west edge of the southwest cell and the east edge of the northeast cell have displacement $v_H = \mathrm{ne}(E) - \mathrm{sw}(E) + (0,1)$. The vertical reading has $v_V = v_H - e$ and is converted by the genuine gauge below. Hypothesis V is not imposed.

For completeness, the cells in a composition can be described directly. If the outer diagonal j has row interval $[\ell_j, h_j]$, the copy indexed by $d = (i, i + j)$ has origin

$$L_\omega(d) = \begin{cases} jv + ie, & \text{(a)},\\ jv - ie, & \text{(b)},\\ jv + (i - \ell_j)e, & \text{(c)},\\ jv - (h_j - i)e, & \text{(d)}. \end{cases} \tag{2.7}$$

In case (c), respectively (d), a vertical pair on the same northwest, respectively southeast, boundary ribbon contributes the supplemental copy $W_+ + L_\omega(d) + e$, respectively $W_+ + L_\omega(d) - e$,

with d the lower cell. These copies are part of the definition. A proper chart has connected D, E with $|D|, |E| \geq 2$.

Proposition 2.2. *For a positive-width legal mark, $W_- = E \cap (E - v)$. The middle is thin on the marked diagonals, and the private diagonals w and $p - 1$ are singletons. Its middle content interval has the form $[s, p + t - 1]$, with $1 \leq s \leq w$, $0 \leq t < w$; in the mixed cases $t < s$. A composition of outer span m has span $mp+w$. Its first and last p diagonals determine the inner, with the final copy translated by $(m - 1)v$ in the mixed cases.*

Proof. These are the local geometry and copy-placement statements of [9, Lemma (local geometry)]. Here are the points used in subsequent constructions. The intersection $E \cap (E - v)$ is skew and contains the supplied connected end copy on every marked diagonal; maximality gives equality. If a middle tail first has two cells on a marked diagonal, its northwest cell is hatted-middle and is edge-adjacent to the hatted mark on the preceding diagonal. This contradicts IV. At the first private diagonal either III gives a new extreme singleton, or the same adjacency excludes a second cell; rotation gives the other private singleton. In a mixed mark the two thin tails cannot overlap at a relative marked content, by the full-intersection assertion, and the endpoint attachment rules exclude equality of their total span with p. Thus $t < s$. Finally, (2.7) gives the span and the two extreme windows. The private singleton in their overlap fixes the joining translation. □

We use the coding theorem of [9, Boundary coding lemma] with its endpoint convention. The only possible nonperiodic residues of the words A, B are

(2.8)

	A	B
(a)	t	s
(b)	s	t
(c)	$\varnothing$	t, s
(d)	t, s	$\varnothing$.

Their actual initial and terminal values are

(2.9)

	initial	terminal
(a)	$A_t = 1,\ B_s = 0$	$A_{N-w+t} = 0,\ B_{N-w+s} = 1$
(b)	$B_t = 0,\ A_s = 1$	$B_{N-w+t} = 1,\ A_{N-w+s} = 0$
(c)	$B_t = B_s = 0$	$B_{N-w+t} = B_{N-w+s} = 1$
(d)	$A_t = A_s = 1$	$A_{N-w+t} = A_{N-w+s} = 0$.

Only genuine internal positions are read in this table. Empty composition has one joining residue, zero, in both words. Antitransposition changes (A, B) to $(1 - B, 1 - A)$; whole rotation also reverses the content order and transports the complete mark.

2.3. **Fibers, gauges, and source minimality.** The forward I–IV composition and reversal theorems are those of [3]. At an ordinary double marking of one inner, the two displacements differ by e, and the cell formula gives

$$D \circ_{\omega_a} E = \mathrm{AT}(D) \circ_{\omega_b} E. \tag{2.10}$$

The same equation holds between horizontal and vertical empty readings. Thus a gauge changes the outer function by ι. Rotation of a full mark rotates both end copies, interchanges their roles, and transports the attachment data.

For a connected outer function Q, the prescribed full fiber $\mathcal{F}(\omega, Q)$ consists of all proper compositions through ω or its transported rotation, with outer function Q, together with their equivalent gauge descriptions (2.10). It is contained in one Schur class. The distinction between equal outer functions and literal equal outers is retained.

Proposition 2.3. *The following facts hold for actual legal charts.*

(1) *Two proper charts of the same parent with equal periods define the same full fiber. On a fixed inner and at a fixed positive width, at most two full legal marks occur; a double consists of the two ordinary gauges.*

(2) *If $U = V \circ_\omega E$ has period p, and a legal mark η on U has displacement of content divisible by p, there is an actual descended full mark on V. Composition through η reassociates to a composition through ω, with the genuine empty gauge when needed.*
(3) *If a chart of a parent is source-minimal, every proper-root fiber through that parent is contained in its prescribed fiber.*
(4) *If the inner of a source-minimal chart is not a line, changing its outer to any connected proper outer preserves source minimality. Its period divides every proper root period of the new parent.*

Proof. Items (1)–(3) are the equal-period comparison, aligned-composition, and source-minimal domination results of [9]. The aligned statement includes the full descended mark and all mixed supplements. Item (4) is the outer-independent minimality lemma of [7]. Its hypothesis is minimality of a chart already on its source. None of the four statements supplies a chart on an arbitrary Schur-equal diagram. □

Example 2.4. The inner $E = (4, 4, 2)/(2)$ has two legal width-two, period-four ordinary marks. Their displacements differ by $e = (1, 1)$, and

$$\mathrm{Row}_2 \circ_{\omega_a} E = \mathrm{Col}_2 \circ_{\omega_b} E.$$

The outer functions are h_2 and $e_2 = \iota(h_2)$, not the same polynomial in a fixed gauge. Each end mark has two cells adjacent to the middle, so this example uses I–IV without V. These are the two full markings in the component-key example of [9]. In zero-based row and column coordinates their full copies are

$$W_a^- = \{(1,0),(2,0)\}, \quad W_a^+ = \{(0,3),(1,3)\}, \quad v_a = (-1,3),$$
$$W_b^- = \{(2,0),(2,1)\}, \quad W_b^+ = \{(0,2),(0,3)\}, \quad v_b = (-2,2).$$

Figure 1 displays both marks and the corresponding domino composites. Their equality is up to translation; both normalize to $(7, 7, 5, 2)/(5, 2)$, of area 14.

2.4. **The smaller-area package.**

Definition 2.5. A diagram is rooted if it has a proper legal chart. For every connected Schur class of occupied area less than n, the hypothesis $\mathsf{J}_{<n}$ asserts:

(1) if one member is rooted, every member is rooted;
(2) if the class contains two rotation orbits, some member is rooted;
(3) any two rooted members have the same source-minimal marked key, modulo transported full-mark rotation and the gauge (2.10).

The key consists of the full marked source-minimal inner together with the outer Schur function. No current-area assertion is included.

Lemma 2.6. *Assume $\mathsf{J}_{<n}$, and let G be connected with $|G| < n$. Write $[G]$ for its connected Schur class.*

(1) *If G has a prescribed source-minimal chart $\chi = (A, K, \kappa)$, then $[G] = \mathcal{F}(\kappa, s_A)$.*
(2) *If G is chartless, then $[G] = \{G, G^*\}$.*
(3) *If G and a nonrotation mate have equal whole functions, then G has a proper source-minimal chart and the mate belongs to its prescribed full fiber.*

Proof. For (1), rootedness transfer makes every mate rooted, and key invariance, followed by source-minimal domination, gives membership in the prescribed fiber. Forward sufficiency gives the reverse inclusion. For (2), a second rotation orbit would give a rooted member by root existence, and then a root of G by transfer. Part (3) follows by the same two assertions and then (1). This is also the lower full-fiber consequence in [10, Smaller-area full-fiber lemma]. □

Lemma 2.7. *In a proper legal composition $P = A \circ_\kappa K$, both factors have area strictly smaller than P. An actual connected child obtained by deleting a nonempty outer strip, or by one of the interior cuts used below, has smaller occupied area whenever it is a strict cell subset. A Schur-equal mate of that child has the same strict area.*

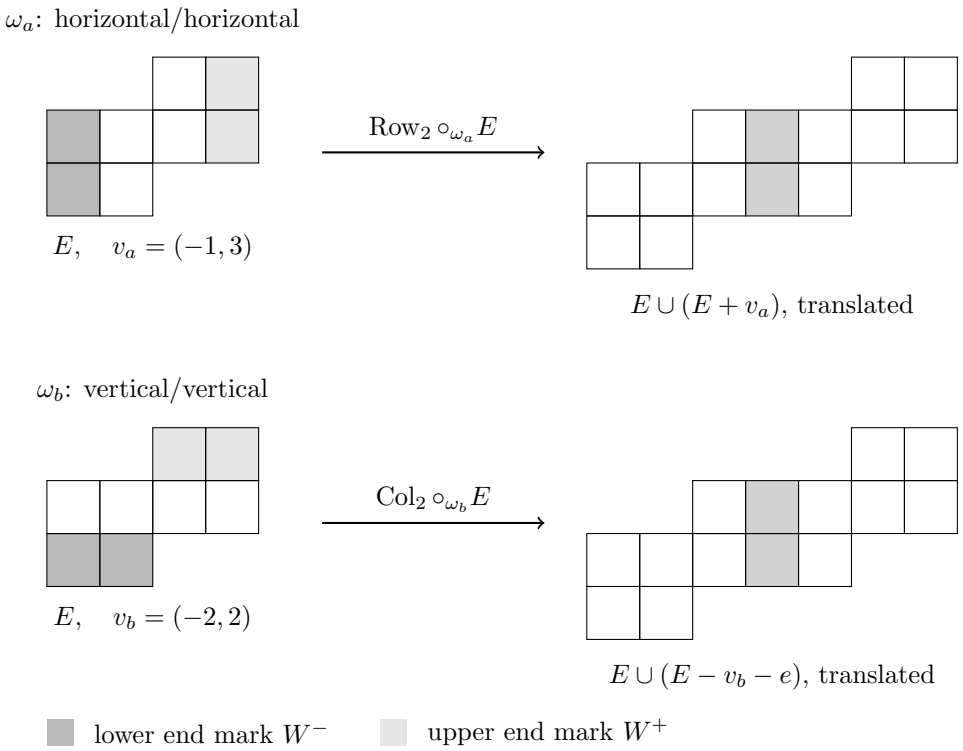


FIGURE 1. Two full ordinary marks on $E = (4, 4, 2)/(2)$, each of width 2 and period 4, and their equal 14-cell composites after translation. On each inner, dark and light cells are the complete lower and upper marked copies; on each composite, shading indicates the two cells shared by its placed copies of E. The gauge changes the outer function from h_2 to $e_2 = \iota(h_2) \neq h_2$. This is a geometric I–IV example, not an assertion of source minimality or scalar activity.

Proof. The extreme copy of K is contained in P, and a second outer diagonal contributes contents outside that copy, proving $|K| < |P|$. For the other factor, put $\alpha = |W|$, $\beta = |\widehat{W}|$, $\gamma = |\widehat{O}|$. The full modified-composition identity and its generator-span normalization give

$$|K| = p + \alpha + \beta + \gamma, \qquad |P| = p|A| + \alpha r(A) + \beta c(A) + \gamma N_A.$$

An ordinary gauge exchanges the indicated row and column conventions together. All three coefficients are nonnegative and a proper inner has $p \geq 2$, so $|P| \geq 2|A| > |A|$. These are the full all-type identities, including the mixed supplements [3, 7]. A strict finite cell subset has smaller area. Homogeneity of a skew Schur function transfers the area equality to every mate. We invoke this lemma only after the actual cell object and its proper inclusion have been established. □

3. Factor readout and the two-copy reduction

Apply [10, Ribbon-outer period rigidity] to the original charts in Theorem 1.1. It gives $p = q$. Throughout this section p denotes this common original period. The span law gives equal widths and outer spans. By corner transport, a nonempty component of C^- would contribute a connected prime of span at least p to the common corner, whereas the ribbon-outer chart contributes only factors of span below p. Thus C is a ribbon. These uses require no current-area instance of $\mathsf{J}_{<n}$.

3.1. **Ordinary and mixed readout.** Put

$$\delta_k = p_k^{\perp} = \sum_{j \geq k} h_{j-k} \frac{\partial}{\partial h_j}, \qquad \mathcal{V}_k = \sum_{j \geq 1} j h_j \frac{\partial}{\partial h_{j+k}}.$$

The two derivations have different roles. In particular, $\mathcal{V}_k h_k = 0$, whereas $\delta_k h_k = 1$. A mixed period-p chart has $\delta_{kp} s_F = 0$ for every $k \geq 1$: its complete fixed boundary word makes every shifted Jacobi–Trudi row either duplicate or zero. A proper ordinary chart has $\delta_p s_F \neq 0$, by the ordinary readout theorem [8, Factor recovery at a prescribed period and mixed-vanishing lemma]. Thus the two original equal-period charts have the same ordinary or mixed category. Empty marks belong to the ordinary category.

For a nonzero rational multiple of a skew Schur polynomial, prim clears denominators, divides the integer content, and makes the coefficient of the minimum-square monomial positive. The ordinary readout gives

$$a = \max\{j : \delta_p^j P \neq 0\} = |D|, \qquad s_E = \operatorname{prim}\left(\frac{\delta_p^{a-1} P}{\gcd(\delta_p^{a-1} P, \delta_p^a P)}\right), \tag{3.1}$$

and the ratios $\delta_{p\lambda} P / \delta_p^a P$, for $\lambda \vdash a$, recover the complete outer character. The recovered outer is s_D in case (a) and $\iota(s_D)$ in case (b), exactly as required by (2.10).

For the mixed case, rotate the whole chart to case (d). In the notation of the mixed weighted covariance and mixed ribbon factor recovery theorems of [8], put

$$\rho = s - t, \quad A = s_W, \quad B = s_{\widehat{W}}, \quad C_0 = s_{\widehat{O}}, \quad \chi = (-1)^{-v_1} ABC_0, \quad \mathcal{B}_k = p\mathcal{V}_k + \rho\delta_k,$$

where $0 \leq t < s \leq w < p$. The map is $\phi(h_j) = A^{j-1} s_{\mathrm{Row}_j \circ_\omega E}$ for $j \geq 1$, with $\phi(1) = 1$. The following displayed specializations will also be used by the two-copy argument. The weighted beta-move identity of [8] is

$$\mathcal{V}_r s_{X/Y} = \sum_{x \in X} (x - r) \det \mathrm{JT}_{X : x \mapsto x-r, Y} - \sum_{y \in Y} y \det \mathrm{JT}_{X, Y : y \mapsto y+r}\,. \tag{3.2}$$

The lists remain in their original positions; subsequent sorting contributes its determinant sign, including at zero and negative entry indices. For $1 \leq k \leq m$, the column move at $(m-k)p + s$ in the actual row power leaves

$$(\mathrm{Row}_{m-k} \circ_\omega E) \;\sqcup \bigsqcup_{j=m-k}^{m-1} (\widehat{O} + jv + e) \;\sqcup \bigsqcup_{j=m-k+1}^{m} (\widehat{W} + jv + e). \tag{3.3}$$

This is the mixed strip remainder of [8], with the first term interpreted as W at $m = k$, not as the algebraic unit. Its marked intervals have heights differing from those of $\widehat{W}$ by one precisely on residues $t, \ldots, s-1$. For a homogeneous Q of weight m, the mixed terminal evaluation of the same reference gives

$$\mathcal{B}_1^m Q = m! p^m Q(1^{\rho/p}). \tag{3.4}$$

It is nonzero when Q is a ribbon Schur function. Thus the mixed ribbon inverse uses $m = \max\{j : \mathcal{V}_p^j P \neq 0\}$, the terminal quotient $s_E/(\rho\chi)$, and the fixed-order triangular system for the outer character.

Applied to the two actual mixed charts in (1.2), that inverse gives $s_E = s_U$ and $s_D = s_C$, with genuine outer functions. In the ordinary case (3.1) gives $s_E = s_U$ and equality of the recovered outers in the indicated gauge. Both inners have area less than n, by Lemma 2.7.

If $U = E$ or E^*, transport its full upper mark to E. The fixed-width atlas in Proposition 2.3(1) identifies the marks, with only the ordinary gauge as a possible double. In the mixed case the mark is unique. The recovered outer functions then give exactly (1.3). A chartless E forces this case by Lemma 2.6(2).

3.2. **Why the remaining original outer has two cells.**

Lemma 3.1. *If the original ribbon outer has at least three cells, then Theorem 1.1 holds. For a two-cell original outer, it also holds when the opposite domino probe is strictly smaller than the original parent. In the mixed case only the row-two selector remains.*

Proof. Use the composition covariance of [8] on the actual original chart. For an ordinary mark put $A = s_W$, $B = s_{\widehat{W}}$, $C_0 = s_{\widehat{O}}$, and let $\mathcal{M}_\omega(h_j) = A^{j-1} s_{\mathrm{Row}_j \circ_\omega E}$. Its algebraic value on one is one. In the read-outer convention,

$$(\Phi, Q) = \begin{cases} (\mathcal{M}_\omega, s_D), & \text{(a)}, \\ (\mathcal{M}_\omega \circ \iota, \iota(s_D)), & \text{(b)}. \end{cases}$$

For $a = |D|$, $r_D = \operatorname{rows} D$, $c_D = \operatorname{cols} D$, and $M_D = N_D$, the entire ordinary normalization is

$$\Phi(Q) = A^{a-r_D} B^{a-c_D} C_0^{a-M_D} s_{D \circ_\omega E}.$$

Write $e = s_E$, $\xi = \delta_p e$, and $\theta = \mathcal{V}_p e/\xi$. The cited covariance gives $|\theta| < p$ and

$$\mathcal{V}_{kp} \Phi(f) = \xi^k \Phi((p\mathcal{V}_k + \theta\delta_k) f).$$

In case (b), the coarse involution in Φ, the physical signs, and the whole normalizer are part of this identity. At either genuine empty edge $\theta = 0$. For a mixed chart use the positive-phase convention above, with $\theta = \rho$, $\Phi = \phi$, $Q = s_D$, and ξ replaced by χ. Its full modifier has the ribbon sign $(-1)^{r_D - 1}$, retained below in $\mathcal{N}_D$.

Let Q be the read outer function in this convention, and let $m = |D| \geq 3$. Write

$$(p\mathcal{V}_{m-2} + \theta\delta_{m-2}) Q = \gamma p_2 + \lambda p_1^2.$$

Let χ_Q be the skew character with Frobenius characteristic Q. Its p_2-coefficient is

$$\gamma = p\chi_Q(m) + \frac{\theta}{2} \chi_Q(m-2, 2) \neq 0.$$

The read ribbon is D or $\mathrm{AT}(D)$, according to the displayed convention. If it has l rows, then $\chi_Q(m) = (-1)^{l-1}$. Its interval-coarsening expansion gives

$$\chi_Q(m-2, 2) = (-1)^{l-1} \big(1 - \mathbf{1}_{2 \text{ a proper partial sum}} - \mathbf{1}_{m-2 \text{ a proper partial sum}}\big).$$

The occurrences are counted separately at $m = 4$. Its absolute value is at most one, so $|\theta| < p$ proves nonvanishing, also at $m = 3$.

Retain $A = s_W$ and $B = s_{\widehat{W}}$ for the actual current mark. Let ζ denote ξ or χ, and write the complete composition normalization as $\Phi(Q) = \mathcal{N}_D P$. The scalar-normalization factor $S = \zeta^m/\mathcal{N}_D$ is read by

$$S = \frac{\delta_p^m P}{\delta_1^m Q} \quad \text{in the ordinary case,} \qquad S = \frac{\mathcal{V}_p^m P}{(p\mathcal{V}_1 + \rho\delta_1)^m Q} \quad \text{in the mixed case.}$$

The ordinary denominator is the positive number of standard tableaux of the ribbon, and the mixed denominator is nonzero by (3.4). Hence

$$\zeta^2 \mathcal{V}_{(m-2)p} P/S = \gamma \Phi(p_2) + \lambda e^2 \tag{3.5}$$

reads $\Phi(p_2)$. Newton's identity reads $\Phi(h_2)$ and $\Phi(e_2)$. In case (a), these are $A s_{\mathrm{Row}_2 \circ E}$ and $B s_{\mathrm{Col}_2 \circ E}$; case (b) swaps their assignments. In case (d), they are

$$A s_{\mathrm{Row}_2 \circ E}, \qquad -B s_{\mathrm{Col}_2 \circ E}.$$

The minus sign is the column-domino specialization of the full mixed modifier in [8, Mixed weighted covariance]. Each expression has a unique connected prime factor of span greater than p, namely the actual domino probe of span $2p + w$. Thus both original parents read the same two probe functions.

Choose a domino contained in D. Its composition is a translated strict cell subset of F. For ordinary marks this is immediate from the affine origins (2.7). For mixed marks, a ribbon has one cell on each diagonal, so the rank correction in (2.7) vanishes; the supplemental mark of a contained vertical pair is exactly the one present in the whole composite. The probe has smaller span and hence is a strict subset. Its mate has the same smaller area. Since E is not a line, outer-independent minimality makes the displayed probe chart source-minimal. Lower full-fiber transport now puts the mate probe in its prescribed fiber. The mate has two actual period-p charts, so same-parent equal-period comparison identifies the full marked inners up to the permitted gauge and rotation. The recovered outer functions give (1.3).

At $m = 2$, (3.5) would require a zero-index operator, which gives no smaller probe. Instead the recovered outer and terminal derivative determine $\mathcal{N}_D$; Newton's identity then reads the opposite domino. It may be used for descent exactly when it is strictly smaller. The ordinary domino areas are $2|E| - |W|$ and $2|E| - |\widehat{W}|$. In the mixed case, the marked intervals in (3.3) give $|W| - |\widehat{W}| = s - t > 0$. Hence the mixed column-two parent is strictly larger than its row-two probe and is discharged. Only the mixed row-two selector remains. □

3.3. **The actual common finer source.**

Proposition 3.2. *Under the hypotheses of Theorem 1.1, either the prescribed-fiber conclusion already holds, or, after a permitted whole rotation, there are actual proper source-minimal charts*

$$E = V \circ_\mu J, \qquad U = V' \circ_\mu J, \qquad s_V = s_{V'}, \tag{3.6}$$

with the same literal connected inner J and the same full legal mark μ. The minimum is held on E and transported to U; no minimum is imposed on J.

Proof. The literal and rotation cases were settled before Lemma 3.1. Otherwise the lower class has at least two rotation orbits. Root existence and rootedness transfer make E rooted. Choose a genuine source-minimal proper chart $E = V \circ_\mu J$. The full-fiber consequence Lemma 2.6(1) gives a chart of U with equal genuine outer function and with marked inner μ or its transported rotation. In the latter case rotate the entire original chart of H, including its upper mark and supplements. This gives (3.6); the target fiber is rotation-stable, so the final conclusion transports back. Lower key invariance identifies the source-minimal score on U. The construction uses only classes of area $|E| < n$. □

3.4. **The remaining geometric domain.** At the nonliteral landing (3.6), put

$$N_J = r + v, \qquad N_V = N_{V'} = M, \qquad S = N_E = N_U = Mr + v, \qquad Q = S - w. \tag{3.7}$$

Here r, v are the finer period and width, and Q is the original upper period. In particular, Q is not a stationary exponent. If $V = V'$, then $E = U$ and the direct fiber argument applies. Equal coarse functions with $M \le 2$, or with a line as one realization, force literal equality. Thus every remaining finite comparison has $M \ge 3$ and nonline coarse outers.

Proposition 3.3. *At* (3.6)*, the theorem is reduced to*

$$J \text{ nonline}, \qquad 0 \le w < r, \qquad w \ne v, \tag{3.8}$$

with the same actual full upper mark and the same actual domino selector on E, U. The proof includes the empty-width case and the original selector and gauge. Every excluded case either already gives the prescribed fiber or constructs an actual proper chart of a held source or original parent with strictly smaller score.

Proof. If $r \mid Q$, apply the complete aligned theorem to the actual finer chart and the actual upper mark. It gives a proper period-r chart of the original parent. Since $Q > w$, $M \ge 3$, and $w \equiv v \pmod r$, its score $(v, r + v)$ is strictly below $(w, Q + w)$. This contradicts the original source minimum.

The wide case $w > r$, the boundary case $w = r$, and the literal line-finer case are proved in Lemmas A.2 to A.4. Their source constructions precede every comparison of scores. They leave (3.8); the equality $w = v$ is already aligned.

For $0 < w < r$, the first and last r source windows in (3.6) come from the same literal J. Their final-copy origins agree: in the ordinary cases the origin also reads the equal coarse row count, and in the mixed cases it is $(M-1)v_\mu$. Supplements do not enter these extreme windows. The complete windows $[0, w]$ and $[S-w-1, S-1]$ therefore agree as cell sets, and $2w < S$. The mark-locality lemma [9, Locality of legal marks in the two end windows] transfers the full upper mark, including both hatted clauses at every integer translate. The fixed-width atlas permits only the ordinary double; applying its genuine gauge also changes the actual upper domino.

After this transfer the selectors agree. If the source has t_E rows and the mark a rows, put $d = t_E - a$. The row-two parent has $t_E + d$ rows. An ordinary(a) column-two parent has

$t_E + d + 1$; an ordinary(b) one has $t_E + |d - 1|$. Both differ from the row count of the row-two parent, including $d = 0$. In a mixed upper, the column-two parent has the same two copies as the row-two parent and a strictly positive supplemental patch. The patch is determined by the common two end windows, so its area is the same on both sources. Equal whole functions thus force the same selector by row count in the ordinary cases and by area in the mixed cases. At $w = 0$, use the actual horizontal edges: the two selectors have different row counts, and converting a vertical edge applies (2.10). No positive-width argument is used for this empty case. □

The remaining ordinary and empty comparisons must yield full marked reflection or an improving aligned source chart; their gcd-one certificates retain the actual ordinary positive, empty-upper, or mixed sign endpoint alternatives. A mixed comparison must instead give a literal or whole-parent fiber finish, or a legal divisor alignable through the held source charts. Only a private active mixed second response requires stationary separation.

4. Finite comparisons on actual composition charts

The reduction leaves two actual compositions through the same full marked inner,

$$E = V \circ_\mu J, \qquad U = V' \circ_\mu J, \qquad s_V = s_{V'}, \qquad V \neq V', \tag{4.1}$$

where the displayed chart of E is source-minimal. Write

$$N_J = r + v, \qquad N_V = N_{V'} = M \geq 3,$$
$$S = N_E = N_U = Mr + v, \qquad Q = S - w = Mr + \delta, \quad \delta = v - w.$$

The original upper width satisfies $0 \leq w < r$, $w \neq v$. This section extracts information from the actual whole-parent comparison. In particular, the span M of the finite algebra below is supplied by V; it is not replaced by a smaller test outer.

4.1. **A complete raised current.** For a connected source with full Jacobi–Trudi matrix $A = (h_{x-y})$, set

$$e = \det A, \qquad \kappa = \partial_{h_S} e, \qquad K_Q = (h_{Q+x-y}), \qquad T_Q(A) = \operatorname{tr}(\operatorname{adj}(A) K_Q).$$

The raised matrix includes entries that were negative-index zeros in A. A raised index zero is the actual unit $h_0 = 1$. Let $\mathcal{G}_Q = (\operatorname{adj}(A) K_Q \operatorname{adj}(A))_{0,S}$, with the indicated entry indexed by its actual column and row.

Lemma 4.1. *At the matched narrow upper comparison in* (4.1)*, the actual whole-parent equality supplies*

$$T_Q(E) = T_Q(U). \tag{4.2}$$

Both sources also satisfy the open mate implications

$$\begin{aligned} x \in X_E, \quad 0 < x < w \implies Q + x \in X_E, \\ y \in Y_E, \quad Q < y < S \implies y - Q \in Y_E, \end{aligned} \tag{4.3}$$

and the corresponding implications on U. The assertion holds for all four positive upper attachment types, and for the genuine ordinary empty upper when the finer is ordinary or empty. A mixed finer with an empty upper is already disposed of by Lemma F.1.

Proof. Form the actual two-copy diagram $\mathrm{Row}_2 \circ_\omega E$. Its first and last Q diagonals are the actual end windows of E. The complete coding table changes initial to terminal bits only as follows:

	column changes	row changes
(a)	$t : 1 \longmapsto 0$	$s : 0 \longmapsto 1$
(b)	$s : 1 \longmapsto 0$	$t : 0 \longmapsto 1$
(c)	none	$t, s : 0 \longmapsto 1$
(d)	$t, s : 1 \longmapsto 0$	none.

A low row is never lost and a high column is never new, giving (4.3), including endpoint contacts by the strict displayed ranges. This uses the actual supplemented composition formula; it is not an implication of a height word alone.

For a positive ordinary upper, Lemma B.2 gives the whole $\mathcal{G}_Q$ comparison. At an empty ordinary upper the row-mark splice in Lemma B.1 gives that comparison directly. In the vertical reading, the complete structured dual identity Lemma E.5 retains the two additional terms of the inverse-alphabet expansion:

$$\sum_{0<x<w} L_x(HR)_{Q+x}, \qquad \sum_{Q<y<S} R_y(HL)_{S+Q-y}.$$

Here L, R are the complete global null records. The actual mate implications make each numerator in these sums a source null equation; thus both sums vanish. The remaining sign is $(-1)^{w+1}$, common on the matched pair. This proves the same raw $\mathcal{G}_Q$ comparison in both ordinary readings, while rebuilding both fields.

For a mixed upper, the complete compound expansion is proved in Lemma B.3. If K_t is the actual top-row/returned-column deletion, it gives

$$k_t T_Q(A) - e T_Q(K_t) = (-1)^{a+b}\, b_t\, s_{\mathrm{Row}_2 \circ_\omega E}, \tag{4.4}$$

where a, b are the actual upper mark and private ranks. For that upper overlap W, of width w, put $b_t = \det \mathrm{JT}(X_W \setminus \{w\}, Y_W \setminus \{t\}) \neq 0$, with both lists decreasing. This is the unsigned proper mark minor of Lemma B.3. The complete corner has factorization $k_t = b_t n \widehat{A}$, with the entire private determinant n. The common top coefficient of e, together with the common end data, fixes n; hence k_t is common. Its connected factors have smaller weight than e. Subtraction of (4.4) and connected primality give $e \mid \Delta T_Q$. A complete current has ordinary moment degree at most the number of source rows, equal to the ordinary degree of e. Its surplus weight is $Q > 0$, so the quotient must be zero. Whole rotation proves the other mixed reading.

For completeness, the same degree argument converts the ordinary $\mathcal{G}_Q$ comparison into (4.2). Differentiate the entire adjugate, including the variation of K_Q, to obtain

$$\mathcal{G}_Q = \kappa T_Q - e\, \partial_{h_S} T_Q + e\, U_w(e), \qquad U_w(e) = \begin{cases} \partial_{h_w} e, & w > 0, \\ n_E e - \sum_{d>0} h_d \partial_{h_d} e, & w = 0, \end{cases} \tag{4.5}$$

For $w > 0$ the derivative is at a genuine positive index. For $w = 0$, n_E is the actual row rank and the displayed expression is the entire unit-cofactor sum, earned by homogeneous Euler identity. The common whole function and row rank make this term common. Subtraction gives $\kappa \Delta T_Q = e \partial_{h_S} \Delta T_Q$. The connected polynomial e is prime and coprime to its proper corner κ; the preceding degree and weight argument again gives zero. No individual cofactor summand has been observed separately. □

4.2. **The two actual returned coordinates.** Normalize an ordinary finer to case (a) by the whole-chart transport of Lemma A.1. For an original case (b) comparison, transport both sources and their complete upper presentations first, rebuild the fields, and apply Lemma 4.1 on those image charts. The current and global scalar are thereby earned in the normalized field, not identified with an unmarked bare antitranspose scalar. If the finer contacts are s, t, its complete flags are

$$\begin{aligned} X_J &= (r + X_W) \sqcup B_- \sqcup (X_W \setminus \{s\}), \\ Y_J &= (r + (Y_W \setminus \{t\})) \sqcup C_- \sqcup Y_W, \end{aligned} \tag{4.6}$$

where $B_-, C_- \subset (v, r)$. Its returned coordinate is a *numerator* coefficient. In the actual normalized rational chart,

$$H = P'/G', \qquad G'_t = 1, \qquad P'_s = R_0.$$

The outside coordinates are precisely the available P'_i and G'_j with $v < i, j < N_J$; a numerator coordinate is available at a missing row, and a denominator coordinate at a present column.

For mixed case (d), with $0 \le t < s \le v$, the full flags are

$$\begin{aligned} X_J &= X_W \sqcup B_- \sqcup (r + X_W), \\ Y_J &= (Y_W \cup \{s\}) \sqcup C_- \sqcup (r + (Y_W \setminus \{t\})). \end{aligned} \tag{4.7}$$

The returned coordinate is now a *denominator* coefficient:

$$G'_t = 1, \qquad G'_s = -R_0, \qquad R_0 = (-1)^{|Y_W \cap (t,s)|} \frac{s_W}{\det \mathrm{JT}(X_W, (Y_W \setminus \{t\}) \cup \{s\})}.$$

Its coherent degree is zero and its weight is $\rho = s - t > 0$. This distinction explains why the ordinary finite argument does not kill the mixed stationary responses by homogeneity alone.

Lemma 4.2. *For each of these two cases, the returned coefficients $R_1, \dots, R_M$, all genuine outside high coefficients at depths $1, \dots, M-1$, and the complete neutral fine coordinates give a rational coordinate chart of the actual whole source field. All returned coefficients remain fixed when an outside coefficient is varied.*

Proof. Write $R(u) = \sum_{j=0}^{M} R_j u^j$. For an ordinary finer use

$$\begin{aligned} P_{\mathrm{full}} &= P' + z^s (R(z^r) - R_0) + \sum_{a,i} \alpha_{a,i} z^{(M-a)r+i}, \\ G_{\mathrm{full}} &= G' + \sum_{a,j} \beta_{a,j} z^{(M-a)r+j}, \qquad H_{\mathrm{full}} = P_{\mathrm{full}} / G_{\mathrm{full}}. \end{aligned} \tag{4.8}$$

For a mixed finer replace these first two lines by

$$P_{\mathrm{full}} = P' + \sum_{a,i} \alpha_{a,i} z^{(M-a)r+i}, \qquad G_{\mathrm{full}} = G' - z^s (R(z^r) - R_0) + \sum_{a,j} \beta_{a,j} z^{(M-a)r+j}. \tag{4.9}$$

Every outside wave has its actual label strictly between N_J and S. In the ordinary case, the source-clipped records $z^{kr} G_{\mathrm{full}}$ annihilate every fixed row and return R_{j-k} at the actual returned row. The full fixed-word solve therefore recovers exactly these coefficients. Conversely it recovers G_{full}, and multiplication by H_{full} recovers all available numerator coefficients. This is an inverse rational map.

For the mixed case use the square baseline containing every fixed column and all columns $jr + s$, $0 \le j \le M$. Its determinant is

$$\widehat{A}^{M+1} C_{\mathrm{priv}}^{M}, \qquad \widehat{A} = \det \mathrm{JT}(X_W, (Y_W \setminus \{t\}) \cup \{s\}), \quad C_{\mathrm{priv}} = \det \mathrm{JT}(B_-, C_-). \tag{4.10}$$

Both factors are nonzero weak skew determinants; empty blocks have determinant one. Its actual inverse sends the column $kr+t$ to the $jr+s$ coefficient R_{j-k}. The final partial mark block gives this coefficient also at $j = M$: its constant inverse row has no private part. Complementary-column expansion thus gives the complete coarse Toeplitz determinant, with the actual sorting sign. Equality of the source and coarse functions fixes this sign on the two sources. The inverse solve recovers the whole denominator null record and all its outside coefficients. Before recovering the neutral seed, add back $z^s(R(z^r) - R_0)$. This is necessary when the R_1 term occurs at $r + s < N_J$. It cannot be confused with an outside denominator coordinate, since $r + s$ is absent from the actual finer column list. These solves prove the assertion on the nonempty open set where the complete fixed determinants and the normalizing coefficients are nonzero. □

For an empty finer, the actual chart is different:

$$\begin{aligned} G_0 &= 1, \qquad P_0 = R_0 = \nu, \\ P_{\mathrm{full}} &= P + \sum_{j=1}^{M} R_j z^{jr} + \sum \alpha_{a,i} z^{(M-a)r+i}, \\ G_{\mathrm{full}} &= G + \sum \beta_{a,j} z^{(M-a)r+j}. \end{aligned} \tag{4.11}$$

There is one coherent unit, and $P_r = R_1$ is returned. The proper fixed matrix has M copies of the full proper corner of J, including all crossing blocks. The same direct and inverse solve proves (4.11); no outside derivative of P_r is used.

4.3. **Why two finite waves suffice.** Let $\mathcal{A}_V(j,k)$ be the complete keyed coarse adjugate, zero at absent flags, and let $\Delta\mathcal{A} = \mathcal{A}_V - \mathcal{A}_{V'}$. A Schur-complement differentiation of the entire source gives

$$\sum_{j,k} \Delta\mathcal{A}(j,k)(WK_QU)_{j,k} = 0, \qquad W = (I, -BD^{-1}), \quad U = \begin{pmatrix} I \\ -D^{-1}C \end{pmatrix}. \tag{4.12}$$

The common fixed trace is multiplied by the common entire coarse determinant and cancels as a whole. The fixed block D retains both partial end blocks and all private factors.

Here are the response series used in (4.12). Put $\mathcal{H}_d(u) = \sum_{n\geq 0} h_{nr+d}u^n$. For an ordinary finer let

$$D(u) = (\mathcal{H}_{b-c}(u))_{b\in B_0, c\in C_0}, \quad L = \mathcal{H}_{s-C_0}D^{-1}, \quad \Theta_d = \mathcal{H}_{s-d} - L\mathcal{H}_{B_0-d},$$

where $B_0 = (X_W \setminus \{s\}) \sqcup B_-$ and $C_0 = (Y_W \setminus \{t\}) \sqcup C_-$. Set $\Lambda_b = L_b$ at an actual fixed row, $\Lambda_s = -1$, and zero at absent rows. Extend by $\Lambda_{b+r} = u\Lambda_b$. The responses are

$$F_{P_i} = \Lambda_{i-\delta}, \qquad F_{G_j} = \Theta_{j-\delta}.$$

For mixed (d), use

$$D_s(u) = (\mathcal{H}_{b-c})_{b\in X_W\sqcup B_-, c\in (Y_W\setminus\{t\})\sqcup C_-\sqcup\{s\}}, \quad \ell = e_s^T D_s^{-1}, \quad \Theta_d = \ell\mathcal{H}_{(X_W\sqcup B_-)-d},$$

and

$$F_{P_i} = \ell_{i-\delta}, \qquad F_{G_j} = -\Theta_{j-\delta}.$$

The exact distinguished values are $\Theta_c = 0$ on fixed columns, $\Theta_s = 1$, and $\Theta_t = R(u)$. At an empty finer take the ordinary formulas with actual returned phase zero and its independently constructed chart (4.11).

Proposition 4.3. *At the actual comparison above, write $F_n = [u^n]F$. At the supplied M,*

$$F_M = 0, \qquad \partial_x F_n = 0 \ (2 \leq n < M), \qquad F_1 = Ax + b, \quad A^T = -A, \tag{4.13}$$

where x lists every outside coordinate. For a positive ordinary finer all coefficients of A, b, and all stationary responses, depend only on the actual returned clock. The ordinary and empty cases satisfy $F_2 = \cdots = F_M = 0$. For a mixed finer,

$$F_{n,P} = 0 \ (1 \leq n \leq M), \qquad F_{1,G} = A_{GG}(R_0)x_G + b_1(R_0), \quad F_{n,G} = b_n(R_0) \ (2 \leq n < M). \tag{4.14}$$

The assertion is obtained independently in both own fields: each is the full rational coordinate field rebuilt from that chart's complete source matrix and returned normalization. Equal whole functions do not identify their pivots or inverse matrices.

Proof. The neutral primitive numerator contributes to a raised record only at $k = M - 1$ and $0 < x < w$. Its index is $r + \delta + x$. The actual mate implication makes this a proper finer row, either in the last private band or in the upper mark band; its numerator coefficient is zero. This proves neutral cancellation for both signs of δ.

On the simultaneous high-wave chart, a numerator pulse of cutoff a meets row $(k-a)r+i-\delta$. Its exact prefix is $k \geq a$. An apparent wrapped row at $k = a - 1$ would have phase $b < w$; (4.3) would force the input i to be a present finer row, contrary to its availability. In the ordinary returned phase it is the absent coarse row zero; in the mixed phase the actual repeated row word makes the same contradiction. A denominator pulse lies beyond the source exactly for $k \geq a$ and gives the complete omitted-column residual. Thus the live crossing is

$$-\sum_{a,i} \alpha_{a,i}[k \geq a]F_{j-k+a,i},$$

up to the single consistent choice of response sign. Every inverse is computed after the pulses are inserted.

Scale all positive coarse returns by an auxiliary parameter. Let μ be the first nonzero grade of the *entire* array $\Delta\mathcal{A}$. Lemma B.4 proves that, at that same grade, there is $d \geq \lceil M/2 \rceil$, $a_0 = M - d$, for which

$$C_{1,d} = C_{a_0,d} \neq 0, \qquad C_{a,d} = 0 \quad (a > a_0), \qquad C_{a,D} = -\sum_{j-k=D, k\geq a} \Delta\mathcal{A}^{[\mu]}(j,k).$$

All higher source corrections multiply a lower entire zero array. Taking one or two actual waves and then separating genuine coarse weights gives

$$C_{a,D} F_{D+a,i} = 0, \tag{4.15}$$

$$C_{a,D}\partial_j F_{n,i} + C_{b,D}\partial_i F_{n,j} = 0, \qquad n = D + a + b - M. \tag{4.16}$$

The second equation contains both pulse assignments and the variation of the inverse against the old crossing. The high variation is exactly u^{M-b} times the complete fine variation, by $dD^{-1} = -D^{-1}(dD)D^{-1}$.

Use (4.15) at a_0 to obtain $F_M = 0$. For $2 \leq n \leq d$, use (4.16) at $(a,b) = (a_0 + n - 1, 1)$; for $d < n < M$, use (n, a_0). The first coefficient is zero and the second nonzero, giving outside constancy. The choice $(a_0, 1)$ gives the Killing equation. Differentiating its three cyclic versions gives all second derivatives zero in characteristic zero, hence (4.13).

Every nonreturned positive low coordinate has its actual high mate and $\partial_{\mathrm{low}} F_n = \partial_{\mathrm{high}} F_{n+1}$. At zero, if $t > 0$, varying G_0' also varies $P_0' = \nu G_0'$, giving the combined high derivative $\partial_{G_r'} + \nu\partial_{P_r'}$. Homogenizing the *whole* equal-rank source identity gives the true coherent-unit alias $\partial_\nu F_n = G_0'\partial_{P_r'} F_{n+1}$. For $t = 0$, $G_0' = 1$ is fixed and only the latter alias is used. For positive ordinary finers, descent from $F_M = 0$ leaves each stationary coefficient a function of R_0. Its coherent degree makes its P part constant and its G part proportional to R_0. Its actual weight $nr+s+\delta-i$, $n \geq 2$, exceeds s, whereas the two possible weights are zero and $s - t$; hence it vanishes. For empty finers the outside coordinates already exhaust the fine field over the single weight-zero unit ν, and the positive weight $nr - w - i$ gives the same conclusion without an alias of P_r.

For mixed finers the stationary responses are outside-constant before using these aliases, so all their nonreturned mark derivatives vanish. The same holds for the coefficients of F_1. The mixed clock has coherent degree zero, while a numerator response has degree -1; therefore every positive-order numerator response vanishes. The remaining GG coefficients and stationary values are rational functions of R_0, as asserted. The generic endpoint is $F_M = 0$, so at $M = 3$ it is $F_3 = 0$. No coefficient beyond the original horizon has been used. □

4.4. **The ordinary and mixed first responses.** The finite block is not yet a geometric equality. Its exact further outputs are collected here; the complete algebra is in Lemmas C.2 to C.4. The first-response matrix is the matrix A of (4.13):

$$A_{\alpha\beta} = \partial_{x_\beta}[u] F_\alpha(u).$$

Its rows and columns have the same outside-coordinate labels: P_i' at missing rows $v < i < N_J$, and G_j' at present columns $v < j < N_J$. The response vector is the one defined after (4.12), in that chart's own normalized rational field. All returned and remaining neutral coordinates are held fixed; the complete inverse matrices are differentiated. At an empty finer use the unprimed coordinates of (4.11). An *active pair* means a nonzero entry of the relevant PG or GG first-response matrix, not merely a present coordinate or a nonzero intercept. A private fine index lies in (v, r). For an ordinary pair the inputs and their opposite-chart inputs must lie in that band, with targets $i - \delta, j - \delta$ checked separately. For the mixed second response, the full condition is $v < j, k, j - \delta, k - \delta < r$, as in (4.22).

Proposition 4.4. *At the actual narrow ordinary or empty finer comparison, the first matrix is a constant pure PG skew matrix in each own returned chart. All genuine returned intercepts remain. At negative δ, and at an empty finer, every active pair has four private inputs. In the principal ordinary range, Lemma E.4 gives the complete terminal-or-private alternative, including the actual head and transpose checks.*

At a positive mixed finer the entire GG matrix is constant. If it is nonzero, the actual coarse diagrams satisfy $V' = V^*$, and every active first row is free of its unwanted intercept. For an active pair j, k,

$$j + k = r + s + t + \delta, \qquad \eta Q_{k+\epsilon} = cG_k/g_t, \qquad \eta Q_{j+\epsilon} = -cG_j/g_t, \qquad \epsilon = v - s - t. \tag{4.17}$$

The constants and all proper factors are in their own fields. Complete coface factorization gives two actual connected equal-child pairs and two complete component-product equalities, with their sorting signs and multiplicities. The entire-zero-matrix case has its separate native-word conclusion in Lemma D.6; it does not require $V' = V^*$.

Proof. For an ordinary finer, substitute the full moving returned series in the one-wave equation and take its outside curl. The exact equations are $H_a C_{PP} = 0$, $J_a C_{GG} = 0$, together with $K_a C_{PG} = 0$ when the coarse pair has not yet been reflected. Lemma C.2 proves that the J family is nonzero, and either the K family is nonzero, when the opposite own field excludes PP, or every K vanishes, when the same-cap and linearized same-cap arguments make $H_2 \neq 0$. Thus the first matrix is pure PG. The actual zero-order vector is zero at negative δ and at empty width; its genuine low aliases remove every terminal active input.

For a mixed finer, two actual wave depths first give the whole ordered sum, and the actual curl gives the whole product, of the two endpoint polynomials. The full own-atlas argument in Lemma C.3 excludes every nonconstant clock power in a matrix entry. At power zero, the crossed factor assignment is exactly the actual coarse rotation. The full uncurled first-response equation then gives the finite Leibniz defect of Lemma C.4. Its positive-power exclusion removes the unwanted active intercepts. The permitted constant and linear returned terms are not active: their putative partners would be fixed low returns. The actual one-period inverse row gives (4.17). Clearing its full prefix and suffix denominators and using connected primality identifies the largest connected numerators, then the remaining *complete* products. The two equations give opposite actual signs. The physical form and area inequalities are recorded in Lemma D.1. □

4.5. **The remaining mixed stationary response.** The mixed first matrix may be nonzero. Its active pairs supply the physical child comparison, while a second response is still retained. The complete finite identity now provides precisely the input to whole-field stationary descent [6].

We use the complete coframes of [6] in the actual coarse alphabet. For a board with full lists $X = \{M\} \cup I$, $Y = S \cup \{0\}$, put

$$T = (H_{i-s})_{i \in I, s \in S}, \qquad G = 1 - \sum_{s \in S} (T^{-1}(H_i)_{i \in I})_s z^s.$$

All zero-index entries are the same returned clock $H_0 = \vartheta = R_0$; negative indices are zero. The inverse is taken in the whole rational field, with every unit and crossing retained. The empty proper matrix gives $G = 1$. The left coframe C is the right coframe of the normalized rotation in that same alphabet. For these application coordinates write

$$\mathcal{T}_{G,C}(u) = G[uC]_{<M} - C[uG]_{<M}, \qquad D = \mathcal{T}_{G,C}(H). \tag{4.18}$$

The signed proper corner is the actual coefficient of the top letter, so the complete whole determinant is $\kappa[HG]_M$. The coequal-data lemma of [6] gives $\operatorname{ord}_z D \geq M+1$, $\deg_z D \leq 2M-2$, and $D \neq 0$ for a distinct pair. Here M is the supplied coarse span, never a smaller test span. The whole returned series may retain its higher coefficients: they do not affect the clipped terms or the required finite horizon.

Proposition 4.5. *On the nonzero mixed branch, let $V' = V^*$, and let f be any actually supplied active second response. Write $\vartheta = R_0$ for the actual returned clock and $H = \vartheta + \sum_{i>0} H_i z^i$. The positive coarse letters are independent over the neutral fine field. There is a potential ψ_0 in that field and the full moving homomorphism $\mathcal{R}$, sending ϑ to H, for which*

$$G[zf(H)C]_{<M} - C[zf(H)G]_{<M} + [z\{G[HC]_{<M} - C[HG]_{<M}\}\mathcal{R}(\psi_0)]_{<2M} = 0. \tag{4.19}$$

Every coefficient through $2M-1$ is supplied. The response f is rational in the true returned clock before any integer-power extraction.

Proof. Put $\tau = \mathbf{1}_{s<\delta}$, $\beta = s-\delta+\tau r$, and use the mixed square inverse and residual Θ_β defined after (4.12). Write H_R for the whole physical alphabet reconstructed by (4.9) with returned series $R(u) = R_0 + \sum_{j=1}^{M} R_j u^j$, the neutral fine coordinates fixed, and the outside waves zero. Define the returned response by

$$T(u; H_R) = \Theta_\beta(u; H_R), \qquad T_m(H_R) = [u^m]T(u; H_R).$$

This is substitution in the entire rational solve before coefficient extraction: the clock, every dependent raw coordinate, and every inverse entry move. Equivalently, if $T(u;\vartheta)$ denotes the neutral rational response with the other neutral coordinates suppressed, then $T(u; H_R) = T(u; R(u))$. The same returned substitution on a neutral rational coefficient, with coarse variable z, is the whole map $\mathcal{R}$. Its finite precision here is $z^{2M} = 0$; it fixes rational constants and sends $\vartheta = R_0$ to H.

Use the ungraded complete prefixes

$$C(a,D) = -\sum_{j-k=D,k\geq a} \Delta\mathcal{A}(j,k), \qquad J_b = \sum_D C(b,D)H_{D+b-1}, \quad J_1 = J_M = 0.$$

The full denominator-return baseline is

$$\sum_{m=2+\tau}^{M-1+\tau} J_{M-m+\tau+1} T_m(H_R) = 0, \qquad \tau = [s<\delta],$$

and its one-wave potential is $S_a = \sum_{m=0}^{a} J_{a+\tau+1-m} T_m(H_R)$. Here each T_m is evaluated after the whole returned substitution. Grade independent coarse letters by $\mathrm{wt}_c(H_i) = i$, holding the neutral fine coordinates fixed. The component containing the second response gives, for every $1 \leq a < M$,

$$\sum_D C(a,D)[z^{D+a-2}]f(H) + \sum_{b=2}^{a-1} J_b [z^{a-1-b}]\mathcal{R}(\psi_0) = 0. \tag{4.20}$$

The derivative of the same homogeneous component of the whole baseline supplies

$$\sum_{b=2}^{M-1} J_b [z^{M-1-b}]\mathcal{R}(\psi_0) = 0. \tag{4.21}$$

All other stationary responses remain in their separate homogeneous components; none was set to zero.

Neutralize the unique coarse top letter using its nonzero proper corner. The full adjugate becomes

$$\mathcal{A}_V(j,k) = \kappa G_k C_{M-j}, \qquad \mathcal{A}_{V^*}(j,k) = \kappa C_k G_{M-j}.$$

For any input series $a(z)$, direct finite multiplication gives

$$[z^{M+b-1}]\{G[aC]_{<M} - C[aG]_{<M}\} = \sum_{i\geq b,\ j<b} (G_i C_j - C_i G_j) a_{M+b-1-i-j}.$$

Thus the first term of (4.20) is $-\kappa[z^{M+a-1}]\mathcal{T}(zf(H))$, and $J_b = -\kappa[z^{M+b-1}]D$, where $D = G[HC]_{<M} - C[HG]_{<M}$. The true numerator support gives $\operatorname{ord} D \geq M+1$. The cutoff equations supply the coefficients below $2M-1$, and (4.21) supplies exactly that last coefficient. This proves (4.19). Rationality of f follows from the actual outside constancy and all three aliases in Proposition 4.3; it precedes the use of fine weight. □

Corollary 4.6. *Suppose the active mixed pair satisfies*

$$v < j, k, j-\delta, k-\delta < r. \tag{4.22}$$

Every actually supplied active second response is zero, separately in both independently rebuilt fine fields.

Proof. The complete aliases in Proposition 4.3 put each response f in $\mathbb{Q}(\vartheta)$, where the clock has positive fine weight $\rho = s - t$. The denominator response at order n and actual input j has weight $nr + s + \delta - j$. The constant active entry satisfies $j + k = r + s + t + \delta$, so the second response has weight

$$w_f = 2r + s + \delta - j = r + k - t > 2(s - t) = 2\rho.$$

A nonzero homogeneous rational function of one variable of positive weight is $c\vartheta^a$, with $c \in \mathbb{Q}^\times$, $a \in \mathbb{Z}$. Indeed, its scaling identity permits no nonzero finite zero or pole; the remaining rational function is a monomial, and its weight gives $a\rho = w_f$. Thus here $a \geq 3$.

The supplied stationary equation retains the whole moving potential and every coefficient through $2M - 1$, at this same actual M. Its neutral fine field contains the transcendental returned clock, and the positive coarse letters are independent over it. The homomorphism of Proposition 4.5 fixes $\mathbb{Q}$, is the identity at constant term, and moves every dependent coordinate and complete inverse entry; each nonzero denominator retains its original nonzero constant term. The amplitude c lies in the fixed field $\mathbb{Q}^\times$, and the actual exponent $a \geq 3$ satisfies the imported requirement $a \geq 2$. Thus [6, Whole moving potential] applies to this one response. It would make the actual coarse diagrams equal, contrary to the nonself branch of Proposition 4.4. Hence $f = 0$. The swapped active row has the same proof with its own weight and field. At $M = 3$ the generic endpoint used to obtain the equation is $F_3 = 0$, not $F_2 = 0$. No inactive or higher response is asserted to vanish. □

The role of Corollary 4.6 is now geometric. Its zero is transported on an unrestricted actual two-copy source chart, with both global compensators present, to the equal connected children in Lemma D.4. Their full words, not a rational factor of a coframe, are the input to the next section.

5. Actual children and recovery from the two boundary words

A boundary identity becomes useful for recognition only after its factors have been realized as diagrams. We use recovery from the generic words [9, Recovery from the generic words] on an already legal full marked inner and the two words of an actual connected parent. The phase and native-slot hypotheses are established at each use below.

We use the closed membership words

$$X_P(k) = \mathbf{1}_{k \in X_P}, \qquad Y_P(k) = \mathbf{1}_{k \in Y_P} \quad (0 \leq k \leq N_P).$$

In particular $X_P(0) = Y_P(N_P) = 0$ and $Y_P(0) = X_P(N_P) = 1$. Set

$$h_P(k) = \sum_{i=0}^{k} \big(Y_P(i) - X_P(i)\big), \qquad h_P(-1) = h_P(N_P) = 0.$$

Thus $h_P(k) > 0$ for $0 \leq k < N_P$. Internally these words are $X_P(k) = B_k$, $Y_P(k) = A_k$, retaining common row and column positions and the four actual closed caps.

For a fixed full legal (K, κ) of period $f \geq 2$ and width $a < f$, the cited decoder applies to a connected P of span $mf + a$ only after every nonvariable slot in both words has its native K-value. A slot includes its word label, even when the row and column phases coincide. It then recovers the unique connected outer, with heights $h_P(jf + a) = h_P((j+1)f - 1)$, both true end windows and every mixed supplement. At $m = 1$ it gives $P = K$; at $m \geq 2$ both actual factors are proper and strictly smaller in occupied area. For two cut windows we prove full-mark phase agreement before restoring only variable slots and applying this result.

5.1. **The aligned finish.** We record the two uses of alignment separately. This distinction prevents a source factor from being mistaken for a factor of an unaligned parent.

Lemma 5.1. *Suppose $J = A_J \circ_\kappa K$ is an actual proper chart, where κ has period $d \geq 2$, and μ is an actual legal mark on J of period r, with $d \mid r$. For every connected V, full aligned gluing gives*

$$V \circ_\mu J = \big(\gamma(V) \circ_\lambda A_J\big) \circ_\kappa K, \tag{5.1}$$

with the actual descended mark λ and gauge $\gamma \in \{\mathrm{id}, \mathrm{AT}\}$, with $\gamma = \mathrm{id}$ unless the descended mark is empty. The same (λ, γ) works for a second outer V'. In particular, equality of their outer functions is preserved.

If the constructed width is $v \bmod d$, $d < r$, and the minimum $(v, r+v)$ is held on the actual source $E = V \circ_\mu J$, then (5.1) contradicts that minimum. No minimum on J or K is needed.

For a whole-parent conclusion, suppose instead that actual sources E, U already have proper charts through the same (K, κ), and that both actual upper periods are divisible by d. Then both whole parents have proper charts through this same full marked inner. If their whole Schur functions agree, their genuine outer Schur functions agree. Source-minimal domination then gives the originally prescribed fiber.

Proof. The first assertion is the full aligned gluing lemma [9, Aligned-composition lemma], applied to the two actual charts on J. Its equality is a cell equality, including both full mark copies, all hatted clauses and the supplements. Applying the same descended operation to equal outer functions preserves equality by the I–IV composition identity. The new score is $(v \bmod d, d + v \bmod d) < (v, r+v)$.

For the last assertion, apply aligned gluing independently at E and U. Their descended upper marks and involutions need not coincide; the final marked inner does. The reconstructed outer span is at least two. The inner is a retained copy, and the full nonnegative composition area law makes each new outer strictly smaller than its parent. Only now use same-full-mark injectivity of the composition map, recalled in Proposition 5.2, and then [9, Source-minimal domination theorem]. □

Proposition 5.2. *If $A \circ_\kappa K$ and $A' \circ_\kappa K$ are actual compositions through the same full legal marked inner, then equality of their whole Schur functions implies $s_A = s_{A'}$.*

Proof. Write the full I–IV identity in its actual gauge as $\Phi_\kappa(s_A) = \epsilon(A) N(A) s_{A \circ_\kappa K}$. The normalization $N(A)$ retains every power of $s_W, s_{\widehat{W}}, s_{\widehat{O}}$ and the actual mixed sign. All these factors use raw generators below the period d: the marked bands have width $a < d$, and the hatted middle has singleton private ends and no cells on the thin marked tails, so each of its components has span below d.

For every $j \geq 1$, the genuine complete-row image $y_j = \Phi_\kappa(h_j)$ is affine in the fresh pivot h_{jd+a}, with a nonzero coefficient in lower raw labels. This is the top-pivot expansion of the actual connected row composition; its proper cofactor is nonzero. At a finite cutoff containing both coarse polynomials, let L be the fraction field of all nonpivot raw variables. Invert the successive affine equations in increasing pivot order. Each step is a rational coordinate isomorphism, so no nonzero coefficient becomes zero. Thus the y_j are algebraically independent over L, which contains the entire normalization field.

Equality of the parent functions now gives

$$\Phi_\kappa(s_{A'}) = \frac{\epsilon(A') N(A')}{\epsilon(A) N(A)} \, \Phi_\kappa(s_A).$$

Both coarse Schur polynomials have rational coefficients in the independent y_j. Comparing a nonzero coefficient makes the displayed quotient a rational constant. Their integral primitive normalization makes that constant ± 1: the least-square diagonal monomial in each genuine Jacobi–Trudi polynomial has coefficient one. Positivity on a positive alphabet fixes the sign to $+1$. Any prescribed common fundamental involution is first carried on both outers and then inverted. This proves matching without assuming equal normalization exponents or constructing an inner from an algebraic factor. □

6. Boundary comparisons and actual marked factors

There are two entrances to the boundary argument. A vanishing mixed response gives a zero-row comparison with two column defects. A terminal ordinary coface gives one signed defect in each word. Their cap accounting is different. They share the actual decoder and aligned finish, but we keep the physical words visible throughout. The physical consumers Propositions 6.5, 6.10 and A.8 and Theorem 6.9 retain the actual common finer charts (3.6), the narrow domain

(3.8), and $\mathsf{J}_{<n}$ only at strictly smaller original occupied area. Their two full rational fields are independently rebuilt; the finer minima are held on E, U, and the original minima remain on the whole parents. Each consumer uses the active response, complete coface products, current or endpoint identity, and real word comparisons supplied at its own entrance. Actual connected equal children and strict areas precede every lower-FULL call. In contrast, the local word and fold statements below, including Propositions 6.7 and 6.8, use only their stated legal charts and finite comparisons once those objects exist; they acquire no induction or minimum hypothesis from this paragraph.

6.1. **Zero-row comparisons and two-window sewing.**

Lemma 6.1. *Let two actual charts of span $H = mf + a$, $m \geq 2$, use one full marked inner of period f, carried or wholly rotated. Let u be an integer with $0 \leq u \leq H$. Suppose $X_T(x + u) = X_S(x)$ for every $0 \leq x \leq H - u$, where $H - u \geq f - 1$. In an ordinary or EMPTY chart the two row-port phases align. In the crossed ordinary case the full marked inner is self-rotational and the chart becomes carried. Crossed mixed orientations are impossible.*

Proof. Fill each actual row-port slot by one. Let $P_0, Q_0 \subseteq [0, a]$ be the source and target port sets and B_S, B_T the filled native words. Native-to-native positions are all observed. At a source-only port use its genuine initial value zero; at a target-only port use its genuine terminal value one, whose preimage is $H - u - a + q$. Both positions lie in the observed interval. Thus, residue by residue,

$$B_T(z + u) - B_S(z) = -\mathbf{1}_{P_0 \setminus (Q_0 - u)}(z). \tag{6.1}$$

For ordinary charts the two filled row words have equal cardinality, including after rotation, so their unique holes align. The reflected native identity is $B(z) = C(\sigma + \tau - z \bmod f)$, where σ, τ are the row and column contacts. The full-mark conclusion follows from Lemma 6.2 below.

For type (d), the fixed row word has, say, b_0 ones; the initial column period has $b_0 + 1$ ones by the private singleton. Its rotation has two row ports and filled row count $b_0 + 1$. Applying (6.1) in the two crossed directions would give, respectively, $1 = 0$ and $-1 = -2$. Both are impossible. Reversing the comparison proves the corresponding statement for a full overlap window. □

Lemma 6.2. *For an actual ordinary inner of period f, width a, and contacts σ, τ, let B, C be its row and column native words with the respective return slots filled by one. If*

$$B(z) = C(c - z \bmod f), \qquad c = \sigma + \tau,$$

then $c = a$, and rotation identifies the full *marked inner with itself in its actual gauge.*

Proof. Put $\Phi(z) = \sum_{i=0}^{z} (C(i) - B(i))$. The actual first-period and final partial heights, including their different cap corrections, are

$$\Phi(z) + \mathbf{1}_{z \geq \sigma} > 0 \quad (0 \leq z < f), \qquad 1 + \Phi(z) - \mathbf{1}_{z \geq \tau} > 0 \quad (0 \leq z < a), \qquad \Phi(a) = 0. \tag{6.2}$$

Antisymmetry gives $\Phi(c \bmod f) = 0$. If $c \geq f$, then $c - f < \sigma$, contradicting the first inequality. Hence $c < f$, and partial-sum reflection gives $\Phi(z) = \Phi(c - 1 - z)$ for $0 \leq z < c$. If $a > c$, the second height at c is zero. If $a < c$, reflection takes $\Phi(a) = 0$ to $c - 1 - a < \sigma$, contradicting the first height. Thus $a = c$.

In type (a), the lower mark is obtained on each low marked diagonal by removing its southeast cell from σ onward; the upper mark is obtained by removing its northwest cell below τ. Under rotation, $z \mapsto a - 1 - z$, these deletions match because $a = \sigma + \tau$. Both full copies and their displacement therefore match, not just their functions. Actual antitransposition gives type (b). At $a = 0$ both contacts are zero and the genuine horizontal EMPTY edge is preserved; any prior permitted gauge is retained. □

Theorem 6.3 (Two-window construction)**.** *Let P, S, T be connected, with*

$$N_P = R + V = H + L, \qquad N_S = N_T = H, \qquad D = R - L = H - V > 0, \quad 0 < L < R, \quad V \geq 1.$$

Let distinct $\alpha, \beta \in [L, H]$ be respectively absent and present columns of P. Suppose the four actual cut identities are

$$\begin{aligned} X_S &= X_P \cap (0, H], & Y_S &= (Y_P \cap [0, H]) \setminus \{\beta\}, \\ X_T &= (X_P \cap [L, N_P]) - L, & Y_T &= ((Y_P \cap [L, N_P)) \cup \{\alpha\}) - L. \end{aligned} \tag{6.3}$$

Suppose also

$$X_T(x + D) = X_S(x) \qquad (0 \le x \le V). \tag{6.4}$$

The children have actual proper carried or crossed charts through the same full legal marked inner K, of period $f \ge 2$, width $a < f$, and $H = mf + a$, $m \ge 2$. For a mixed root require, in addition,

$$Y_T(x + D) \le Y_S(x) \qquad (0 \le x \le V). \tag{6.5}$$

Put $b = \min(L, D)$, $c = \max(L, D)$, $e = \gcd(f, b)$, and assume

$$H - b \ge f - 1, \qquad H - c \ge e - 1. \tag{6.6}$$

Then P, S, T have actual proper charts through one same full legal marked inner K', of period

$$d = \gcd(f, L, D) = \gcd(f, L, R), \qquad 2 \le d \le L < R,$$

and width $H \bmod d = N_P \bmod d$. All factors are connected and strict in occupied area, and both genuine end windows and all supplements are recovered. In the ordinary or EMPTY case $d = f$ and the original full mark is retained. In the mixed case only carried type (d) survives, the new width is positive, and at most one physical fold is needed. If $s_S = s_T$, the genuine final outer functions in the constructed charts of S and T agree.

Proof. The cut identities give the complete positive overlap comparison

$$X_T(x - L) = X_S(x), \qquad Y_T(x - L) - Y_S(x) = \mathbf{1}_{x=\alpha} + \mathbf{1}_{x=\beta} \quad (L \le x \le H). \tag{6.7}$$

Distinct edits imply $L < H$. The first bound in (6.6) supplies one complete native period in the comparison with smaller shift. In an ordinary chart, Lemma 6.1 aligns the holes and identifies a crossed full mark with its carried version. Thus $f \mid b$, so $e = f$. The second bound then makes the other comparison complete and gives $f \mid c$. This proves the ordinary assertions without a column-sign hypothesis.

For a mixed root, the same complete window excludes crossed orientations. Carried type (c) has a fixed periodic column word. If the positive window is complete, its nonnegative cyclic differences must all vanish, contrary to the two edits. If the negative window is complete, its nonpositive differences would likewise vanish, whereas their actual sum is

$$\sum_{x=0}^{V} \big(Y_T(x + D) - Y_S(x)\big) = -h_T(D - 1) - h_S(V) < 0. \tag{6.8}$$

Here the zero-row comparison and the true closed caps give the equality; connectedness gives strictness. Hence only carried type (d) remains.

We next justify the single fold without adding unobserved comparisons. On the residues modulo f, the observed smaller-shift comparisons include every edge of addition by b, since their starts include $0, \dots, f - 1$. They give complete finite cycles modulo $e = \gcd(f, b)$. The larger-shift window has starts representing $0, \dots, e - 1$. On the quotient by those cycles its observed edges are all edges of addition by c. Their cycles are the classes modulo $d = \gcd(e, c)$. Lifting a quotient path through the already observed b-cycles gives paths in both directions between any two residues in one d-class.

The row equalities therefore make every native row d-class constant. On a class foreign to both projected column ports, every comparison uses only genuine native columns. Both windows orient a decreasing comparison in the positive shift direction. A binary value cannot decrease strictly along a closed finite path; hence each foreign column class is full or empty. The actual row values at residues zero and a exclude $d = 1$ and $d \mid a$. Native rank excludes coincident projected ports. The full all-width construction in [9, Mixed residue fold] constructs

K' on the actual K, or retains K when $d = f$. Aligned gluing through its original period f then puts the same full K' on both children.

Since $d \mid L$, their phases agree in P. Each edit in (6.7) must be a genuine final column-port position: a generic position has equal fixed bits and could not carry its unit difference. Restoring (6.3) changes only these variable strings. The two cut windows cover P, so both complete generic words agree with K'. Apply [9, Recovery from the generic words]. Its cap argument includes $\alpha = H$ and $\beta = L$; no untouched adjacent step is needed. The child outer spans are at least two and the parent outer span is larger by $L/d \geq 1$. The area proof in that theorem applies to all three charts. Finally, if child functions agree, use same-full-mark matching only after the charts have been constructed. □

Lemma 6.4. *Let J carry a legal type-(d) mark of period r, width v, and let $P = A \circ_\mu J$ be an actual composition, allowing a one-cell A. Suppose $0 < L < r$, all displayed positions are real, and*

$$X_P(x + L) = X_P(x), \qquad Y_P(x + L) \leq Y_P(x) \quad (0 \leq x \leq H), \qquad H \geq r - 1.$$

Then J has an actual proper full marked factor of period $d = \gcd(r, L)$, where $2 \leq d < r$ and $d \nmid v$. Both factors are strict in occupied area. It aligns through the original period r on every supplied source.

Proof. Every native residue has a real representative in the window. The fixed mixed row word is constant along addition-L cycles. On each class foreign to the two projected ports the column inequality compares native bits; following its finite cycle forces equality at every edge. The native row values at zero and v exclude $d = 1$ and $d \mid v$, and $L < r$ gives $d < r$. Apply [9, Mixed residue fold] to J, including its small-width case, and then Lemma 5.1. □

6.2. **A zero mixed second response produces a source factor.**

Proposition 6.5. *At the actual mixed finite-comparison interface, suppose an active second response vanishes and its supplied pair is private: $v < j, k, j - \delta, k - \delta < r$. Then the actual finer source J has a proper full legal factor of period $d \mid r$, $2 \leq d < r$, and width $v \bmod d$. It gives an improving aligned chart at the minimum held on E and, separately, on U.*

Proof. We spell out the passage from the finite equation to actual geometry. Put $b = j - \delta$, let h be the genuine complete private-block head, and set

$$Z = \mathrm{Row}_2 \circ_\mu J, \quad H = r + h, \quad L = r + v - h, \quad D = h - v, \quad N = r + v.$$

The supplied incidence gives

$$v < h < r, \quad v < L < r, \quad 0 < D < r - v, \quad L + D = r,$$
$$L \leq b < r, \quad v < k \leq h, \quad b + k = r + s + t.$$

The full compensated equation of Lemma D.4 is

$$\xi \sigma_Q N_W s_T \Pi_{G,2} - c\sigma_G D_W s_S \Pi_{Q,2} = -b_{2,j}(\vartheta) g_t D_W \Pi_{G,2} \Pi_{Q,2}. \tag{6.9}$$

All products here are complete, including units, multiplicities and private factors. The same physical computation supplies $N_W \Pi_{G,2} = D_W \Pi_{Q,2}$. At $b_{2,j} = 0$, cancellation of these actual nonzero products, followed by primitive positive normalization, gives $s_S = s_T$, for the actual connected children

$$\begin{aligned} X_S &= X_Z \cap (0, H], & Y_S &= (Y_Z \cap [0, H]) \setminus \{r + k\}, \\ X_T &= (X_Z \cap [L, 2r + v]) - L, & Y_T &= ((Y_Z \cap [L, 2r + v)) \cup \{b\}) - L. \end{aligned} \tag{6.10}$$

Their strict block cuts make them actual strict connected cell subsets of Z or its rotation. The full composition area calculation gives

$$|S| = |T| < |Z| = 2|J| - |W| < |E|, |U| < n.$$

Only at this point is lower FULL applied. Its alternatives are literal children, literal rotations, or proper charts through one full legal marked inner, carried or crossed, with equal genuine outer functions. The child-root period satisfies $f \leq H/2 < r$.

The complete closed comparisons are

$$X_T(x-L)=X_S(x), \qquad Y_T(x-L)-Y_S(x)=\mathbf{1}_{x=b}+\mathbf{1}_{x=r+k} \qquad (L\le x\le H), \tag{6.11}$$

$$X_T(x+D)=X_S(x), \qquad Y_T(x+D)-Y_S(x)=-\mathbf{1}_{x=t}-\mathbf{1}_{x=r+s} \qquad (0\le x\le N). \tag{6.12}$$

A literal rotation is impossible: reflect the row equality in the interval $[L,H]$, invariant under $x\mapsto 2r+v-x$. It makes the column difference there zero, contrary to the two positive edits. If $S=T$, their full cut words give a nonpositive L-comparison on Z through every start $0,\dots,H$; since $H>r$, Lemma 6.4 folds J directly.

Otherwise apply Theorem 6.3 with parent Z, $(R,V)=(r,r+v)$, and edits $b, r+k$. Its coverage bounds follow because $H\ge 2f$ and $H>L+D-1=r-1$. Explicitly, if $b_0=\min(L,D)$, $c_0=\max(L,D)$, then $H-b_0\ge f-1$ and $H-c_0\ge b_0-1\ge \gcd(f,b_0)-1$. The theorem constructs a full marked root on Z of period $d\mid L,D,r$, with $2\le d<r$.

It remains to restrict the root to J, not merely to Z. The original type-(d) compiler gives

$$X_Z(x+r)=X_Z(x), \qquad Y_Z(x+r)-Y_Z(x)=-\mathbf{1}_{x=t}-\mathbf{1}_{x=r+s} \quad (0\le x\le N), \tag{6.13}$$

$$X_J=X_Z\cap(0,N], \qquad Y_J=(Y_Z\cap[0,N])\setminus\{r+s\}. \tag{6.14}$$

Since $d\mid r$, the defect at $r+s$ in (6.13) must occur at a genuine root column port. Thus its deletion changes no generic word slot. The whole generic words of J agree with the already constructed root. The first d-diagonal window is unchanged. On the entire true last window one has

$$X_J(x)=X_Z(x+r), \qquad Y_J(x)=Y_Z(x+r) \quad (N-d\le x\le N). \tag{6.15}$$

Indeed $N-d>v\ge t$, and the only possible defect in that window is canceled by the deletion of $r+s$ in (6.14). This includes $s=v$, when the deleted column is the true top. The identities retain the actual translations of both end copies, not merely their height words. Apply [9, Recovery from the generic words] to J. Its outer span is at least two since $r/d\ge 2$, and its proper factors are strict in occupied area. The score is $(v \bmod d, d+v \bmod d)$. Now Lemma 5.1 constructs the improving charts on the held sources. No minimum on Z, J, a coface, or a polynomial was used in this argument. □

6.3. **Signed cap routing.** We next retain the two word labels in the ordinary comparison. A column may be complemented for a calculation, but it remains the supplied column word. Whole rotation is

$$X_{Q^*}(x)=Y_Q(H-x), \qquad Y_{Q^*}(x)=X_Q(H-x), \tag{6.16}$$

without a complement. Actual antitransposition is a different operation: it sends the proper pair (X,Y) to $(1-Y,1-X)$.

Lemma 6.6. *Let g,m,a,p,q,u,e be integers with*

$$g\ge 2, \qquad m\ge 2, \qquad 0\le a<g, \qquad 0\le p,q\le a,$$
$$H=mg+a, \qquad 0<u<H, \qquad 0\le e\le V:=H-u.$$

Let U_S,U_T be binary words on the integer interval $[0,H]$, native g-periodic away from their respective variable phases p,q, with the actual phase caps

$$U_S(p)=U_T(q)=0, \qquad U_S(mg+p)=U_T(mg+q)=1.$$

Suppose the complete observed comparison is

$$U_T(x+u)-U_S(x)=\mathbf{1}_{x=e} \qquad (0\le x\le V).$$

If $p\not\equiv q-u \pmod g$, the following alternatives hold. For $e\ge g$, necessarily $e=V-a+q$; the zero-filled native words agree under the shift, and every proper source port in $[0,V]$ and target port in $[u,H]$ has value zero. For $e\le V-g$, necessarily $e=p$; the one-filled native words agree under the shift, and those observed proper ports have value one. Either alternative earns equality of the filled native cardinalities; no such equality is assumed. If $p\equiv q-u \pmod g$, either of the same error bounds instead gives agreement of the one-filled native words, without prescribing the error position or the proper variable strings.

Proof. For unaligned phases, each stated error bound gives $V \geq g$. A native error would repeat at $e-g$ or $e+g$, respectively, at a different real point of the window. Thus every native-to-native comparison is clean. Fill each hole by one. At the source-only phase use its actual initial point p, and at the target-only phase its actual terminal preimage $V-a+q$. The two filled differences are $-1+\mathbf{1}_{e=p}$ and $\mathbf{1}_{e=V-a+q}$. Their sum is the exact signed balance

$$|F_T^1| - |F_S^1| = -1 + \mathbf{1}_{e=p} + \mathbf{1}_{e=V-a+q}. \tag{6.17}$$

If $e \geq g$, the initial source cap is clean and fixes its target native bit to zero. A positive error cannot lie on that phase. If the terminal target cap were also clean, its source native bit would be one, and a positive error on the remaining phase would require a bit two. Hence the terminal cap is the error. Removing the two filled holes gives the zero-filled identity. The same actual comparisons force all observed proper port values to zero. For $e \leq V-g$, the terminal target cap is clean; the reversed argument forces the initial source cap to be the error and all observed proper values to one. Formula (6.17) gives equal filled counts. Aligned holes have identical filled values, while all remaining native comparisons are clean. This also treats ports at zero or a. □

Here is the exact small-period signed geometry. Write

$$H = R+V-L = V+D, \quad D = R-L > 0, \quad g \leq L \leq p \leq V-q, \quad L \leq \ell \leq V, \quad R \leq j \leq H. \tag{6.18}$$

The two real comparisons, in their normal orders, are

$$X_T(x+D) - X_S(x) = \mathbf{1}_{x=p}, \qquad Y_T(x+D) - Y_S(x) = -\mathbf{1}_{x=q} \qquad (0 \leq x \leq V), \tag{6.19}$$

$$X_T(x-L) - X_S(x) = -\mathbf{1}_{x=\ell}, \qquad Y_T(x-L) - Y_S(x) = \mathbf{1}_{x=j} \qquad (L \leq x \leq H). \tag{6.20}$$

The two error positions in either comparison can instead be exchanged by the actual physical operations described below. These are finite comparisons, not permission to extend a word periodically across a cut. Normalize a positive lower ordinary(b) chart by antitransposing the whole comparison. The complete four-order table is

finer	lower	transport	$W:(+X,-Y)$	shared : $(-X,+Y)$	D	L
(a)	(a)	id	(p,q)	(ℓ,j)	0	1
(a)	(b)	AT	(q,p)	(j,ℓ)	1	0
(b)	(a)	id	(q,p)	(ℓ,j)	1	1
(b)	(b)	AT	(p,q)	(j,ℓ)	0	0.

(6.21)

The last two columns are the common row and column fills earned when the respective shift is unaligned. The cap lemma is applied to X and $1-Y$, so a fill λ in the complemented column means fill $1-\lambda$ in the physical column. Actual AT sends (X,Y) to $(1-Y,1-X)$ and exchanges both defect positions in each window. Whole rotation instead reverses and exchanges the words without complement, as in (6.16); its crossed phases are $\sigma^* = a-\tau$, $\tau^* = a-\sigma$. A vertical EMPTY reading is converted on both charts by its genuine edge gauge, including the outer involution.

Proposition 6.7. *Suppose $S = A \circ_\kappa K$, $T = B \circ_{\kappa^*} K^*$ are actual crossed ordinary or EMPTY charts, with $H = mg + a$, $m \geq 2$, and satisfy* (6.18)–(6.20) *in one of the four physical orders. Then the crossed full mark becomes carried, or $T = S^*$ and $B = A^*$ literally. No equality of outer functions or minimum is used in this local assertion.*

Proof. Normalize the lower ordinary root to type (a) by actual simultaneous antitransposition when necessary. Its row and column ports are σ, τ, and the crossed ports are $\sigma^* = a-\tau$, $\tau^* = a-\sigma$. The possible errors in the first window have real second occurrences $p-g$ and $q+g$, by (6.18). If the holes align, the full native comparison and Lemma 6.2 identify the same full mark.

If they do not align, apply Lemma 6.6 to X and $1-Y$. For order $+p/-q$, the clean initial row cap and clean terminal column cap give $p = V-\tau$, $q = \tau$, and every proper row and column port of S in $[0,V]$, and of T in $[D,H]$, has value zero. For order $+q/-p$, they give $q = \sigma$, $p = V-\sigma$, and every such bit has value one. Both cases give $p+q = V$.

If $D \le g$, those windows contain every proper port of both children, so their actual outers are the same row or the same column, and (6.16) follows. Otherwise $D > g$. The second window is then complete, and its defects have distinct real second occurrences $\ell + g < H$, $j - g > L$. Aligned holes again identify the full marked root. In the unaligned case the same cap calculation, now from T to S, gives

order	error positions	proper bits
$-\ell/+j$	$\ell = L + \sigma^*,\ j = H - a + \tau$	1
$-j/+\ell$	$j = H - a + \sigma,\ \ell = L + \tau^*$	0.

These values hold for both words on $S \cap [L, H]$ and $T \cap [0, H - L]$. The two pairs of intervals are exact reflections:

$$[0, V] \longleftrightarrow [D, H], \qquad [L, H] \longleftrightarrow [0, H - L].$$

They cover all proper ports because $L \le V$. Moreover

$$H - (kg + \sigma) = (m - k)g + \tau^*, \quad H - (kg + \tau) = (m - k)g + \sigma^*.$$

Thus every proper row port equals its reflected column port, and vice versa. Generic bits already reflect because the inner is crossed; the initial caps reflect to the actual terminal caps with their correct values. Hence both whole child words satisfy (6.16). Decoding the returned strings gives $B = A^*$. No hook-function comparison is needed. □

Proposition 6.8. *Suppose the charts in the preceding proposition are carried ordinary or EMPTY, with $S \neq T$. Then $g \mid L$ or $g \mid D$. Set $d = \gcd(g, L, D)$. If $d \ge 2$, one physical fold of K, or no fold when $d = g$, constructs a same full legal marked root of period d on S, T and on their actual parent, whenever its cut words are those of* (6.22) *below. Its width is $a \bmod d$, and all factors are proper and strict in occupied area. If $d = 1$, the constant completions give a row inner, an EMPTY column inner, or an impossible positive one-filled inner. The actual cap certificates are retained in the proof; no proper period-one factor is claimed.*

Proof. Apply Lemma 6.6 separately to the row and complemented column. If $g \nmid D$, the first window gives

order	completion	cap equations
$+p/-q$	0	$p = V - a + \sigma,\ q = \tau$
$+q/-p$	1	$q = \sigma,\ p = V - a + \tau.$

If $D < g$, the forced bits cover every proper port of both charts; they are the same row or column and $S = T$. Thus a remaining case has $D \ge g$. The shared defects now have real second representatives. If $g \nmid L$, the second window gives

$-\ell/+j$	1	$\ell = L + \sigma,\ j = H - a + \tau$
$-j/+\ell$	0	$j = H - a + \sigma,\ \ell = L + \tau.$

An aligned shift imposes no filling condition. With different fills, write $F^1 = F^0 + \delta_p$. If neither shift is divisible by g, commuting the two translations gives

$$(T_D - I)(T_L - I)\delta_p = 0, \qquad 1 + \mathbf{1}_{D+L=0\ (g)} = \mathbf{1}_{D=0\ (g)} + \mathbf{1}_{L=0\ (g)},$$

an impossibility. With the same fill, the two forced-port windows cover every proper port and make both complete outers the same line, again $S = T$. Hence one shift is divisible by g, and the other makes both completed native words constant on d-classes.

For $2 \le d < g$, restore the actual positive mark's native row and column returns to one. This changes only their respective exceptional d-classes, which are nonempty because they contain those returns; the native ranks remain equal. The ordinary residue fold [9, Ordinary residue fold] therefore constructs an actual root of a positive-width K, including a zero remainder. If K is EMPTY, the cap table above has earned one common physical row and column fill, so its completed words satisfy [9, Empty residue fold]. For $d = g$, retain the original full mark without folding. The full aligned lemma puts this same root on both children. Undo any lower-root antitransposition before restoring the parent cuts. Because $d \mid L$, the child phases agree. Their signed defects force the restored row and column positions to be the corresponding real ports;

a generic bit could not carry the defect. The two cut windows cover the parent, so [9, Recovery from the generic words] gives the full chart and strict areas.

At $d = 1$, a zero-filled positive inner is Row_{g+a}, with $\sigma = a, \tau = 0$. A positive one-filled inner is impossible because its canonical row phase zero is absent. At EMPTY width the two possibilities are Row_g and Col_g. The first-window equations then give $p = V, q = 0$, or the shared-window equations give $j = H, \ell = L$. These exact certificates are the ones used in the original and amplified consumers below. □

6.4. **The original terminal coface and its genuine endpoint exits.**

Theorem 6.9 (The signed terminal comparison)**.** *Let J be the actual ordinary finer inner, of span $N = r + v$, $0 < v < r$, in the held source chart $E = V_0 \circ_\mu J$. Retain the complete terminal coface supply, including activity, full products, actual whole-child equality and strict areas, and its two physical word comparisons. After the lawful own-field swap write*

$$\epsilon = v - s - t \geq 0, \quad j = r + c, \quad k = j + \epsilon \leq h, \quad L = N - h, \quad D = h - v = r - L, \quad \ell = N - k,$$

where $1 \leq L \leq s \leq v - t$, $\ell \leq v$, $j \geq r$, $h \geq r > N/2$. The complete child cuts are

$$\begin{aligned} X_S &= X_J \cap (0, h], & Y_S &= (Y_J \cap [0, h]) \setminus \{j\}, \\ X_T &= ((X_J \cap [L, N]) \setminus \{\ell\}) - L, & Y_T &= (Y_J \cap [L, N)) - L. \end{aligned} \tag{6.22}$$

Their comparisons are (6.19)–(6.20) *with $(H, R, V, p, q) = (h, r, v, s, t)$, in the actual physical order. The original upper is either ordinary of positive width $0 < w < v$, genuinely EMPTY with $w = 0$, or mixed of positive width $0 < w < v$. The EMPTY case concerns the upper mark; the finer still has $0 < v < r$. In the mixed case retain the full finite comparison in both own fields. If an endpoint argument gives $s = v, t = 0$, then $s \geq \delta$ and $t \leq w$, with $\delta = v - w$, hold automatically, so Lemma E.3 excludes that endpoint. At these entrances J is literally self-rotational with its full original mark transported, or an actual aligned strict chart improves the minimum held on E; the same construction is available on U. The ordinary positive and EMPTY endpoints use Proposition A.8, whereas the mixed sign restriction excludes $s = v, t = 0$. Finer (b) and terminal numerator inputs are transported with their whole charts and rebuilt fields as in Lemmas D.2 and E.4. Negative-δ terminal activity has already been excluded by Proposition 4.4; it is not a further case assumed away here. No unconditional divisor conclusion is asserted at gcd one.*

Proof. The coface supply gives actual connected whole children with $|S|, |T| < |J| < |E| < n$, before lower FULL is invoked. If $T = S^*$, the unrotated cofaces agree with their sources below the deleted columns $j, k \geq r > N/2$. Those genuine initial windows and their rotations cover J, so $J = J^*$, with its full mark carried by rotation. If $T = S$, the full cut identities give

$$X_J(x + L) - X_J(x) = \mathbf{1}_{x=\ell-L}, \qquad Y_J(x + L) - Y_J(x) = -\mathbf{1}_{x=j} \quad (0 \leq x \leq N - L).$$

The true block cut has $h_J(L - 1) = 1$. The finite rectangle construction, Lemma A.7, gives an actual period-L chart when $L \geq 2$; at $L = 1$, J itself is a rectangle. For $L \geq 2$, let $a_0 = N \bmod L$ and (σ, τ) be the contacts of the constructed root. If $L \mid r$, its chart aligns immediately. Otherwise compare this chart with itself under the original shift r, on the actual closed interval $[0, v]$. The original two mark words give one row error at s and one column error at t, in their actual ordinary order. The inequalities $s \geq L$, $t \leq v - L$ give real second representatives. By Lemma 6.6, the normal order yields complete zero-filled native words and cap equations $s = v - a_0 + \sigma, t = \tau$; the exchanged order yields complete one-filled words and $t = \sigma, s = v - a_0 + \tau$. Both completed words are r-periodic modulo L. Thus $d = \gcd(L, r) \geq 2$ gives an actual fold by [9, Ordinary residue fold] when $a_0 > 0$: restoring the true native returns changes only the nonempty exceptional classes. At $a_0 = 0$, the same cap calculation has supplied one common physical fill in both words, so use [9, Empty residue fold] instead. The genuine aligned lifts through L and r follow. At $d = 1$, zero fill gives Row_{L+a_0} with $\sigma = a_0, \tau = 0$; one fill can only give an EMPTY column. Both cap equations give $s = v, t = 0$, and the endpoint conclusions cited in the statement apply. This literal case imposes no minimum on J.

In the remaining case choose an actual proper source-minimal chart of S. Lower FULL supplies its mate over the same or fully rotated marked inner, with equal genuine outer functions. A line outer gives a literal or rotational child alternative already treated. Thus the outer span m is at least three. Suppose its period $g > L$, and write its width as a, so $h = mg + a$. The source proper row and column returns are

$$kg + \sigma, \quad kg + \tau \qquad (1 \le k < m).$$

They lie in $[L, h]$, since their least possible position is $g > L$ and their greatest is $h - g < h - L$. The target returns, with its actual phases σ_T, τ_T, are observed at the comparison starts $L + kg + \sigma_T, L + kg + \tau_T$, also in $[L, h]$. The window has length $mg + a - L > 2g$, so every native phase has two representatives. For carried ordinary roots the unmatched source and target return residues are distinct; the one row error can meet at most one of these complete proper return strings. On the untouched string every bit equals the fixed native bit of the other word. A constant complete outer row string is a row or column. The genuine outer-function equality earned by the preceding lower-FULL call makes its mate that same literal line, returning the literal or rotation alternative already treated. For crossed ordinary roots the same argument applies unless the holes align. Aligned holes give full marked reflection by Lemma 6.2, and then $g \mid L$, impossible. A carried mixed root has a fully periodic word at one singleton defect, which would repeat. A crossed mixed root has two distinct proper return strings in the nonfixed word; both are fully observed in the displayed ranges, and at most one meets its singleton error. The other is constant, again giving a line and then the same line mate by the previously earned genuine outer equality. Consequently $g \le L$.

For ordinary or EMPTY roots, Proposition 6.7 resolves the crossed case geometrically. A carried root is handled by Proposition 6.8. Its $d \ge 2$ chart on J aligns through $r = L + D$; its width is less than $d \le L \le v$, so Lemma 5.1 contradicts the held source minimum. If the original subsidiary minimum on S is retained, a strict refinement would also improve that chart; thus one recovers the original stronger output $d = g$, $g \mid L, D$, whenever that output is wanted.

For clarity we retain the actual gcd-one cap tests. If $g \nmid L$, the shared equations give $j = h, \ell = L$. Positive ϵ is impossible since $k = j + \epsilon \le h$. In the balanced case, for original finer (a) the true missing row $X_J(s) = 0$ forces $g \mid s$ in the surviving EMPTY-column reading. Since $g \nmid L$ and $L \le s$, we have $s > L$; the error-free comparison at s then reads a nonreturned one at $s - L$, a contradiction. In the actual lower antitranspose the same argument uses $Y_{\mathrm{AT}(J)}(s) = 1$ against a nonreturned zero; it includes a positive-width row inner. For original finer (b), away from $s = v, t = 0$, the true private cap is $Y_J(v) = 0, X_J(v) = 1$. The EMPTY-column case forces $g \mid v$ and fails at $v - L$; its actual antitranspose forces $v \equiv a \pmod g$ and fails in the row word there. The excluded endpoint goes to its genuine upper-specific consumer. If instead $g \mid L$ but $g \nmid D$, the first-window cap equations give that same endpoint $s = v, t = 0$.

For mixed lower roots, a carried type (d) has a periodic row word and cannot support the first-window row singleton, which repeats at $s - g$ or $t + g$; type (c) uses its periodic column word. For a crossed pair normalize S to type (d). Let B be its fixed row word and C its initial column period with both ports filled one. The private singleton gives $|C| - |B| = 1$. The first full source period and final full target period in the normal first window give the complete residue identities

$$B(v - x) = C(x) - \delta_t(x), \qquad B(v - x) = C(x) - \delta_{v-s}(x).$$

The count first forces $t < g$ and $v - s < g$, then $s + t = v$. In the exchanged order it would force $s < g$, contrary to $g \le L \le s$. At the balanced value the generic words give $Y_T(y) = X_S(h - y)$ globally. The shared comparison gives

$$X_T(y) - Y_S(h - y) = -\mathbf{1}_{y=\ell-L} + \mathbf{1}_{y=h-j} \quad (0 \le y \le h - L),$$

and the first comparison gives

$$X_T(y) - Y_S(h - y) = \mathbf{1}_{y=D+s} - \mathbf{1}_{y=h-t} \quad (D \le y \le h).$$

Both vanish, since $\ell + j = h + L$ and $s + t = v$. Their actual ranges cover $[0, h]$; the exchanged order is the same calculation with its two positions exchanged. Thus $T = S^*$, already treated.

Finally, a terminal P-input is transported by the actual whole antitranspose and its two rebuilt own fields, not by relabeling a row comparison. With $\mathcal{T}(F)(z) = \iota(F(-z))$, the complete coframes satisfy

$$G^\dagger = \mathcal{T}(P),\ P^\dagger = \mathcal{T}(G), \qquad C^\dagger = \mathcal{T}(Q),\ Q^\dagger = \mathcal{T}(C).$$

The returned normalizers give $G_a'^\dagger = (-1)^{a-s}\iota(P'_a/R_0)$, with the same nonzero paired factor and shift on the two sides. If $\epsilon < 0$, the true singleton cut at $r-1$ ensures that a common head at least r has both inputs terminal; the legal own-field swap therefore makes $\epsilon \geq 0$. All cells, marks, units, normalizers and areas are transported. This accounts for the original terminal, opposite-sign and both-gauge callers. □

6.5. **Amplified ordinary and EMPTY comparisons.**

Proposition 6.10. *At the wholly private ordinary or EMPTY active-edge interface, the actual finite two-copy coface equation forces literal marked reflection of J or an improving chart at the original held source minimum. It uses the same supplied finite comparison; the constructed two-copy diagram is an actual smaller object, not a new observer.*

Proof. Normalize the *original* ordinary chart to type (a), or to the horizontal EMPTY convention, by the permitted whole-chart transport, then rebuild both fields and use the supplied finite identity. Swap the two actual own orientations when necessary so that $\epsilon = v - s - t \geq 0$. The paired identities and their full normalizers are transported by this swap. Only now form the actual $Z = \mathrm{Row}_2 \circ_\mu J$; it is not a row composite silently identified with the antitranspose of an earlier row composite. The source-minimal propagation theorem [7, Outer-independent minimality lemma] (applied at its nonline-inner gate) says that every proper root period of Z is divisible by r. For a positive ordinary mark write $\alpha = |W|$, $\beta = |\widehat{W}|$, $\gamma = |\widehat{O}|$. The full area law gives

$$|E| - |Z| = r(|V| - 2) + \alpha(\mathrm{rows}\, V - 1) + \beta(\mathrm{cols}\, V - 2) + \gamma(M - 2) > 0.$$

All terms are nonnegative and $|V| \geq M \geq 3$; a nonline connected outer has at least two rows and columns. For EMPTY, $|Z| = 2|J|$ and $|E| = r|V| + (|J| - r)M \geq M|J|$, where $M \geq 3$. Thus $|Z| < |E| < n$ in both cases.

The actual finite coface supply Lemma D.3 gives strict connected equal whole children with cuts (6.22), now on Z, and parameters

$$H = r + h, \quad V = r + v, \quad L = V - h, \quad D = h - v = r - L, \quad p = r + s,\ q = t, \quad j_2 = r + j.$$

Here h is the complete connected head at the chosen private column j, $\ell = i - \delta$, and $k_2 = r + j + \epsilon$. The actual paired head identities give

$$L \leq \ell < r \leq V < j_2 \leq k_2 \leq H, \qquad L < p \leq V - q, \qquad q < L, \qquad \ell + j_2 = H + L - \epsilon.$$

Here $r < V = r + v$ at positive finer width, whereas $r = V$ for the genuinely empty finer $v = 0$. The genuine missing row is $X_Z(p) = 0$, with $L < p < H$. The complete enlarged children are

$$\begin{aligned} X_S &= X_Z \cap (0, H], & Y_S &= (Y_Z \cap [0, H]) \setminus \{j_2\}, \\ X_T &= ((X_Z \cap [L, 2r + v]) \setminus \{\ell\}) - L, & Y_T &= (Y_Z \cap [L, 2r + v)) - L. \end{aligned} \tag{6.23}$$

Their complete word comparisons are

$$X_T(x - L) - X_S(x) = -\mathbf{1}_{x=\ell}, \qquad Y_T(x - L) - Y_S(x) = \mathbf{1}_{x=j_2} \qquad (L \leq x \leq H), \tag{6.24}$$

$$X_T(x + D) - X_S(x) = \mathbf{1}_{x=p}, \qquad Y_T(x + D) - Y_S(x) = -\mathbf{1}_{x=q} \qquad (0 \leq x \leq V). \tag{6.25}$$

The second comparison follows from the actual two-copy lists: $X_Z(x + r) - X_Z(x) = \mathbf{1}_{x=r+s}$ and $Y_Z(x + r) - Y_Z(x) = -\mathbf{1}_{x=t}$ throughout $[0, r + v]$. Neither child edit meets that second range. Thus the missing middle return is included, not inferred from a comparison on the first copy. At EMPTY these formulas are $v = 0, V = p = r, q = 0, L = r - h, D = h$, and the missing row is the actual middle row r. The comparisons are exactly (6.19)–(6.20), with $j = j_2$. Their strict cuts are $h_Z(L - 1) = h_Z(H) = 1$.

A reflected child pair gives $Z = Z^*$, since the deleted columns lie past half its span. The two genuine period-r end windows then give $J = J^*$. Literal children give the finite rectangle construction. At $L \geq 2$ this is a proper period-L root of Z, contrary to $0 < L < r$. The EMPTY possibility $L = 1$ makes Z itself a rectangle, again giving $J = J^*$; it is not called a proper cell root.

Otherwise choose an actual source-minimal child chart and use lower FULL. The long-window argument in Theorem 6.9 applies verbatim to its four actual cut words and gives $g \leq L < r$. The crossed ordinary case is Proposition 6.7. Its proof needs only the real inequalities here, not $V < r$. For a carried mixed root, its fully periodic row word in type (d), or column word in type (c), would have a singleton defect with the real second representative $p - g$ or $q + g$; this is impossible. For a crossed mixed root normalize the source child to type (d). Let B be its fixed row word and C its first native column period with both true ports filled one. The private singleton gives $|C| - |B| = 1$. The whole final target row period is $C(a - y \bmod g)$, on the interval $(H - g, H]$, retaining both terminal caps. Summing the normal second comparison over the first source period and final target period gives, respectively,

$$B(V - x) = C(x) - \mathbf{1}_{x=q}, \qquad B(V - x) = C(x) - \mathbf{1}_{x=V-p} \quad (0 \leq x < g).$$

The counts force both defect positions into this range. Hence $p+q = V$, so $\epsilon = 0$. The exchanged order instead would force $p < g$, contrary to $p \geq r > g$.

In the balanced case the generic words already give $Y_T(y) = X_S(H - y)$. Write the shared defects as row $-\mathbf{1}_{x=\alpha}$, column $\mathbf{1}_{x=\beta}$, where $(\alpha, \beta) = (\ell, j_2)$ or its exchanged pair. Then $\alpha + \beta = H + L$, and the shared comparison gives

$$X_T(y) - Y_S(H - y) = -\mathbf{1}_{y=\alpha-L} + \mathbf{1}_{y=H-\beta} = 0 \quad (0 \leq y \leq H - L).$$

The second comparison gives

$$X_T(y) - Y_S(H - y) = \mathbf{1}_{y=D+p} - \mathbf{1}_{y=H-q} = 0 \quad (D \leq y \leq H),$$

with the two positions exchanged in the other physical order. The ranges cover $[0, H]$, since $L < V$, and include every closed cap. Thus both complete words give $T = S^*$, an earlier exit. Only a carried ordinary or EMPTY root remains.

For that root Proposition 6.8 gives a proper root of Z at period $2 \leq d \leq L < r$, unless its completed words are constant. We include the constant boundary rather than importing a positive-width endpoint theorem. If the first window is unaligned, its constant completion gives either $K = \mathrm{Row}_{g+a}$ with $p = V, q = 0$, or $K = \mathrm{Col}_g$ with EMPTY mark and the exchanged order. If $g \mid L$, the shared phases already agree and [9, Recovery from the generic words] gives a proper period-g root of Z. If $g \nmid L$, use the true shared caps. For the row inner, normal shared order would require $\ell = H$, impossible; exchanged order forces $j_2 = H, \ell = L$ and all observed proper source ports zero. Together with the first-window zeros, these cover every proper port, making both outers the same row and the children literal. For the EMPTY column, exchanged shared order would require $j_2 = L$, impossible; normal order forces the same endpoint equalities and all proper ports one, again giving literal children.

If the first shift is aligned and the shared shift is unaligned, a constant completion gives $j_2 = H, \ell = L$. Positive ϵ is impossible from the actual opposite head. At balance the genuine missing row $X_Z(p) = 0$, with $L < p < H$, forces $g \mid p$ in the EMPTY-column case. The source value at this clean point is zero, but the target at $p - L$ is a nonreturned one, a contradiction. Under the actual lower antitranspose, the same cap is a column one against a nonreturned zero; this includes a positive-width row inner. For EMPTY, this is precisely the actual middle cap $p = r$.

Every remaining case therefore constructs an actual proper root of Z of period strictly below r, contradicting its inherited period divisibility. The argument uses no minimum on a coframe or a factor polynomial and no upper divisibility inferred from a source-only root. The ordinary and EMPTY two-copy supplies remain separate algebraic inputs; setting $v = 0$ in the former is not used to obtain the latter. □

7. Completion of marked-factor recognition

The final argument uses minimality only after the preceding constructions have supplied a full legal chart. We record the empty-cycle case separately, because it uses an actual returned edge in place of a positive marked overlap.

Lemma 7.1. *Let J carry its genuine horizontal empty mark, with span r, $X_J = \{r\} \sqcup I$, $Y_J = C \sqcup \{0\}$. Suppose $0 < w < r$ and*

$$(I - w) \bmod r \subseteq I \cup \{0\}, \qquad (C + w) \bmod r \subseteq C \cup \{0\}.$$

Put $g = \gcd(r, w)$. If $g = 1$, the full marked J is self-reflecting. If $g \geq 2$, there is an actual proper period-g empty chart of J, with both factors strictly smaller in occupied area. It aligns through every actual period-r empty source chart and improves that held source minimum.

Proof. Add the returned zero to both native sets. Each nonzero residue cycle modulo g is full or empty. The exceptional zero cycle is an occupied segment ending at its returned zero, possibly the entire cycle. Let $R = r/g$. Its two occupancies lie in $[1, R]$. Equal row and column rank forces equal numbers of full nonzero cycles and equal exceptional occupancies, since their difference has absolute value less than R.

If $g = 1$, the two oppositely directed segments have equal length and end at zero. They are negatives of one another. Thus $I = r - C$, and the complete flags give $X_J = r - Y_J$. The actual empty end edges are interchanged by the same half-turn.

For $g \geq 2$, let $I_K, C_K \subset \{1, \dots, g-1\}$ be the full nonzero classes, and define

$$X_K = \{g\} \sqcup I_K, \quad Y_K = C_K \sqcup \{0\}, \qquad X_D = g^{-1}(X_J \cap g\mathbb{Z}), \quad Y_D = g^{-1}(Y_J \cap g\mathbb{Z}).$$

In the first g-diagonal strip, the divisible rows and columns contribute net one; later full background classes cancel. Hence its actual positive heights and singleton ends are exactly the heights of K. Equal background threshold counts give positivity and connectivity of D. This also covers a full exceptional cycle: its row zero is replaced by the true top r, whereas column zero is retained. The empty interval-union formula on content $jg + z$ is

$$h_{D \circ K}(jg + z) = h_D(j) + h_K(z) - 1 \qquad (0 \leq z < g).$$

Both endpoint words are the displayed full flags of J, so it constructs $J = D \circ_{\mathrm{EMPTY}_H} K$ literally. The empty marked set has no enlargement on its empty content band, the period is positive, deleting it leaves connected K, and its hat is empty. These prove I–IV in the genuine edge convention.

The exact area is

$$|J| = g|D| + R(|K| - g).$$

Since $R \geq 2$, $|D| \geq R$, and $|K| \geq g \geq 2$, both factors are proper and strict. Empty-edge reassociation, or the aligned theorem at $g \mid r$, constructs the chart on the held source. Its score is $(0, g) < (0, r)$. A vertical original edge is transported with the outer involution and then transported back. □

Proof of Theorem 1.1. The ribbon-outer period-rigidity theorem of [10] applies because the original periods satisfy $p \geq q$, the chart of the larger-period parent has ribbon outer, and all its original source and lower-area hypotheses are exactly those of the present theorem. It gives $p = q$, not a marked-fiber conclusion. The Euclidean span formula at this common period gives equal widths and equal outer spans. Corner transport [7, Corner transport theorem] shows that every connected prime of the proper corner of the ribbon-outer parent has generator span less than the common period p. If C^- had a nonempty connected component C_i, its actual transported child $C_i \circ_\eta U$ would contribute a connected prime of span $pN_{C_i} + w \geq p + w$ to that same corner. Unique factorization excludes this. Hence C is also a ribbon.

Read the two factor functions by mixed ribbon factor recovery [8] or (3.1). The literal or rotated inner is settled by the direct full-mark argument in Section 3, with its actual gauge. Before constructing any finer chart, apply Lemma 3.1: an original ribbon of at least three cells, or a strictly smaller opposite domino probe, already proves the result on actual smaller diagrams.

Only a nonliteral two-copy comparison remains, with the mixed row-two selector when the category is mixed. Now Proposition 3.2 constructs $E = V \circ_\mu J$, $U = V' \circ_\mu J$, at area less than n. The original source minima remain on the whole parents; the newly chosen finer minimum is held on E, and on U by its lower key. No minimum is placed on J. Preprocessing Proposition 3.3 excludes alignment, wide and boundary widths, and line finers. It also matches the actual full upper mark and the original selector. We are left with

$$J \text{ nonline}, \quad M \geq 3, \quad 0 \leq w < r, \quad w \neq v, \quad s_V = s_{V'}, \quad V \neq V'.$$

Every transport so far has been a whole marked rotation, antitransposition with its rebuilt field, or a genuine edge gauge.

Ordinary and empty finers. Use the complete finite comparison and its first-response alternatives, Propositions 4.3 and 4.4. If the whole first matrix is zero, Lemma D.7 supplies both native shift conditions. For a positive finer put $g = \gcd(r, v - w)$. If $g \geq 2$, then $g < r$, every native cycle away from its own returned phase is full or empty, and both exceptional sets are nonempty at the true returns s, t. Their ranks are equal by the actual flags. Thus [9, Ordinary residue fold] constructs a proper period-g chart of J. Because $g \mid r$, Lemma 5.1 places it on the held sources with smaller score, a contradiction. If $g = 1$, Lemma A.5 proves full marked self-reflection. For an empty finer, Lemma 7.1 gives the corresponding actual root or full reflection, without a positive-mark specialization. The reflected cases finish by Lemma E.8.

If the first matrix is nonzero, choose an actual active PG edge, retaining its two coface equations and complete component products. The full endpoint and head analysis in Lemmas D.1, D.2 and E.4 puts a terminal edge in the signed comparison of Theorem 6.9; it includes a terminal numerator after whole antitransposition and either sign of the reflection parameter after the lawful own-field swap. The principal ordinary and empty upper endpoints use their actual cap-splice supply; a mixed upper uses its actual sign restriction. Thus that theorem gives marked reflection or an improving actual aligned chart. When all active inputs and both targets are private, the complete two-copy identity Lemma D.3 and Proposition 6.10 give the same alternatives. In particular this includes negative $v - w$ and every empty finer. The strict factors are constructed before the held minimum is invoked. Reflection again finishes by Lemma E.8. This exhausts the ordinary first matrix without dropping a returned intercept.

Mixed finers. Normalize to case (d) by whole rotation. If $w = 0$, Lemma F.1 recovers $V' = V$ using the actual empty selector comparison, and the literal-inner finish applies. Assume $0 < w < r$. Whole-current activation is available in each of the four actual upper types. The finite block first supplies the complete native row cycles, Lemma D.5. If the entire GG matrix is zero, Lemma D.6 and Corollary A.6 constructs the legal divisor factor, retaining both exceptional column classes and arbitrary mark-only intercepts. Its period divides r and both actual upper periods $Mr + \delta$. Both whole parents are constructed through the same full mark before their genuine outer functions are compared. Source-minimal domination gives the original prescribed fiber. The same construction also improves the held finer-source score. No single zero response is substituted for this zero entire matrix.

Otherwise Proposition 4.4 makes the matrix constant, earns $V' = V^*$, and gives the complete active-row identities. If $\delta = v - w > 0$, take the larger genuine connected head of an active pair. A head cannot equal r, which is an absent native row. Above r, Proposition A.9 constructs an actual divisor factor with its full marked windows. If both heads are below r, their actual coface bounds earn

$$v < j, k, j - \delta, k - \delta < r.$$

For $\delta < 0$, the zero-order and coherent-unit aliases in Lemma D.5 earn exactly the same private conditions. These are conclusions of the actual packet, not additional hypotheses on the original theorem.

At that private gate, the fixed-horizon stationary equation includes its last coefficient $2M - 1$. The complete aliases and the actual fine weight make a nonzero active response a positive integer power with constant rational amplitude. Corollary 4.6 applies whole-field stationary descent to this actual response and excludes it. Each actually supplied active second response is therefore

zero in its own field. The full physical two-copy identity then constructs actual connected equal whole children of strictly smaller occupied area. Only now is the lower full-fiber consequence used. Proposition 6.5 treats its literal, rotated, carried, and crossed outcomes and restricts the resulting full legal factor to both true end windows of J. Its period divides r, so Lemma 5.1 constructs the improving charts on the held sources. This is the final contradiction.

Every surviving branch has therefore already returned the original prescribed fiber. In any branch using a whole-parent construction, both actual original charts over the same full legal factor are made first; Proposition 5.2 then earns the genuine outer function equality, and source-minimal domination gives that same prescribed fiber. In the source route, the actual divisor, both end windows, proper factors and alignment precede the comparison of scores. All lower-class calls above are on the actual strict-area objects identified in the corresponding lemmas. Thus only $\mathsf{J}_{<n}$ has been used. Undoing the full transports proves $H \in \mathcal{F}(\omega, s_D)$, with exactly its permitted full-mark rotations and genuine gauges. □

8. Concluding remarks

Under the stated strict-lower-area hypothesis, a source-minimal ribbon-outer chart is recognized by its original full marked fiber, not merely by its period or its factor functions. The finite algebra produces whole equalities of actual smaller diagrams, and the two complete boundary words convert those equalities into legal aligned charts. This is where the marked information enters the conclusion.

The algebraic coframe separation is supplied by [6]; it is not itself chart recognition. Here the physical comparison supplies its whole finite input and turns the resulting vanishing into equal whole functions of actual smaller diagrams. The terminal coefficient and the independently rebuilt fields remain essential. The joint package at the current area is not proved here.

Appendix A. Boundary constructions and exceptional entrances

A.1. **Whole-mark transports.**

Lemma A.1. *Antitransposition carries a positive ordinary(b) source to an ordinary(a) source with the same coarse outer, and a mixed(c) source to mixed(d) with the same coarse outer. All upper marks, supplements, and both hatted clauses are transported. For empty marks,*

$$V \circ_{\mathrm{EMPTY}_V} J = \mathrm{AT}(V) \circ_{\mathrm{EMPTY}_H} J, \qquad \mathrm{AT}(V \circ_{\mathrm{EMPTY}_H} \mathrm{Col}_r) = \mathrm{AT}(V) \circ_{\mathrm{EMPTY}_H} \mathrm{Row}_r\,. \tag{A.1}$$

These identities preserve properness, area, period, width, and source scores.

Proof. Write $\mathcal{A}(i,j) = (-j,-i)$, so $\mathcal{A}(e) = -e$. For positive ordinary(b), $\mathcal{A}(jv - ie) = j\mathcal{A}(v) + ie$, the ordinary(a) origin with the same coarse cell. Mixed ranks are interchanged with their complementary boundary ranks, and $\mathcal{A}(W + L - e) = \mathcal{A}(W + L) + e$, so every supplemental copy is carried as well. Both hats obey

$$\widehat{O}' = \mathcal{A}(\widehat{O}) - e, \qquad \widehat{W}' = \mathcal{A}(\widehat{W}) - e.$$

Indeed the substitution $y = \mathcal{A}(x + e)$ interchanges the two marked clauses and reverses every middle pair. Their common translation preserves adjacency at every integer translate. I–III are preserved by the cell isometry.

For the first empty identity, $v_V = v_H - e$ and $jv_V - ie = jv_H - (i + j)e$; the antitransposed coarse cell has row $-i - j$ and unchanged content j. For the second, the horizontal empty displacement of Col_r is $(1-r, 1)$, whose image is $(-1, r-1) = (0, r) - e$. Hence $\mathcal{A}(j(1-r, 1) + ie) = j(0, r) - (i + j)e$. These are cellwise identities before taking unions. On functions, the whole change is the invertible involution $h_j \mapsto e_j$. Every field, normalization, and inverse used after a nontrivial transport is rebuilt on the image board. □

A.2. **A line finer and the complete grouped cofactor deck.**

Lemma A.2. *In the landed comparison of Proposition 3.2, after the original outer has been reduced to a domino by Lemma 3.1, suppose the common finer inner is a line. The original whole parents are in the prescribed marked fiber, or an actual proper chart on a held source or on both original whole parents has strictly smaller score. This includes a positive or empty upper mark and both original domino selectors.*

Proof. Use Lemma A.1 to make the finer horizontal. Thus $J = \mathrm{Row}_{r+v}$, its marked overlap is Row_v, and with $M = N_V = N_{V'}$,

$$X_E = rX_V + v, \qquad Y_E = rY_V, \qquad S = Mr + v, \quad Q = S - w.$$

The aligned case $r \mid Q$ is already disposed of in Proposition 3.3; assume otherwise. Apply the following argument separately to the two actual upper marks, whose width and period agree. For a positive upper, the actual bit $A_Q = 0$ excludes ordinary(b) and mixed(c), and $B_w = 0$ excludes mixed(d). The remaining ordinary(a) comparison has upper phase $t = 0$. Translation by Q moves each nonexceptional residue support off itself, so all internal bits of its lower marked window are zero. Its positive height, singleton ends, and actual upper cap then force $s = w$. The upper mark is therefore the literal Row_w, with its two actual end copies. Its full periodic hatted mark $\widehat{W}$ is empty: only the two neighboring copies meet a marked band, in the same row, and neither southeast-neighbor test has a cell. This proves all translates of both hatted clauses as well as the visible attachment.

The common source function fixes the number m of rows. If $m = 1$, the coarse diagram is a row, determined literally by its Schur function. Thus $V' = V$, and the direct marked-fiber argument in Section 3 applies. Assume henceforth that $m \geq 2$. The genuine row companion has $2m - 1$ rows; the column companion, whose second copy is at displacement $\nu - \mathbf{e}$, has $2m$ rows. Consequently the original selectors agree. The actual companion identity is

$$s_E^2 = h_w s_{F_R} + s_{F_C}. \tag{A.2}$$

At $w = 0$, its coefficient is the genuine unit 1, and the same identity follows from the two empty edge gluings. Thus equality for either actual selector, together with $s_E = s_U$, gives equality of the row companions. This passage uses no positive-mark formula at an empty mark.

Put $g = \gcd(r, Q)$. If $g > 1$, every row flag of either original whole parent has residue $t = v \bmod g$, and every column flag has residue zero; the column selector has these same residues because its actual extra displacement is $-\mathbf{e}$. Divide the strict flags by writing $x = g\bar{x} + t$, $y = g\bar{y}$. Their connectivity inequalities give $\bar{x}_{i+1} \geq \bar{y}_i$, and their positive diagonal differences give an actual connected coarse diagram. The literal row-inflation formula reconstructs each whole parent through the same full marked Row_{g+t}, with a genuine empty edge when $t = 0$. Injectivity of the row substitution gives equal genuine coarse functions. Both charts are proper, and $(t, g + t) < (w, S)$, contradicting the original source minimum. In particular, this construction is made on the original column parent itself when that was the selector; a factor of the row companion has not been transferred to it.

It remains to treat $g = 1$. Put $A = h_v$, with $A = 1$ for $v = 0$, and set $f_j = h_{rj+v}/A$, $f_0 = 1$. Neither endpoint row can be fully covered by the upper mark: its length has the form $ra + v$, so equality $w = ra + v$ would give $r \mid Q$. Thus both endpoint rows extend beyond their marks. For a coarse diagram V, let $J_V = \mathrm{JT}_f(V)$, let N_V be its first-row/last-column deletion, and write

$$e = \det J_V, \quad c = \epsilon \det N_V, \quad \epsilon = (-1)^{m-1}.$$

In particular $c = [f_M]e$, because the maximum label M occurs only at the first-row/last-column entry. For an integer q, let C_q be the sum of all signed cofactors of J_V at original label q, and let D_q be ϵ times the analogous sum for N_V. In the native sum the physical row is its old row $i + 1$. Thus for $k > 0$ define the complete group

$$W_k = cC_{-k} - eD_{-k}.$$

Expansion of the rectangular splice in Lemma B.1, followed by the original Jacobi adjugate identity, gives

$$s_{F_R(E)} - s_{F_R(U)} = A^{2m-2} \sum_{k>0} h_{Q+v-rk}\big(W_k(V) - W_k(V')\big). \tag{A.3}$$

Here is the cancellation in this formula. For a nonnegative original label, the adjugate product is $cC_q - eD_q$; its entire sum is fixed by the common e, c and their positive moment derivatives. At label zero the two groups are given by the two ordinary-degree Euler identities, so all actual units cancel in the pair as well. The remaining labels are exactly the negative native positions, and restoring label $-k$ in the row-inflated board gives the raw letter h_{Q+v-rk}, with the displayed common source/native scale. This proves (A.3) without observing any individual tied cofactor. Let a_E, b_E be the actual first and last row lengths of E. Its end copies of Row_w give $a_E, b_E \geq w$, and both rows are nonempty. The least original native label is $a_E + b_E - S$. Thus every restored index is at least $Q + a_E + b_E - S = a_E + b_E - w > 0$, also when $w = 0$. Its residue $Q + v$ is different from $v \bmod r$. These distinct raw letters are therefore algebraically independent over the whole source alphabet. The row-companion equality implies equality of every W_k.

The common coarse function fixes e and c. As $\mathrm{wt}(c) < \mathrm{wt}(e)$, connected primality gives $\gcd(e, c) = 1$. Thus $c\,\Delta C_{-k} = e\,\Delta D_{-k}$ implies $e \mid \Delta C_{-k}$. In the physical ordinary(a) moment alphabet, all actual units have value one. The positive row-degree lemma of [6] gives degree m for e, whereas every cofactor has degree at most $m - 1$. Thus both differences vanish. For positive q, the groups are the derivatives of e, c. For $q = 0$ they are

$$C_0 = me - \sum_{j>0} f_j \partial_{f_j} e, \qquad D_0 = (m-1)c - \sum_{j>0} f_j \partial_{f_j} c.$$

We have recovered the entire grouped source and native decks.

The least original label $q_* = x_m - y_1$ occurs uniquely, at the last-row/first-column entry. Its two cofactors are the genuine cross minor and, with the native sign, minus the middle minor. Dodgson's identity yields

$$s_{T_+} s_{T_-} = cC_{q_*} - eD_{q_*}, \tag{A.4}$$

where T_+, T_- are the connected first-row and last-row deletions; for two rows the middle determinant is 1. Connected primality, with repeated factors retained, identifies their unordered Schur pair. After rotating the mate if necessary, the first-row deletion functions match. The boundary-reconstruction theorem of [5, Reconstruction from two boundary observations] now gives $V' = V$ or $V' = V^*$.

The line inner and its full mark are self-reflecting. Composition rotation therefore gives literal or fully marked whole-source rotation. The actual row or column domino is self-reflecting too; transport its whole mark and, when necessary, the genuine ordinary/empty gauge. This puts the original whole mate in its original prescribed fiber. □

A.3. **Wide upper marks.** The following construction concerns an actual mark on an actual source. The upper mark is not treated as a second proper chart of that source.

Lemma A.3. *Let $E = V \circ_\mu J$ be a proper chart with period r, width v, and span $S = Mr + v$. Suppose E carries a legal upper mark of width $w > r$, period $Q = S - w > w$, and $r \nmid Q$. Put $g = \gcd(r, Q)$, $u = v \bmod g$. Then one of the following occurs:*

1. *there is an actual proper full legal chart of E of period $g \geq 2$, width u, and strictly smaller score than $(v, r + v)$;*
2. *the finer inner J is a literal line, with its actual marked reading.*

No equality of parent functions or smaller-area induction is needed for this geometric alternative.

Proof. By Lemma A.1, it suffices to treat ordinary(a), mixed(d), and horizontal empty finers. We first obtain the exact comparisons supplied by the upper mark. Let $[a_i, b_i]$ be its lower

marked interval, and subtract the upper displacement from the upper interval. The four rows of the following table give the complete endpoints:

(A.5)

	ℓ_i	h_i	$\ell_{Q+i} - v_{\omega,1}$	$h_{Q+i} - v_{\omega,1}$
(a)	a_i	$b_i + \mathbf{1}_{i \geq s_\omega}$	$a_i - \mathbf{1}_{i < t_\omega}$	b_i
(b)	$a_i - \mathbf{1}_{i \geq s_\omega}$	b_i	a_i	$b_i + \mathbf{1}_{i < t_\omega}$
(c)	a_i	$b_i + \mathbf{1}_{i \geq s_\omega}$	a_i	$b_i + \mathbf{1}_{i < t_\omega}$
(d)	$a_i - \mathbf{1}_{i \geq s_\omega}$	b_i	$a_i - \mathbf{1}_{i < t_\omega}$	b_i.

For $1 \leq i < w$, consecutive differences give

(A.6)

	$A_i - A_{Q+i}$	$B_i - B_{Q+i}$
(a)	$\mathbf{1}_{i=t_\omega}$	$-\mathbf{1}_{i=s_\omega}$
(b)	$\mathbf{1}_{i=s_\omega}$	$-\mathbf{1}_{i=t_\omega}$
(c)	0	$-\mathbf{1}_{i=t_\omega} - \mathbf{1}_{i=s_\omega}$
(d)	$\mathbf{1}_{i=t_\omega} + \mathbf{1}_{i=s_\omega}$	0.

Thus the A comparisons are nonincreasing and the B comparisons nondecreasing. The physical representatives $1, \ldots, r$ all occur because $w > r$. On a complete addition-by-Q residue cycle not containing a finer hole, these directed comparisons force equality throughout. Endpoints $0, S$ are used below only with their actual absent or present flag values.

Mixed finer. The finer B word is fully r-periodic with $\beta_0 = 0$, $\beta_v = 1$. The cycles make it fully g-periodic, hence $g > 1$ and $g \nmid v$. This includes the actual global values $B_0 = 0, B_S = 1$. Every ordinary upper or opposite mixed upper would require a cap pair $B_a = 0, B_{Q+a} = 1$, with the same interpretation when $a = 0$ or $a = w$. Periodicity excludes it. The upper is mixed(d).

The variable word A is g-periodic off the two finer phase classes. On a generic class its full residue cycle is monotone by (A.6); an upper charge cannot lie on such a constant cycle. Set the generic A values to γ, putting zero at the two possible exceptional classes for bookkeeping, and let $\rho = \sum \beta$, $c = \sum \gamma$, $r = k_r g$. The first and last actual fine private windows have singleton ends. Their exceptional-one counts are

$$Z_- = k_r(\rho - c) + \gamma_0, \qquad Z_+ = k_r(\rho - c) - 1 + \gamma_u. \tag{A.7}$$

The actual initial caps force $\rho - c \geq 1$. If the two classes coincided, the two terminal zero caps would give $Z_+ \leq k_r - 2$, contradicting (A.7). For two distinct classes the bound is $Z_+ \leq 2k_r - 2$, which forces $\rho - c = 1$. These counts omit the global endpoint when a phase is terminal; they count the internal initial cap when the other phase is zero.

Let the two distinct classes be $t_0 < s_0$. If zero is generic its value is $\gamma_0 = 1$, read from the first joining band; if u is generic its value is $\gamma_u = 0$, read from the last marked band. The same count on a single g-window shows that all its initial exceptional bits are one and all its final exceptional bits zero, and $H_E(g-1) = H_E(S-g) = 1$. Put

$$P_a = \sum_{i=1}^{a} (\gamma_i - \beta_i), \qquad \epsilon = \mathbf{1}_{t_0 > 0}, \qquad P_{g-1} = -1 - \epsilon.$$

The first-window heights are $1 + P_a + \#\{b \in \{t_0, s_0\} : 1 \leq b \leq a\}$. Writing $S = Lg + u$, the last-window heights are

$$\begin{cases} 1 + P_a - P_u, & u \leq a < g, \\ P_a - P_u, & 0 \leq a < u. \end{cases} \tag{A.8}$$

Positivity at its actual class $g - 1$, and positivity of the first-window height at u, force

$$0 \leq t_0 < s_0 \leq u, \qquad P_u = -1 - \epsilon, \qquad H_E(u) = 1.$$

In particular $u > 0$. Define

$$\Omega_a = 1 + P_a + \mathbf{1}_{t_0 > 0,\, a \geq t_0} \quad (0 \leq a < u), \qquad \nu = (-\rho, g - \rho).$$

Before t_0, positivity follows from the first window; after t_0, it follows from the last. The two extreme values of Ω are one. Construct G from the first g diagonals of E and the last g

diagonals translated by $-(L-1)\nu$. Their private overlap agrees in both endpoints, and its complete heights are

$$\begin{aligned} H_G(a) &= \Omega_a + \mathbf{1}_{a\ge s_0}, && 0 \le a < u,\\ H_G(a) &= 2 + \epsilon + P_a, && u \le a < g,\\ H_G(g+a) &= \Omega_a + \mathbf{1}_{a<t_0}, && 0 \le a < u. \end{aligned} \tag{A.9}$$

Use the actual first-window southeast endpoints and their drift $-\rho$ on the upper mark. The diagonal criterion proves connectedness. The overlap of G and $G-\nu$ is the southeast-justified interval of height Ω_a. Its thin extra tails occupy the disjoint ranges $[s_0, u-1]$ and $[0, t_0-1]$, so the overlap is the full maximal mark. Singleton extreme cells fix the displacement of any larger candidate. Removing either mark leaves a thin tail joined to the private singletons and the opposite end window; the inherited steps are binary. This proves I–III. The hatted middle is confined to $[u+1, g-2]$, while both clauses of the periodic hatted mark have content in $[0, u-1] \bmod g$. Their content distance is at least two, proving IV for every translate. The attachment is mixed(d), with phases t_0, s_0.

To reconstruct the outer, set

$$M_j = H_E(jg+u), \qquad S_j = H_E(jg+t_0) - \Omega_{t_0} + 1.$$

The actual exceptional bits give

$$S_j - M_{j-1} = A_{jg+t_0}, \qquad S_j - M_j = 1 - A_{jg+s_0}.$$

The heights M_j are positive and the true first and last windows give $M_0 = M_{L-1} = S_0 = S_L = 1$. These binary steps therefore construct a connected outer K. The entire source heights are

$$H_E(jg+a) = \begin{cases} \Omega_a + M_{j-1}, & 0 \le a < t_0,\\ \Omega_a + S_j - 1, & t_0 \le a < s_0,\\ \Omega_a + M_j, & s_0 \le a < u,\\ H_G(a) + M_j - 1, & u \le a < g. \end{cases} \tag{A.10}$$

Here $M_{-1} = M_L = 0$, and there is no terminal private band. The middle line includes precisely the mixed supplemental copies for all four pairs of coarse bits. The southeast endpoints agree by full g-periodicity. Thus $E = K \circ_\nu G$ on every cell, not just at the level of heights.

Ordinary finer, $g > 1$. Each word has one finer hole. First exclude a mixed upper. Its fixed boundary word becomes constant on every punctured residue cycle. In case (d), the actual initial B cap has value zero and the terminal cap value one. They force contradictory generic values on the same gcd class. If the finer southeast phase equals v, use the real upper private bit $B_w = 1$ in place of the omitted terminal cap. In case (c), use the analogous A caps, and when the finer northwest phase is zero use the real bit $A_Q = 1$. For an empty finer, the contradictory generic pairs are $B_Q = 0, B_w = 1$ or $A_Q = 1, A_w = 0$. All these positions are actual. Thus the upper is ordinary.

Write T for the A-hole class and B_0' for the B-hole class, and put $k_r = r/g$. The two classes may coincide. Put $\alpha_T = 0, \beta_{B_0'} = 1$ for bookkeeping and retain both actual streams. If $\Delta = \sum(\alpha - \beta)$, the exceptional charges are A_i on the first channel and $1 - B_i$ on the second. The first and last fine private windows give

$$Z_- = -k_r\Delta + \epsilon_0, \qquad Z_+ = -k_r\Delta - 1, \qquad \epsilon_0 = 1 - \mathbf{1}_{T=0} - \mathbf{1}_{B_0'=0}.$$

Initial caps force $\Delta \le -1$; each terminal channel has capacity at most $k_r - 1$, forcing $\Delta \ge -1$. Thus $\Delta = -1$. The argument counts two channels even when they share a phase. The one-window counts then force the first caps, last caps, and both private singleton ends. With $P_a = \sum_{i=1}^{a}(\alpha_i - \beta_i)$, the last-window formula is (A.8). If $u > 0$, positivity gives

$$P_u = P_{g-1} = -1 - \epsilon_0, \qquad T, B_0' \in [0, u], \qquad H_E(u) = 1.$$

Both phases cannot be zero. If $u = 0$, positivity at the last class $g-1$ forces both phases zero and $P_{g-1} = 0$.

For $u > 0$, use the actual antitransposition if needed so that $T \le B_0'$. The two word roles are complemented and exchanged; all cells and marks are transported. If both phases are positive,

put $\Omega_a = 1 + P_a + \mathbf{1}_{a\ge T}$; if $T = 0$, put $\Omega_a = 1 + P_a$. Before the first switch and between the switches, these are actual lower-window heights; after the second switch they are actual upper-window heights. Hence they are positive. Their ends are singletons; in particular $T < u$. With $b = \sum \beta$, put $\nu = (1-b, g+1-b)$. Match the actual private singleton to align the two end windows, and construct G with lower and upper marked heights $\Omega_a + \mathbf{1}_{a\ge B'_0}$ and $\Omega_a + \mathbf{1}_{a<T}$. The lower tail is southeast and the upper tail northwest. The genuine marked southeast word is β, because removing the lower tail restores its missing switch. The two marked intervals intersect in exactly the positive interval Ω. The same singleton, complement, and hatted-content checks used above prove I–IV, now with ordinary(a) attachments.

Read both endpoints of each whole-source private interval at $jg + u$, subtract the private coordinate of G and $j\nu_1$, and call the results ℓ^K_j, h^K_j. Summing the actual word over a block gives

$$\ell^K_{j-1} - \ell^K_j = A_{jg+T}, \qquad h^K_{j-1} - h^K_j = B_{jg+B'_0}. \tag{A.11}$$

Positive heights and singleton outer ends construct K. On a marked content $jg + a$, the literal copy union has endpoints

$$\ell(jg+a) = j\nu_1 + \ell_W(a) + \begin{cases} \ell^K_{j-1} - 1, & a < T, \\ \ell^K_j, & a \ge T, \end{cases}$$

$$h(jg+a) = j\nu_1 + h_W(a) + \begin{cases} h^K_{j-1}, & a < B'_0, \\ h^K_j + 1, & a \ge B'_0. \end{cases}$$

Only existing copies are used at the two ends. The positive overlap and binary coarse steps leave no interval gap. These formulas and (A.11) reproduce both source endpoints on every content. Pulling back an antitransposition gives the ordinary(b) chart when that reading was needed.

For $u = 0$, take the first g actual source diagonals as G, with its genuine horizontal empty displacement. The generic within-block height drift is zero, and both joins are the actual streams at residue zero. Reading both source endpoints at jg gives the coarse bits A_{jg}, B_{jg}. The same interval calculation gives $E = K \circ_{\mathrm{EMPTY}_H} G$. No positive unit mark is inserted. For an originally empty finer, the first and last channel counts give the same drift -1 with bookkeeping $\alpha_0 = 0, \beta_0 = 1$, so this construction applies unchanged to the actual empty edges.

In every constructed case, $g < r$, $u \le v$, and the proper supplied source has at least two new coarse diagonals. The new factors are connected and proper; the new inner is an actual first copy with extra source contents beyond it. Both factors have strictly smaller occupied area. Its score $(u, g+u)$ is strictly below $(v, r+v)$.

The ordinary coprime case. Here a period-one construction would not be proper. For positive ordinary(a), a constant generic A word forces the finer northwest phase to be zero: otherwise the actual generic bits at r and v are one and zero. The constant is then zero, and the first fine private height sum forces all generic B bits to be zero and its phase to be v. The inner is a literal row. Starting from a constant generic B word gives the same conclusion. If the inner is not a line, both punctured addition-by-Q cycles are nonconstant binary monotone paths. Each therefore contains its unique actual upper charge, with both endpoints outside that word's finer hole. Both upper phases lie in $[1, r]$.

The actual upper height identity is

$$H_i - H_{Q+i} = \mathbf{1}_{i\ge s_\omega} - \mathbf{1}_{i<t_\omega}.$$

At $i = r - 1$, positivity and $H_{r-1} = 1$ force the larger phase to be r. At $i = w - r$, the singleton $H_{S-r} = 1$ forces the smaller phase k to satisfy $k \le w - r$. Its strict generic comparison repeats at $k + r$ whenever $k + r < w$, which would create a second charge in the same word. Thus $w = r + k$. Both phases are strictly below w, and the actual ordinary upper private cap gives $A_w = 0, B_w = 1$ in either ordinary reading. The charge at k, however, propagates by the original generic r-word to $w = k + r$, forcing $A_w = 1$ or $B_w = 0$. This contradiction leaves only the literal row. For a horizontal empty finer, a constant generic word makes both internal

words identically zero or identically one, hence the inner a row or column. Otherwise both strict generic charges avoid residue zero and lie below r; the height identity at $r-1$ already gives zero height, a contradiction. The transports in Lemma A.1 recover every original reading. □

A.4. **The upper width equal to the finer period.**

Lemma A.4. *Under the notation of Lemma A.3, suppose $w = r$ and $r \nmid Q$. A mixed finer cannot carry the upper mark. A positive ordinary finer is either a literal line or has an actual proper empty-overlap chart on E with strictly better score. An empty finer is already aligned.*

Proof. For a mixed(d) finer the complete fixed word has $B_r = 0$ and $B_Q = B_v = 1$. The full end-window cap argument gives no legal width-r mark: if x is the private singleton on $r-1$ and y the one on $S-r$, its displacement would have to equal both $y - \mathrm{sw}(E)$ and $\mathrm{ne}(E) - x$. Their row components differ by $B_v - B_r = 1$, a contradiction. This is the width-equals-period exclusion in [9, Mark-transfer corollary for mixed marked inners], whose domain is a legal mixed inner and connected equal-span outers, as supplied here. Transport handles mixed(c).

For a positive ordinary(a) finer, the actual end windows give $H_{r-1} = H_Q = 1$, $A_Q = 0$, and $B_r = 0$. The private caps exclude mixed(c), mixed(d), and ordinary(b) uppers, leaving ordinary(a). Its marked end heights force upper phases $t_\omega = 0, s_\omega = r$, so the marks are the entire first and last r-bands. Removing the lower mark gives a connected complement beginning at r, whose extreme diagonal is singleton. That diagonal still contains the initial finer copy, so its upper thin tail must be absent: $t_\mu = 0$. Removing the upper mark gives $s_\mu = v$ by the same argument at $Q-1$.

Put $g = \gcd(r, v)$. The actual comparisons are $A_i = A_{Q+i}$, $B_i = B_{Q+i}$ for $1 \le i < r$. The missing representative r removes only the residue edge from zero to v. For the A word this edge meets its own hole zero; for the B word it meets its own hole v. Every remaining cycle is connected, or is a connected path after removing that hole. Hence each generic word is constant on its gcd classes. If $g = 1$, the fine endpoint and height argument in Lemma A.3 makes J a literal row.

If $g > 1$, both joining channels have residue zero. The actual nonexceptional bits are zero there, and the fine private height sum makes the within-block generic drift zero. Take the first g source diagonals as G, with its genuine horizontal empty displacement. Read both endpoints at jg to obtain the coarse bits A_{jg}, B_{jg}. The diagonal criterion gives connected K, and the whole interval union gives $E = K \circ_{\mathrm{EMPTY}_H} G$, as in the zero-remainder construction above. The factors are proper and strict, and $(0, g) < (v, r+v)$. Ordinary(b) is transported before this argument. Finally, $v = 0$ gives $Q = (M-1)r$, which is aligned. □

A.5. **The primitive ordinary cycle.**

Lemma A.5. *Let J be an actual positive ordinary inner of period r, width $v < r$, and contacts $1 \le s \le v$, $0 \le t < v$. In its ordinary(a) reading put $B = X_W \sqcup B_-$, $C = Y_W \sqcup C_-$, with the actual private row and column lists in (v, r). These are the equal-rank completed native sets of the ordinary residue fold* [9, Ordinary residue fold]*; in particular $s \in B$, $t \in C$, $0 \notin B$, and $v \notin C$. Suppose the actual no-eligible implications are*

$$b \in B \setminus \{s\} \Longrightarrow b + \delta \in B, \qquad c \in C \setminus \{t\} \Longrightarrow c - \delta \in C \pmod r,$$

and $\gcd(r, \delta) = 1$. Then the full marked J is self-rotational. This conclusion is reflection, not a proper period-one factorization.

Proof. The unique δ-cycle is broken only at its actual return. Since $0 \notin B$ and $v \notin C$, neither native set is full. Consequently

$$B = \{s - i\delta : 0 \le i < \ell_B\}, \qquad C = \{t + i\delta : 0 \le i < \ell_C\} \pmod r.$$

Equal native ranks give $\ell_B = \ell_C$, whence $C = (s+t) - B$. Applying the full cap-corrected height argument in Lemma 6.2, with the actual mark heights of J, gives $s + t = v$. The two complete words now satisfy $X_J = (r+v) - Y_J$; the full intersection mark is carried by this rotation. The same argument in the actual opposite ordinary gauge proves its corresponding reading. □

A.6. **The whole-parent zero-matrix finish.**

Corollary A.6. *At the original mixed comparison, suppose the entire first response matrix is zero. Put $g = \gcd(r, \delta)$. There is one full legal marked inner of period g, width $v \bmod g$, on J, E, U and on both original whole parents. Their genuine whole-parent outer functions agree, and the mate belongs to the original prescribed fiber. The source charts also improve the minimum held on E and U.*

Proof. The complete native supply of Lemma D.6 gives the full row classes and every foreign column class in both actual fields, for either sign of δ. Its row caps give $2 \le g < r$ and $g \nmid v$; its square-rank identity retains both exceptional column subsets. These are exactly the hypotheses of [9, Mixed residue fold]. That construction covers $g \le v$ and $g > v$, in the latter case keeping the literal original mark. It gives an actual proper chart $J = A_J \circ_\kappa K$, with all four legality conditions, both all-integer hatted tests, both words, and all supplements. Its factors are strict in occupied area.

Since $g \mid r$, full aligned composition constructs both source charts through the same (K, κ). Their score $(v \bmod g, g + v \bmod g)$ is strictly smaller than $(v, r + v)$, so the held finer-source minimum may already be used. For the separate whole-parent conclusion, both actual upper periods are the original common period

$$Q = Mr + \delta,$$

and hence are divisible by g. Apply aligned composition independently to the two actual upper marks. It constructs both whole charts through the same full (K, κ); their descended upper marks need not be identical. Every constructed factor is proper and strict by the retained-copy and composition area arguments in Lemma 2.7 and [9, Recovery from the generic words]. Only after these two charts exist apply Proposition 5.2 to the original whole-function equality, obtaining equality of their genuine outer Schur functions. Source-minimal domination at the original parent F then puts H in $\mathcal{F}(\omega, s_D)$. Neither upper divisibility has been inferred from a source-only root. □

A.7. **A finite closed-word rectangle construction.**

Lemma A.7. *Let P be an actual connected diagram of span $N = mg + a$, where $1 \le g < N$, $m \ge 2$, $0 \le a < g$, and $h_P(g-1) = 1$. Suppose the two closed words satisfy*

$$X_P(x+g) - X_P(x) = \mathbf{1}_{x=s}, \qquad Y_P(x+g) - Y_P(x) = -\mathbf{1}_{x=t} \quad (0 \le x \le N-g).$$

Then P is an actual composition of a rectangle through a legal ordinary or EMPTY inner of period g, width a. When $g \ge 2$, both factors are proper and strictly smaller in occupied area. When $g = 1$, P itself is a rectangle.

Proof. Write $s = k_s g + \sigma, t = k_t g + \tau$. Along each actual residue chain, including its real closed caps,

$$X_P(kg+z) = x_z + \mathbf{1}_{k>k_s}\mathbf{1}_{z=\sigma}, \qquad Y_P(kg+z) = y_z - \mathbf{1}_{k>k_t}\mathbf{1}_{z=\tau}.$$

Here $x_\sigma = 0, y_\tau = 1$ and $0 \le k_s, k_t \le m-1$. Put $B = x + \delta_\sigma, C = y$, $\Phi(z) = \sum_{i \le z}(C(i) - B(i))$, and

$$A(k) = k - \max(0, k - k_s - 1) - \max(0, k - k_t - 1).$$

The actual first singleton gives $\Phi(g-1) = 0$, and direct summation, at each position which actually occurs, gives

$$h_P(kg+z) = A(k) + \Phi(z) + \mathbf{1}_{z \ge \sigma}(1 - \mathbf{1}_{k>k_s}) - \mathbf{1}_{z \ge \tau}\mathbf{1}_{k>k_t}.$$

At positive width, first-block positivity gives $\Phi(z) \ge 1 - \mathbf{1}_{z \ge \sigma}$. The true last occupied complete block has height $A(m) \ge 1$. At the actual top,

$$0 = A(m) + \Phi(a) - \mathbf{1}_{\tau \le a}.$$

It follows that $A(m) = 1, \Phi(a) = 0, \tau \le a$, and then $\sigma \le a$. Thus $k_s + k_t = m - 1$. Final-partial positivity also gives $\Phi(z) \ge \mathbf{1}_{z \ge \tau}$ for $z < a$. The contact pairs $(0,0), (a,a)$ are excluded by these

inequalities and $\Phi(a) = 0$. At $a = 0$, the actual final singleton gives $A(m) = 1$, and the top equation is $1 = -1 + \mathbf{1}_{\sigma=0} + \mathbf{1}_{\tau=0}$. Hence both contacts are zero and again $k_s + k_t = m - 1$.

Construct K by its actual first g-word and the final partial word with both jumps completed. Its heights are the first g and last a heights of P, with both private heights one. At positive width, if $\sigma \geq \tau$, its mark height is $\Phi(z) > 0$ for $z < a$. The excluded equal endpoint pairs give $0 \leq \tau < a$, $1 \leq \sigma \leq a$. Writing A, B for the two endpoint bits of this constructed K, the completed jumps give

$$A_{g+c} = A_c - \mathbf{1}_{c=\tau}, \qquad B_{g+c} = B_c + \mathbf{1}_{c=\sigma} \quad (1 \leq c < a).$$

They also give $h_K(g) = 1 + \mathbf{1}_{\tau>0}$ and $h_K(c) - \mathbf{1}_{c\geq\sigma} = \Phi(c) \geq 1$ for $c < a$, besides $h_K(a) = h_K(g-1) = 1$. At $\sigma = a$ the actual low row bit is $B_a = 0$; at $\tau = 0$ the completed upper column bit is $A_g = 0$. These are all the hypotheses of [9, Ordinary cell criterion], which constructs a full type-(a) mark. If $\sigma < \tau$, perform that construction on the actual antitranspose; its mark height is $\Phi(z) + \mathbf{1}_{z\geq\sigma} - \mathbf{1}_{z\geq\tau} > 0$. Transporting back gives the genuine type-(b) mark. At zero width use the connected first g diagonals and the actual EMPTY convention.

For type (a), the outer words are $X_D(k) = \mathbf{1}_{k>k_s}$, $Y_D(k) = \mathbf{1}_{k\leq k_t}$. Since $k_s + k_t = m - 1$, they are exactly the rectangle of width $k_s + 1$ and height $k_t + 1$, with positive heights $A(k+1)$. For type (b), the actual antitranspose interchanges k_s, k_t in this construction. The full compiler reproduces every one of the two closed words of P, and thus every cell. The root uses its disjoint first g and last a diagonal windows, leaving positive diagonals unused. The outer heights are the distinct samples $h_P(kg + g - 1)$. These prove both strict area inequalities for $g \geq 2$; for $g = 1$, the inner is a cell and the diagram itself is the rectangle. □

A.8. **Endpoint cap splices and the genuine gcd-one exit.** The auxiliary splice notation is that of [6, Fixed-cap rigidity]: for a child A of span c and a cap K,

$$X_{K\triangleleft A} = (c + X_K) \cup (X_A \setminus \{c\}), \qquad Y_{K\triangleleft A} = (c + (Y_K \setminus \{0\})) \cup Y_A.$$

These are complete lists, including common beta positions. This 00-splice is not an overlap-one row extension.

Proposition A.8. *Let J be the actual ordinary finer inner of period r, width $0 < v < r$, and endpoint contacts $s = v, t = 0$. Suppose its actual upper comparison is either ordinary of positive width $0 < w < v$, or genuinely EMPTY. Retain the complete physical endpoint identity and the actual held finer chart on E, with its paired chart on U. Then J is literally self-rotational with its full original mark, or an actual proper aligned chart improves the minimum held on E and U.*

Proof. Put $N = r + v$, and set $\delta = v - w$ in the positive case, $\delta = v$ in the EMPTY case. The physical endpoint construction Lemma E.2 gives actual connected caps B, D of span δ, actual connected strict children

$$S = D \triangleleft J, \qquad T = J \triangleleft B, \qquad s_S = s_T, \qquad \operatorname{sp} S = \operatorname{sp} T = N + \delta =: H.$$

Both occupied areas are below $|E| < n$. It also gives the two complete shared word identities

$$X_S(x) = X_T(x + \delta), \qquad Y_S(x) = Y_T(x + \delta) \quad (0 < x < N), \tag{A.12}$$

the identical first and last δ-diagonal cell windows, and equality of both end words on $0 < x < \delta$ and $N < x < H$. The two true splice seams are

$$(X_T(\delta), Y_T(\delta)) = (0, 0), \qquad (X_S(N), Y_S(N)) = (0, 0). \tag{A.13}$$

The other two seam values are not assumed equal. The original endpoint mark on J has the true 00 seams at v, r.

If $T = S^*$, the last N-diagonal windows give $J = J^*$. Suppose $T = S$. Then the two words of J are internally δ-periodic. For its original completed native sets B_0, C_0, this yields

$$B_0 \setminus \{v\} + \delta \subseteq B_0, \qquad C_0 \setminus \{0\} - \delta \subseteq C_0 \pmod{r}.$$

To see the wrap in the column assertion, use the low original window for a column above δ, and the genuine high window at $r + c$ for $0 < c \leq \delta$; the equality case would otherwise put a column

at the true absent position r. The row assertion uses the opposite window in the same way. Thus every foreign residue cycle modulo $d = \gcd(r, \delta)$ is full or empty. If $d \geq 2$, [9, Ordinary residue fold] and Lemma 5.1 give the claimed strict source improvement. If $d = 1$, Lemma A.5 gives full reflection.

Otherwise lower FULL is applied to the actual equal strict children. Choose a proper source-minimal chart of S, and obtain a chart of T through the same or fully rotated mark, with equal genuine outer functions. Let its period be g, width a, and outer span m, so $H = mg + a$, $0 \leq a < g$. A line outer reduces to the literal or rotational cases already treated. We henceforth have $m \geq 3$.

First exclude carried mixed roots before using any ordinary common-window bound. Since $g \leq H/2 < N$, a carried type-(d) row word and the actual seam give $X_S(N - g) = X_S(N) = 0$. The shared comparison then gives $X_T(H - g) = 0$, contrary to its terminal private row. For carried type (c), its genuine first private column gives $Y_S(g) = 1$, whereas the shared comparison and $Y_T(\delta) = 0$ give $Y_T(g + \delta) = 0$. This is again impossible.

We claim $g \leq \delta$. If $g > \delta$, every proper return $kg + \sigma$ or $kg + \tau$, $1 \leq k < m$, lies strictly between δ and $H - \delta = N$. Thus (A.12) observes the whole source strings at those positions and the whole target strings at starts $kg + \sigma_T - \delta, kg + \tau_T - \delta \in (0, N)$. This comparison is clean, unlike the singleton-error terminal comparison. Its length $mg + a - \delta > 2g$ includes every native phase twice. An unmatched ordinary return string therefore has every proper bit equal to one fixed native bit, so its actual outer is a row or column. The genuine outer equality supplied by the preceding lower-FULL call makes the mate the same line. For crossed ordinary roots the only alternative is aligned holes, giving $B(z) = C(\sigma + \tau - z)$. Full marked reflection by Lemma 6.2 makes the chart carried and forces $g \mid \delta$, impossible. A crossed mixed pair has a fully periodic word on one side and two distinct returned strings on the other. Both whole proper strings are in the displayed ranges; the clean comparison makes each equal its opposite native bit. One constant string already makes that outer a line, and the earned genuine equality gives the same line mate. All these cases return literal or rotational children. Carried mixed roots were excluded at their actual seams above, before this ordinary/crossed argument. Hence $g \leq \delta$.

Since $g \leq \delta$, the identical actual end windows reconstruct the same unmarked K. If its marks were crossed ordinary, rotation would give the same-type displacement on this one inner; equal-period mark uniqueness gives the same full mark. If they were crossed mixed, they would give two incompatible mixed displacements on that same inner, forbidden by the same full gluing theorem [9, Equal-period comparison lemma]. Together with the preceding mixed exclusion this leaves a carried ordinary or EMPTY chart, still with the original equal outer functions.

We next prove $g \mid \delta$, keeping the two missing seam ports. Suppose otherwise. In an ordinary type-(a), or horizontal EMPTY, reading, write its proper row-port positions as $z_k = kg + \sigma$, $1 \leq k < m$. The shared equality makes the proper source row bits equal a constant b for $z_k < N$, and the target bits equal a constant c for $z_k > \delta$. These are the native bits at the two unmatched phases. Put

$$c_0 = \#\{k : \delta < z_k < N\} > 0, \qquad I_\delta = \mathbf{1}_{\delta \text{ is a proper row port}}, \quad I_N = \mathbf{1}_{N \text{ is a proper row port}}.$$

The strict end-window agreements and equal genuine outer row counts give the exact count

$$c_0(b - c) + I_\delta b - I_N c = 0. \tag{A.14}$$

Here $c_0 > 0$ because $N - \delta = r + w > \delta \geq g$, with $w = 0$ in the EMPTY case. If neither seam is a port, $b = c$ and both entire outer row words are constant, so the outers are the same line. If $I_\delta = 1$, the inverse comparison reads native row phase zero, absent in an ordinary type-(a) word, so $c = 0$; then the count gives $b = 0$. If also $I_N = 1$, the native phase-a value gives an immediate opposite value. If instead $I_N = 1, I_\delta = 0$, the forward comparison reads the present native phase a, so $b = 1$, and the count becomes $c_0 = (c_0 + 1)c$, impossible.

For ordinary type (b), apply the actual antitranspose to both children and their common mark. The resulting type-(a) seams have row value one, and the complete count is now

$$c_0(b - c) + I_\delta(b - 1) + I_N(1 - c) = 0. \tag{A.15}$$

If $I_\delta = 1, I_N = 0$, its absent native phase gives $c = 0$, so $(c_0 + 1)b = 1$, impossible. If both are one, the opposite native phase gives the contradictory value $b = 1$. If $I_N = 1, I_\delta = 0$, that phase gives $b = 1$, and the count gives $c = 1$, hence the same line outers. With neither seam a port the line conclusion again follows. Every failure of divisibility therefore returns a literal case. We may assume $g \mid \delta$.

Return to the original 00-seam frame. At positive width, type (b) would have its native row value one at the seam phase a, contrary to $X_S(N) = 0$. In type (a), the seam at δ forces $\tau = 0$, and the seam at N forces $\sigma = a$. At width zero the convention is the canonical EMPTY one. We now restrict an actual chart to J, not merely its words. Write $N = jg + a$; then $j \geq 2$, since $N > 2\delta \geq 2g$. In the coarse outer of S, the source singleton at $N - 1$ and the zero seam at N give consecutive coarse heights one and both coarse seam bits zero. Indeed, if M_i are the coarse heights and S_j the crossing span at this seam, the full endpoint formulas give

$$S_j = 1, \qquad S_j - M_j = B_j = 0, \qquad S_j - M_{j-1} = A_j = 0.$$

The actual coarse prefix on contents $0, \ldots, j - 1$ is therefore connected. Composing this prefix with the same K gives precisely J: the next copy starts at jg and contributes below N only the already retained endpoint mark; there are no ordinary supplements. At EMPTY width the same 00 seam and disjoint edge convention give the identical conclusion. Thus

$$J = A_J \circ_\kappa K \tag{A.16}$$

is an actual proper chart with both true end windows.

If $g \mid r$, its width $a < g \leq \delta \leq v$ immediately improves the held source chart. Otherwise put $\rho = r \bmod g$. For $a > 0$, fill the actual endpoint native words by zeros at their returns:

$$b = B_K - \delta_a, \qquad c = C_K - \delta_0, \qquad b(0) = b(a) = c(0) = c(a) = 0.$$

For EMPTY use the analogous zero-filled residue-zero words. The original mark supplies $X_J(x) = X_J(x + r)$, $Y_J(x) = Y_J(x + r)$ for $0 < x < v$, and the genuine zero seams at v, r. Reading both words in the actual chart (A.16) gives

$$b(z + \rho) = b(z), \qquad c(z + \rho) = c(z) \quad (z \in \mathbb{Z}/g\mathbb{Z}). \tag{A.17}$$

Here are the residues not supplied by a generic interior reading. The initial returned row $z = a > 0$ is actually zero and gives $b(a + \rho) = 0$. The missing source residue zero is supplied by the true row-zero seam at r, giving $b(\rho) = 0$. The possible returned target at $v \bmod g$ is supplied by the true row-zero seam at v. In the column word the seam at r supplies the missing source residue zero. For target residue zero and $a > 0$, use the actual position $x = v - a > 0$, whose translated column at jg is zero. If $a = 0$, this position would be a cap; use the actual column-zero seam at v instead. Thus (A.17) includes every residue in both words.

Let $d = \gcd(g, r)$. The two zero-filled words are d-periodic. At $d \geq 2$, the ordinary residue construction gives an actual endpoint inner K_0 of period d, width $a_0 = a \bmod d$, including genuine EMPTY when $a_0 = 0$. More explicitly, zero complete-block drift follows from the private singleton at $g - 1$, and the actual mark endpoint supplies the singleton at $a_0 - 1$ when positive. For $a_0 > 0$, on the first $d + a_0$ diagonals the endpoint bits satisfy $A_{d+c} = A_c$, $B_{d+c} = B_c$ for $1 \leq c < a_0$. The two actual zero seams give $A_d = B_{a_0} = 0$, and the zero drift gives $h(d) = h(a_0) = h(d - 1) = 1$. On the marked band $c < a_0$, the required retained height is the positive actual height $h(c)$. Thus [9, Ordinary cell criterion] applies with $(\sigma, \tau) = (a_0, 0)$. At $a_0 = 0$, take the connected first d-window with its genuine horizontal edges; its joining bits are both zero, as required by [9, Empty residue fold] when K was already EMPTY, or by the zero-remainder ordinary fold otherwise. Their full words give $K = \mathrm{Row}_{g/d+\lfloor a/d \rfloor} \circ K_0$. Two actual aligned gluings, first through g and then through r, now improve the minimum held on E, and on its paired source U.

At $d = 1$, both zero-filled native words are identically zero and K is a row. Every proper returned row of the original mark maps under $+r$ to a nonreturned zero; every proper returned column maps to the nonzero native phase ρ and is also zero. Thus the original W is Row_v, and

both actual caps are $B = D = \mathrm{Row}_\delta$, with $B^* = B$. The supplied splice equality and whole rotation give

$$s_{B\triangleleft J} = s_{J\triangleleft B} = s_{(J\triangleleft B)^*} = s_{B\triangleleft J^*}, \qquad s_J = s_{J^*}.$$

These are the two observations of [6, Fixed-cap rigidity]. The cap is the same literal connected row B on both sides, and the actual connected children are J, J^*; equality of cap functions alone is not being used. It follows that $J = J^*$. The mark is the entire end intersection, so the rotation carries the full original mark and all its hatted data. This finishes the positive and EMPTY upper cases without asserting a nontrivial divisor at gcd one. □

A.9. **Principal comparisons at every maximal head.**

Proposition A.9. *Let the actual type-(d) principal packet have original period r, width v, $\delta = v - w > 0$, and complete native row period $g = \gcd(r, \delta)$, where $g \geq 2$ and $g \nmid v$. Choose the larger actual connected head H of an active constant pair. Both actual row equations are retained: their heads satisfy $H_k \geq k, N-(j-\delta)$ and $H_j \geq j, N-(k-\delta)$, and $H = \max(H_j, H_k)$. Put $N = r + v$, $L = N - H$, $D = H - v$. Retain its complete compensated child equality and the four actual cuts*

$$\begin{aligned} X_S &= X_J \cap (0, H], & Y_S &= (Y_J \cap [0, H]) \setminus \{k\}, \\ X_T &= (X_J \cap [L, N]) - L, & Y_T &= ((Y_J \cap [L, N)) \cup \{b\}) - L, \end{aligned}$$

where $v < H < N$, $L \leq b < H$, $L < k \leq H$, $b < r$, $k > v$, and $b \neq k$. Suppose the earned maximal head bound is $2H \geq N + \delta + |v - s - t|$. Then every ordinary or EMPTY lower-root outcome gives an actual divisor factor of J and an improving chart at the source minimum. If $H > r$, every outcome gives that exit. If $H < r$, the only unresolved geometric alternatives are literal children or a carried type-(d) common root; they are the actual private pairs consumed by Corollary 4.6 and Proposition 6.5.

Proof. The four cuts give the complete positive comparison on $[L, H]$,

$$X_T(x - L) = X_S(x), \qquad Y_T(x - L) - Y_S(x) = \mathbf{1}_{x=b} + \mathbf{1}_{x=k}.$$

The genuine two original mark bands give, separately,

$$X_T(x + D) = X_S(x), \qquad Y_T(x + D) - Y_S(x) = -\mathbf{1}_{x=t} - \mathbf{1}_{x=s} \quad (0 \leq x \leq v).$$

The inserted column is below r, so it does not enter this second window, and the deleted column is above v, so it does not enter the first reading of that window. The equation includes the true closed caps, also when $s = v$.

The physical construction gives actual connected equal children with $|S| = |T| < |J| < |E|, |U| < n$. Thus lower FULL is legitimate. Literal rotation is impossible: on the reflection-invariant overlap the zero row comparison would force zero column difference, contradicting its two positive defects. If the children are literal and $H > r$, the closed cuts are the hypotheses of Lemma 6.4, with $H \geq r - 1$ and $L < v$, and give the source exit. For a nonliteral pair, take the actual lower chart supplied by FULL, with period f, width a, outer span $m \geq 2$. Line outers return to the literal alternatives.

The maximal-head bound gives

$$A := H - L \geq \delta \geq g, \qquad v > \delta \geq g, \qquad r \geq 2g, \qquad H > 2g, \qquad H - g > H/2 \geq f.$$

Both child row words have full period g on $[0, H - g]$, including their closed caps; for T this uses the actual cut row $X_J(L) = 0$. If the lower root is ordinary or EMPTY, apply Lemma 6.1 to one child's comparison with its own g-translate. The entire native period is observed, so $f \mid g$. The positive overlap has length at least g, and the same lemma aligns the two child row ports. In a crossed chart its full reflection argument makes the mark carried. Thus $f \mid L$. The two real column defects are then at actual column-return positions. Restoring them leaves every generic word unchanged, so [9, Recovery from the generic words] sews J through that same full K. Its two extreme windows are actual: $k > v > g \geq f$ leaves the first intact, and $b < r < N - f$ leaves the last intact. Since $f \mid g \mid r$, Lemma 5.1 gives the held-source exit at every head.

We include the mixed-orientation reduction at low heads, where a full f-period need not occur in the second comparison. Suppose an actual mixed type-(c) chart occurs on either child and $f > g$. Its full row word has period g and punctured period f, with two holes $0 \le \tau < \sigma \le a$. Its real initial values are both zero and its terminal values both one. The internal word on $1, \dots, H-1$ has length $mf + a - 1 \ge 2f + a - 1$. The finite two-hole period lemma [9, Finite-period lemma, part (iii)] gives full internal period $e = \gcd(f, g)$. If $\tau > 0$, the two internal positions $\tau, mf + \tau$ contradict that period. If $\tau = 0$, the actual closed g-period gives $X(g) = X(0) = 0$, whereas the internal position mf has value one; their difference is a multiple of e. This separately handles the missing initial cap. Thus type (c) is impossible when $f > g$, on either child.

If $f \le g$, the positive overlap contains a full native f-period. Carried type (c) has a fixed column word whose nonnegative cyclic difference must vanish, contrary to the defects. For crossed (d)/(c), the first complete target row period has one fewer one than the fixed source row period; for crossed (c)/(d), the last complete source row period has one more one than its fixed target period. Both contradict the zero row comparison. The counts are the actual initial $1, 1$ and terminal $0, 0$ column ports of a type-(d) chart after full rotation. Therefore at every maximal head the only mixed orientation is carried type (d).

When $H > r = L + D$, the sufficient coverage conditions of Theorem 6.3 hold for every actual lower period $f \le H/2$. Indeed, with $b_0 = \min(L, D)$, $c_0 = \max(L, D)$, we have $b_0 < H/2$, hence $H - b_0 \ge f - 1$, and $H - c_0 > b_0 \ge \gcd(f, b_0) - 1$. The second column comparison is nonpositive. That theorem constructs the same full marked factor on J, S, T, with period $d \mid L, D, r$, $2 \le d < r$, strict areas and all supplements. Only then is the original held source chart lifted.

Finally $H = r$ is impossible because the actual head is a present row, whereas the full native row class of zero is absent. If $H < r$, both actual heads of the chosen pair lie below r. The principal row equations give $j, k < r$ and $N - (j - \delta), N - (k - \delta) < r$; the genuine outside inputs already satisfy $j, k > v$. Hence they place $j, k, j - \delta, k - \delta$ in (v, r): they are the complete private pairs of Corollary 4.6. Each actually supplied active second response is therefore zero in its own field, and Proposition 6.5 supplies the required actual divisor root. The orientation reduction is stated separately to identify exactly what geometry has supplied before this scalar return is used. □

Appendix B. Whole currents and the first complete grade

This appendix gives the determinant identities used in Section 4. All row and column lists are the actual complete lists. A common row and column is retained as a genuine unit, and every proper determinant means the entire determinant, with multiplicity.

B.1. **Rectangular splices and current activation.**

Lemma B.1. *Let A have span S, strict flags X, Y, and a legal horizontal row mark Row_w of width $w \ge 0$, with $Q = S - w$. Write $\ell_x = \mathrm{cof}_{x,0}\,\mathrm{JT}(A)$ and $R_y = \mathrm{cof}_{S,y}\,\mathrm{JT}(A)$. For the actual two-copy row composition $F = \mathrm{Row}_2 \circ A$,*

$$X_F = (Q + X) \cup (X \setminus \{S\}), \qquad Y_F = (Q + (Y \setminus \{0\})) \cup Y, \qquad s_F = \sum_{x \in X,\ y \in Y} \ell_x R_y h_{Q+x-y}.$$

Here $w = 0$ means the genuine horizontal empty mark; at positive width both marked copies lie in one actual end row. If X is closed under subtracting r at positive results and has no multiple of r, then the whole substitution $h_i \mapsto h_i + z h_{i-r}$ fixes every R_y and sends

$$\ell_x \longmapsto \sum_{j \ge 0} (-z)^j \ell_{x+jr}.$$

Consequently, for every $L \ge 1$,

$$[z^L] s_F(h_i + z h_{i-r}) = (-1)^{L-1} \sum_{b=1}^{r} \ell_{b+(L-1)r} \sum_{y \in Y} R_y h_{S+b-r-w-y}.$$

Absent cofactor labels have value zero.

Proof. For a one-row source the identity is immediate. Otherwise the last row of the lower copy and the first row of the upper copy merge at the genuine horizontal mark. All other rows remain separate. This gives the two displayed flag lists, including the empty case. The upper and lower end-window conditions give $x_2 \le Q$ and $y_{n-1} \ge w$, strictly in the empty case, so that $x - y - Q < 0$ for $x \ne S$, $y \ne 0$. In the induced orders the full matrix therefore has the rectangular form

$$\begin{pmatrix} A \text{ without column } 0 & (h_{Q+x-y}) \\ 0 & A \text{ without row } S \end{pmatrix}.$$

Laplace expansion on its first $n-1$ columns gives the asserted sum: the sign for omitting row x is canceled by the sign defining ℓ_x, and the remaining top-row expansion gives R_y.

For the substitution, expand every changed row. In an R_y minor the least changed row either duplicates an unchanged preceding row or becomes a strictly negative row; only the unchanged term remains. In an ℓ_x minor, the only surviving changed set is $x + r, \dots, x + jr$. Sorting this chain and including the two cofactor signs gives $(-z)^j \ell_{x+jr}$. Apply these identities to the splice sum. At degree L, the two possible contributions at label x' have coefficient

$$(-1)^{L-1}\big([x' - (L-1)r \in X] - [x' - Lr \in X]\big).$$

Closure and the absence of residue zero make this the indicator of $x' = b + (L-1)r$, $1 \le b \le r$. This proves the last formula at every L, including its real terminal coefficient. □

Lemma B.2. *Work in the type-(a) reading of a positive ordinary upper on E. Here W is the upper overlap, $w = N_W$, $Q = N_E - w$, and s, t are its contacts. Order all flags decreasingly and put*

$$\begin{aligned} A &= s_W, \\ B &= A_{s,t}(W) = (-1)^{i_s + j_t} \det \mathrm{JT}(X_W \setminus \{s\}, Y_W \setminus \{t\}), \\ C &= \det \mathrm{JT}(X_E \cap (w, Q), Y_E \cap (w, Q)). \end{aligned}$$

where i_s, j_t are the positions of s, t in the mark lists. Empty determinants are one. Then

$$B\,\mathcal{G}_Q = \mathcal{W}\, s_{\mathrm{Row}_2 \circ_\omega E} + (\partial_{h_w} B) s_E^2, \qquad \mathcal{W} = B\,\partial_{h_w} A - A\,\partial_{h_w} B \ne 0.$$

The same equality holds with $B = 1$, $\mathcal{W} = 1$ at an empty horizontal upper, interpreted by Lemma B.1.

Proof. Order the complete fixed blocks by increasing period. There are two complete contact-deletion blocks and one private block in one period, and three and two, respectively, in two periods. The one-period fixed determinant is $d_1 = B^2 C$. The two-period returned insertion has parity $(-1)^{i_s + j_t}$; absorbing it gives the signed fixed factor $d_2 = B^3 C^2$. Let

$$w_1 = (-L_1 D_1^{-1}, 1), \qquad z_1 = (-D_1^{-1} Z_1, 1)^T$$

be the actual two port records. The two-period right record agrees with z_1 on all one-period columns. Their difference is confined to the new upper B, C blocks, which the shifted w_1 annihilates. Hence $w_1 K_Q z_1 = R_2$, the actual second returned coefficient.

The complete global left and right records are

$$\ell = \eta_1 d_1 (c_L w_1 + R_1 \alpha), \qquad R = \eta_1 d_1 (c_R z_1 + R_1 \beta).$$

The row α is supported on the initial mark, and the column β on the terminal mark; they vanish when the corresponding extreme is returned. The two mixed contractions $w_1 K_Q \beta$ and $\alpha K_Q z_1$ are actual source null equations by the open mate implications. The seed–seed contraction is $\partial_{h_w} B / B$. The full mark Schur complement gives

$$c_L c_R = \partial_{h_w}(A/B) = \mathcal{W}/B^2, \qquad s_E = \eta_1 d_1 R_1, \qquad s_{\mathrm{Row}_2 \circ_\omega E} = d_2 R_2.$$

Substitution, with the common actual sorting signs, proves the formula. Both A and B are affine in h_w, but B may depend on that letter. The nonzero proper minor B and the nonzero extreme cofactor $\partial_{h_w} A$ have smaller ordinary degree than the connected prime A. Therefore

$$\mathcal{W} \bmod A = B\,\partial_{h_w} A \ne 0.$$

At an empty mark there is no seed and no derivative of h_0; the rectangular splice gives the asserted specialization directly. □

Lemma B.3. *For a positive mixed (d) upper of arbitrary mark rank k and private rank b, let K_t be the actual source minor obtained by deleting its top row and returned column t. Here W is the upper overlap, $w = N_W$, and all flags are ordered decreasingly. Put $h = \det \mathrm{JT}(X_W \setminus \{w\}, Y_W \setminus \{t\})$ and $k_t = \det K_t$, with an empty determinant equal to one. The minor h is unsigned; its cofactor parity is included in the displayed sign. Then*

$$k_t T_Q(\mathrm{JT}(E)) - s_E T_Q(K_t) = (-1)^{k+b} h\, s_{\mathrm{Row}_2 \circ_\omega E}.$$

All raised entries, including raised negative slots and raised units, are included. Whole rotation gives the corresponding (c) identity.

Proof. Put $C_0 = Y_W \setminus \{t\}$, $F = Y_W \cup \{s\}$, and $a = k + b$. Order the first rows as $(Q + X_W) \cup B$, and the last rows as X_W; order the first columns as $(Q + C_0) \cup C$, and the last as F. The whole matrix is

$$M = \begin{pmatrix} A_0 & J \\ 0 & W^+ \end{pmatrix}, \qquad A_0 = \begin{pmatrix} H & P \\ 0 & N \end{pmatrix}, \quad H = \mathrm{JT}(X_W, C_0).$$

Here A_0 has a rows and $a - 1$ columns. Delete the first row of H to obtain determinant h, and put $n = \det N$. Let $B_s = [A_0 \mid j_s]$ and $r = e_a^T \operatorname{adj} B_s$. Then $rA_0 = 0$, $rj_s = \det B_s$, and $r_1 = (-1)^{a+1} hn$. Write L for the actual raised high-column block, v_s for the actual source column vector at s, and $\chi_s = (-1)^{|\{c \in C_0 : c > s\}|}$. Expansion of the full two-copy rectangular matrix gives

$$R_{\mathrm{row}} := s_{\mathrm{Row}_2 \circ_\omega E} = \chi_s \{\det M \, \det B_s - rL \operatorname{adj}(M) v_s\}.$$

Indeed the fixed rectangular columns force one omitted row; their signed maximal-minor vector is r. Expansion on that row leaves the full source adjugate, so this equality contains every mark component.

Set $\rho = r/r_1$, $\alpha = \rho v_s = \chi_s \det G_t / h$, where $G_t = W^+ \setminus t$. There is a unique solution $H\eta = k_s - \alpha e_1$. Extend η to the source vector b_0 on its high fixed columns. Then $Mb_0 = v_s - \alpha e_S$, and with $z = \operatorname{adj}(M) e_S$,

$$\operatorname{adj}(M) v_s = \alpha z + \det(M) b_0.$$

In the low column list let $d_s = 1$, $d_{C_0} = -\eta$, $d_t = 0$. Thus $W^+ d = \alpha e_1$ and $Jd = j_s - Lb_0$. If f_t is the position of t in F, put $\beta = (-1)^{f_t - 1} \det G_t$ and $y = e_t^T \operatorname{adj}(M)$. The top part of y is βr; multiplying the low columns by d yields

$$\beta r J d + \alpha y_{\mathrm{bottom},1} = 0.$$

The bottom rows of $K_Q z$ equal $\det(M) e_1$, by the actual source null equation. Hence the complete off-extreme contraction is

$$(\operatorname{adj} M K_Q \operatorname{adj} M)_{t,S} = \beta r L z + \det(M) y_{\mathrm{bottom},1} = \beta \{rLz - \det(M) rJd/\alpha\}.$$

The previous two-copy expansion becomes $R_{\mathrm{row}} = \chi_s \{\det(M) rJd - \alpha rLz\}$. Consequently the off-extreme contraction is $(-1)^{f_t} h R_{\mathrm{row}}$. Finally, the actual adjugate entry is $(-1)^{a+f_t} k_t$, with $k_t = hn \det G_t$. The directional adjugate identity converts the last contraction exactly to the displayed compound formula. No rank-one hypothesis on the mark has entered. Reversing both complete flag lists transposes both the source and raised matrices and proves the (c) case. □

B.2. **The first grade of the complete coarse array.**

Lemma B.4. *Let V, V' be distinct connected coequal diagrams of span $M \geq 3$. Set $R_0 = \rho \neq 0$ and $R_\ell = \epsilon x_\ell$ for $\ell > 0$. Let μ be the first nonzero grade of the entire absolute adjugate difference. There are d, a with*

$$d = M - a \geq \lceil M/2 \rceil, \qquad 1 \leq a \leq M - 2$$

such that

$$\Delta \mathcal{A}^{[\mu]}(d, 0) = -\Delta \mathcal{A}^{[\mu]}(M, M - d) = U_d \neq 0. \tag{B.1}$$

Both boundary differences vanish for every larger distance $D > d$. For the entire prefixes

$$C_{b,D} = -\sum_{j-k=D,\ k\geq b} \Delta\mathcal{A}^{[\mu]}(j,k),$$

one has $C_{1,d} = C_{M-d,d} \neq 0$ and $C_{b,d} = 0$ for $b > M - d$.

Proof. The positive row-degree lemma of [6] makes the number n of nonempty rows equal. With the same coherent unit $R_0 = \rho \neq 0$, the whole determinants are their degree-n homogenizations and are equal. Remove all common row and column flags I from V, obtaining its actual connected zero-free core B, of rank q. Write $m = |I|$. In native core/unit order the entire matrix is

$$\begin{pmatrix} \epsilon M_B & \epsilon U \\ \epsilon V & \rho I_m + \epsilon T \end{pmatrix}.$$

The off-diagonal unit interaction T is retained. If τ_I is the full row/column sorting sign, its determinant is

$$\tau_I \rho^m \epsilon^q (s_B + O(\epsilon)).$$

Thus coequality fixes q, m, and positivity fixes the common sign and the equality $s_B = s_{B'}$. The complete inverse gives

$$\mathcal{A}_V^{[q-1]} = \tau_I \rho^m \operatorname{adj} M_B$$

embedded on the actual core flags; every other entry at this grade is zero.

Suppose the core flags differ. The boundary entries are

$$\mathcal{A}_B(d,0) = \kappa_B C_B(M-d), \qquad \mathcal{A}_B(M, M-d) = \kappa_B G_B(M-d).$$

If both boundaries agreed for all $d \geq \lceil M/2 \rceil$, the full left and right coframes would agree through $\lfloor M/2 \rfloor$. Their convolutions with H would agree there too. The native-support and numerator-zero assertions of the complete coframe atlas [6], with this same coherent normalization, apply separately at both ends. They recover the lower row and column halves and the upper row and column halves. These exhaust all proper positions, including the even middle. The cores would be literal, a contradiction. Hence $\mu = q - 1$, with a nonzero boundary in the required half-window.

If the cores are literal but the restorations differ, choose $z \in I \setminus I'$, reversing the pair if necessary. The two off-diagonal inverse blocks at leading order are $-\rho^{-1} M_B^{-1} U$ and $-\rho^{-1} V M_B^{-1}$. Consequently

$$\mathcal{A}_V^{[q]}(z,0) = -\tau_I \rho^{m-1} (\operatorname{adj} M_B\, U_z)_0, \qquad \mathcal{A}_V^{[q]}(M,z) = -\tau_I \rho^{m-1} (V_z \operatorname{adj} M_B)_M.$$

Both cofactors are nonzero: they replace an actual boundary by the genuine vacant position z, giving feasible weak Jacobi–Trudi boards. The corresponding entries of V' are absent. Thus $\mu = q$, and at least one of the distances $z, M - z$ lies in the required half-window.

It remains to couple the two boundaries. Let $e = s_V = s_{V'}$ and $\kappa = [R_M]e$. The exact one-return inverse identity is

$$\mathcal{A}_V(j,k) = \kappa G_V(k) C_V(M-j) + e(D_V^{-1})_{k,j}, \tag{B.2}$$

where D_V is the complete proper corner and its inverse is embedded by zero at the missing ports. Summing on a genuine positive diagonal yields

$$[z^{M-D}](G_V C_V) = \partial_{R_D}(e/\kappa), \qquad 1 \leq D \leq M. \tag{B.3}$$

Thus the whole class fixes this product through degree $M - 1$. The first differences of the two coframes have relative grade $\eta = \mu - (q-1)$, equal to zero or one in the two cases above. Let a be their least positive index of difference at that grade. The half-window argument gives $a \leq \lfloor M/2 \rfloor$. At coefficient $\epsilon^\eta z^a$, every internal product term in (B.3) cancels because its indices are smaller. The constant coefficients are one, so $\Delta G_a^{[\eta]} + \Delta C_a^{[\eta]} = 0$. Multiplying by the common leading κ proves (B.1), including its nonzero value and all larger-distance vanishing.

The distance M is the common corner, so $d < M$; this also gives $a \le M-2$. The derivative of the whole equal determinant in a positive letter R_D gives $\sum_{j-k=D} \Delta\mathcal{A}^{[\mu]}(j,k) = 0$. Therefore

$$C_{1,D} = \Delta\mathcal{A}^{[\mu]}(D,0), \qquad C_{M-D,D} = -\Delta\mathcal{A}^{[\mu]}(M, M-D).$$

There is no position with $j-k = D$ and $k > M-D$. These equations give all the asserted prefix identities at the same first nonzero grade. □

Appendix C. First responses and the unshifted defect

C.1. **Whole gradients and the ordinary returned values.**

Lemma C.1. *For distinct actual connected coequal coarse boards, at least one of*

$$J_a = \sum_D C_{a,D} R_{D+a-1}, \qquad 2 \le a < M,$$

is nonzero. More precisely, it is not divisible by their common whole polynomial e.

Proof. Neutralize the unique top letter using $R_M = -(e - R_M\kappa)/\kappa$. The complete proper inverse remains defined. Let G, C be the right and left coframes of the first coarse board, and G', C' those independently rebuilt on the second board. Primes here denote neither derivatives nor the earlier fine normalization. Their exact neutral adjugates are respectively $\kappa G_k C_{M-j}$ and $\kappa G'_k C'_{M-j}$. Consequently

$$J_a\big|_{e=0} = -\kappa[z^{M+a-1}](GQ - G'Q'), \qquad Q = [HC]_{<M}, \qquad Q' = [HC']_{<M}.$$

The omitted terms $k < a$ are exactly zero because their numerator index is at least M. The common class fixes GC below M, so it fixes GQ there. Degree M of each product is zero individually: a nonzero G_k requires a true proper column k, at which $Q_{M-k} = 0$. Thus if every displayed J_a vanished on $e = 0$, the whole polynomial equality $GQ = G'Q'$ would follow.

In the independent own atlas, $G = 1 + \sum g_j z^j$ and Q/R_0 are irreducible when nonconstant. To check this directly, regard one independent coefficient g_d as a variable: the polynomial is linear in g_d, with coprime coefficients z^d and a polynomial of constant one. Gauss's lemma proves irreducibility in the full rational coefficient field. The same argument applies in the independently rebuilt opposite atlas. The constant exceptions are the actual one-row and one-column boards; equal rank makes the pair literal in either exception. Unique factorization leaves either

$$G = G',\ Q = Q', \qquad \text{or} \qquad G = Q'/R_0,\ G' = Q/R_0.$$

The first assignment recovers both complete native supports and is literal.

For the second, actual support gives $j \in Y \iff M - j \notin Y'$, $0 < j < M$. Equal rank yields $2(|Y| - 1) = M - 1$, so M is odd. The complete Cramer signs of the two sides at a positive specialization are

$$(-1)^{|Y \cap (0,j]|}, \qquad (-1)^{|Y' \cap (M-j, M)|}.$$

Their equality forces every positive column j to be even. The same applies to Y'; the support complement now forces $Y = Y' = \{0, 2, \dots, M-1\}$. The least positive column is therefore two. The entire C' solve is independent of R_{M-1}, whence

$$\partial_{R_{M-1}}(Q'_{M-1}/R_0) = 1/R_0.$$

On the other hand the actual G solve has derivative zero when row $M-1$ is absent, and derivative $-1/R_0$ when it is present: in the latter case it is the genuine highest unit and its row equation is $R_0 G_{M-1} + R_{M-1} + \sum_{0<j<M-1} R_{M-1-j} G_j = 0$. Both contradict the crossed assignment. Hence some J_a has nonzero neutral value, proving the stronger assertion. □

Lemma C.2. *The ordinary and empty finite response matrices in Proposition 4.3 are constant pure PG matrices in both own fields. In the positive ordinary principal range, after the actual coarse-star reduction and the actual pinches $s > w$, $t < \delta$, put*

$$\tau = [s < \delta], \quad \beta = s - \delta + \tau r, \quad T = L_\beta$$

at a genuine native row β, and $T = 0$ otherwise. Then

$$T_n = -G'_{(n-\tau)r+\delta+t}, \qquad 0 \le n \le M - 1 + \tau. \tag{C.1}$$

The two returned-column border identities are

$$P^{*'}_{\delta+v-t} = P'_{\delta+s}, \qquad P^{*'}_{\delta+v-t-r} = P'_{\delta+s-r}. \tag{C.2}$$

All absent slots are zero. Each opposite normalization is rebuilt in its own field.

Proof. First suppose $0 \le w < v$, so $\delta > 0$. In the normalized field of (4.6), Lemma 4.1 and Eq. (4.5) supply the whole scalar consumed by Lemma E.7. This gives the actual coarse-star reduction before the following response calculation. Here H_R is reconstructed from the ordinary chart (4.8), or from (4.11) in its separate empty case. As in Proposition 4.5, $T_n(H_R)$ means the coefficient after the whole returned substitution, not a frozen T_n or inverse entry. In the complete coarse array of V, set

$$C_{a,D} = \sum_{j-k=D} \mathcal{A}_V(j,k)([k < a] - [j > M - a]).$$

Every surviving term has $2 \le D + a \le M$. With $U(u) = 1/R(u)$, put

$$H_a = \sum_D C_{a,D} U_{D+a-1}, \quad J_a = \sum_D C_{a,D} R_{D+a-1}, \quad \widehat{J}_a = \sum_D C_{a,D} R_{D+a}.$$

Full column-zero and top-row Laplace expansions give $J_1 = J_M = 0$. Subtracting the adjacent prefixes gives replacement of column a by column zero and replacement of row $M - a$ by row M; both are actual duplicate determinants. Hence $\widehat{J}_a = J_{a+1}$.

The returned monomial $R_\ell z^{\ell r+s}$ in the full chart meets the actual source row $(\ell + k - M - \tau)r + \beta$. It exists precisely for $M - k + \tau \le \ell \le M$, and its entire left coefficient is $-T_{M+j-k+\tau-\ell}$. Thus the full no-wave comparison, after summing the two coarse orientations, is

$$\sum_{n=2+\tau}^{M-1+\tau} J_{M-n+1+\tau} T_n(H_R) = 0. \tag{C.3}$$

Every returned series and inverse in this formula is moving. A genuine high wave at cutoff a shifts the complete native variation by u^{M-a}. Differentiation of the full baseline, including the inverse variation against its old crossing, gives

$$\sum_D C_{a,D} F_{D+a,i}(H_R) + \partial_{x_i} S_a = 0, \qquad S_a = \sum_{m=0}^{a} J_{a+\tau+1-m} T_m(H_R). \tag{C.4}$$

This also follows directly by inserting the actual row just displayed; no discarded baseline term is being declared an independent observation.

The finite theorem and coherent degrees give

$$A_{PP} = C_{PP}/R_0, \quad A_{PG} = C_{PG}, \quad A_{GG} = R_0 C_{GG}, \quad b_{1,P} = c_P, \quad b_{1,G} = R_0 c_G$$

with constant matrices and vectors. The complete neutral F_0 is mark-only: its inverse row has zero private part by the triangular native matrix. Replacing R_0 by $R(u)$ in the whole rational compiler is exact through M. Therefore (C.4) takes the form

$$\operatorname{diag}(H_a C_{PP}, J_a C_{GG}) x + \Phi_a + \nabla_x S_a = 0,$$

where Φ_a is outside-independent. Its whole antisymmetric Jacobian gives $H_a C_{PP} = J_a C_{GG} = 0$.

At cutoff two, duplicate-row and duplicate-column cancellation gives

$$J_2 = s_{V \triangleleft \mathrm{CELL}} - s_{\mathrm{CELL} \triangleleft V}.$$

Let μ be the first complete adjugate grade and $\lambda = \mu + 1$. If the zero-free cores differ, their whole functions agree. Rotating the first cell splice makes its cap the same literal cell as the second. Fixed-cap rigidity [6] therefore makes the degree-λ term nonzero. If the cores agree and the restored unit masks differ, the complete unit expansion gives the constant rational weights $\rho_b = \mathbf{1}_{U_V}(b) - \mathbf{1}_{U_V}(M - b)$, not arbitrary rational functions of the raw alphabet. A zero degree-λ cell-cap term, together with the whole equal-core identity, would make both variations vanish.

The child core and the literal cell cap are actual connected zero-free diagrams, so [6, Zero-free linearized fixed-cap rigidity] makes every vacancy weight zero, a contradiction. The complete-array grade is the first cell-splice grade of that same reference, with the coherent factor R_0 retained. Thus $J_2^{[\lambda]} \neq 0$ in both cases. Since $U_n = -R_n/R_0^2 +$ terms of positive-return degree at least two for $n > 0$, the same first whole grade gives $H_2^{[\lambda]} = -R_0^{-2} J_2^{[\lambda]} \neq 0$. This proves pure PG.

The genuine low/high, combined-zero and coherent-unit aliases now integrate F_0 completely:

$$F_0 = A\ell - e_0 + R_0 d_0, \qquad F_1 = Ax + R_0 d_1.$$

Here $\ell_{P_{r+a}} = P'_a$, $\ell_{G_{r+c}} = G'_c$ at actual nonreturned mates only; e_0 is the actual $P_{\delta+s}$ basis vector, and d_0, d_1 are the actual $G_{\delta+t}$, $G_{r+\delta+t}$ basis vectors. Absent inputs give zero vectors. The signs follow from $\Lambda_s = -1$, $\Theta_t = R(u)$. The only potential P intercept at order one is beyond N_J because $s > w$. Consequently the full value equation is

$$\nabla_x S_a + J_{a+1} d_0 + J_a d_1 = 0. \tag{C.5}$$

We record the finite endpoint argument because its final coefficient is needed below. The exact substitution is $T(u; H_R) = T(u; R(u))$. If the preceding neutral outside gradients are constants, its lower substitution corrections have zero outside gradient. For $\tau = 0$, cutoffs $1, 2, \ldots, M-1$ in (C.5) successively give $d_0 = 0$, $\nabla T_1 = -d_1$, and $\nabla T_n = 0$, $2 \le n \le M-2$. The gradient of (C.3) then gives $J_2 \nabla T_{M-1} = 0$. For $\tau = 1$ the corresponding conclusions are $\nabla T_1 = -d_0$, $\nabla T_2 = -d_1$, $\nabla T_n = 0$, $3 \le n \le M-1$, followed by $J_2 \nabla T_M = 0$. These statements also hold at $M = 3$, when their intermediate ranges are empty.

The aliases for the entire row response T remove the remaining nonreturned mark dependence of all preceding neutral values. Its coherent degree is zero and its weight is $(n-\tau)r+\delta$. Thus the degree-zero function of R_0 remaining after integration is constant and vanishes at this nonzero weight. This gives $T_1 = -G'_{r+\delta+t}$ when $\tau = 0$, and $T_1 = -G'_{\delta+t}$, $T_2 = -G'_{r+\delta+t}$ when $\tau = 1$, with all intervening higher values zero. The last value follows from the first degree λ of the *whole* baseline: every correction produced by substituting $R(u)$ for R_0 has at least one extra positive-return degree, so its degree-λ part is respectively $J_2^{[\lambda]} T_{M-1}$ or $J_2^{[\lambda]} T_M$. Both vanish, proving the terminal value without freezing T_0. Finally the aliases into T_1 give $T_0 = -G'_{\delta+t-\tau r}$. The only zero target is $\tau = 1, \delta + t = r$; there $t > 0$, and the genuine combined alias has derivative -1 in G'_0. No derivative of a fixed physical unit is used. This proves (C.1).

For the column statement use bounded-below bilateral native series. The full compiler $HG' = P'$, multiplied by the complete native row relations, gives

$$\sum_i P'_i \Lambda_{i-\delta} + \sum_j G'_j \Theta_{j-\delta} = 0.$$

Every coefficient is a finite sum. Separating the outside inputs, their real mates and the two fixed returns, and putting $\gamma = r + t - \delta \in (0, r)$, gives exactly

$$\Theta_\gamma = -(ux + \ell)^T F - R_0 u^{1-\tau} T.$$

Skewness cancels the two quadratic terms and their paired cross terms. The P intercepts leave $P'_{s+\delta-r} + uP'_{s+\delta}$. Equation (C.1), through its stated terminal index, cancels every remaining G term. Therefore

$$(\Theta_{\gamma,0}, \Theta_{\gamma,1}, \Theta_{\gamma,2}, \ldots, \Theta_{\gamma,M}) = (P'_{s+\delta-r}, P'_{s+\delta}, 0, \ldots, 0).$$

The actual terminal mark solve has returned-row coefficient one and $-L_{0,b} = G^{*'}_{v-b}$. It identifies $\Theta_{0,d} = P^{*'}_{v-d}$ and $\Theta_{\gamma,1} = P^{*'}_{N_J-\gamma}$, proving (C.2). Its normalization is $c_R = A_{v,t}(W)/A_{s,t}(W) = 1/g^*_t$, not $1/g_t$.

We finish the negative and empty cases, where no coarse-star hypothesis is available or needed. When $\delta = -k < 0$, the genuine order-zero response is zero, and the complete first response is

$$F_1 = Ax - e_1 + R_0 d_1, \qquad e_1 = P_{r+s-k}, \quad d_1 = G_{r+t-k}.$$

Keep the unfurled difference prefix. For $D + a \le 1$ it equals K_{1-D}, where

$$K_a = - \sum_{j-q=1-a} \Delta\mathcal{A}(j,q).$$

Here $K_1 = 0$ by the one coherent-unit derivative of the entire equal-rank identity. Put $\beta = s + k \in (0, r)$. The full returned baseline and potential are now

$$\sum_{n=0}^{M-1} B_n T_n(H_R) = 0, \qquad B_n = - \sum_{j \ge n, q} \Delta\mathcal{A}(j,q) R_{M+j-q-n},$$
$$S_a = \sum_{m=0}^{a} B_{M-a+m} T_m(H_R).$$

The endpoint $B_M = 0$ is the whole top-row Laplace equality. Taking the whole curl gives

$$H_a C_{PP} = 0, \qquad J_a C_{GG} = 0, \qquad K_a C_{PG} = 0.$$

By Lemma C.1, $C_{GG} = 0$, separately in both fields. If some $K_a \neq 0$, then $C_{PG} = 0$. A surviving PP entry would give, at its two genuine coface outputs,

$$G^{*'}_{N_J+\delta-i} = -(c/R_0) P'_j, \qquad G^{*'}_{N_J+\delta-j} = (c/R_0) P'_i.$$

Neither output can be absent or mark-only, by independence of the outside inputs. Thus they are two genuine opposite outside G coordinates. The opposite first-response equation would have a nonzero GG entry, contradicting the already proved opposite $C_{GG} = 0$. Hence $PP = 0$. If all $K_a = 0$, every lower-support exception disappears; all prefixes have $2 \le D + a \le M$. The same full cell-cap and unit-variation argument above gives $H_2^{[\lambda]} = -R_0^{-2} J_2^{[\lambda]} \neq 0$, so again $PP = 0$.

At empty width the separately constructed chart (4.11) has $R_0 = \nu$, $F_0 = 0$, and the identical unfurled argument applies. Positive weights remove the stationary values; no alias for the returned P_r is inserted. In both negative and empty cases the zero-order aliases kill every terminal column of A in each own field. Every active pair and its opposite pair therefore lie in the private interval. This proves all claims. □

C.2. **The complete mixed returned baseline.** In the mixed chart let

$$\tau = \mathbf{1}_{s<\delta}, \qquad \beta = s - \delta + \tau r \in [0, r), \qquad T_n = [u^n]\Theta_\beta.$$

These are the neutral native residuals of the square denominator-return solve, not the ordinary numerator-return residuals. As in Proposition 4.5, H_R denotes the entire alphabet after the returned rational substitution, and $T_n(H_R) = [u^n]T(u; H_R) = [u^n]T(u; R(u))$. The clock and every dependent coordinate and inverse entry move before this coefficient is taken. If k is an actual coarse column, its clipped null record omits the returned columns with coefficients $-R_l$ exactly when $k + l > M$. Multiplying by the full moving baseline inverse gives

$$(\dot{T}_{\text{base}})_{j,k} = \sum_{l=M-k+1}^{M} R_l T_{M+j-k-l+\tau}(H_R). \tag{C.6}$$

The column has level $k + l - M - \tau$ and phase β. Its level is nonnegative, including $\tau = 1$; no column below the actual source is introduced. Put

$$B_n = \sum_{j \ge n+1-\tau,\ k} \Delta\mathcal{A}(j,k) R_{M+j-k+\tau-n}, \tag{C.7}$$

with returns outside $0, \dots, M$ zero. The surviving returned indices are positive. Summing (C.6) over the entire coarse array gives

$$\sum_{n=0}^{M-1+\tau} B_n T_n(H_R) = 0, \tag{C.8}$$

$$-\sum_{D} C(a,D) F_{D+a,i}(H_R) + \partial_i \sum_{m=0}^{a-1+\tau} B_{M-a+m} T_m(H_R) = 0. \tag{C.9}$$

The second line is one actual high-wave derivative of the first, including the inverse variation against its old crossing. The returned coefficients are fixed in this outside variation, whereas every native residual is evaluated in the whole moving field. In the reflected branch, grouping the complete prefixes gives $B_n = -J_{M-n+\tau+1}$. The duplicate-column and top-row identities give $J_1 = J_M = 0$, producing exactly the cutoff and terminal baselines used in Proposition 4.5.

Lemma C.3. *At the mixed finite block, every matrix entry is constant in the returned clock. If the matrix is nonzero, the actual coarse mate is V^*.*

Proof. Fine weight and coherent degree make a nonzero entry cR_0^p with $p \in \mathbb{Z}$. Use the complete all-wave current, before extracting a coefficient:

$$\sum_{n} B_n T_n(H_{\mathrm{full}}) - \sum_{a,D,i} C_{a,D} \alpha_{a,i} F_{D+a,i}(H_{\mathrm{full}}) = 0.$$

The genuine second derivative at cutoffs $a = M - d$, $b = 1$, $1 \le d < M$, kills its entire baseline by order. In the sole borderline case $\tau = 1, d = 1$, the surviving second derivative is that of the actual mark-only T_0, hence is zero. The two explicit wave assignments, with opposite skew entries, give

$$R_0^p (L_d + R_d^{\mathrm{bd}}) + \sum_{D>d} [u^{D-d}] R(u)^p L_D = 0,$$

where $L_D = \Delta\mathcal{A}(D, 0)$ and $R_D^{\mathrm{bd}} = \Delta\mathcal{A}(M, M - D)$. These are whole ungraded equations at the original M.

Let $\bar{H} = R/R_0$. Since $L_D = \kappa(C - C')_{M-D}$ and $R_D^{\mathrm{bd}} = \kappa(G - G')_{M-D}$, they give

$$G + [\bar{H}^p C]_{<M} = G' + [\bar{H}^p C']_{<M}.$$

Independently the full outside curl gives $\sum_D C_{a,D} [u^{D+a-1}] R(u)^p = 0$ for every a. At the actual neutral top the complete adjugate is its coframe product. Finite multiplication identifies the cutoff equations with every coefficient of degrees $M, \dots, 2M - 2$ of $G[\bar{H}^p C]_{<M} - G'[\bar{H}^p C']_{<M}$. Below M the common class fixes the product. Therefore

$$G[\bar{H}^p C]_{<M} = G'[\bar{H}^p C']_{<M}.$$

Whole sum and product in the integral domain imply either $G = G'$, which is literal by native support, or $G' = [\bar{H}^p C]_{<M}$, $G = [\bar{H}^p C']_{<M}$.

In the latter assignment use the independent own reflected atlas $\bar{H} = Q/C$, with $Q_0 = C_0 = 1$. It gives

$$G' = [Q^p C^{1-p}]_{<M}, \qquad P' = [Q^{p+1} C^{-p}]_{<M}.$$

If $p \ge 1$, every nonzero coefficient of Q/C contains a coordinate monomial of Q-degree zero or one. Its coefficient in $Q^{p+1}C^{-p}$ remains nonzero: the binomial coefficients are nonzero, and all terms of a fixed coordinate monomial have the same sign. Every positive P'_n, $n < M$, is therefore nonzero. The actual numerator support makes V' a row; equal rank makes V the same row. If $p \le -1$, use the same argument on C/Q: every positive coefficient of $Q^p C^{1-p}$ is nonzero. Thus every proper column is present, squareness makes V' a column, and equal rank makes the pair literal. Both contradict the hypothesis. Only $p = 0$ remains. Its crossed assignment is $G' = C, C' = G$; the actual full supports recover $V' = V^*$. The zero-matrix case makes no such assertion. □

C.3. **The unshifted first-intercept equation.** Use the actual coarse coframes and operator of (4.18), with the coherent clock $\vartheta = R_0 \neq 0$ and independent positive coarse letters. The opposite-end unshifted exclusion of [6] applies only after the physical comparison has earned $V' = V^*$. It is distinct from shifted stationary separation: commuting z through the clipped operator contributes the whole cap polynomial

$$-z^M\{G[HC]_{M-1} - C[HG]_{M-1}\}.$$

This is the $u = H$ specialization of the exact shift-cap identity in [6]. No separate scalar-cap zero is assumed in the physical argument below.

Lemma C.4. *At the nonzero mixed-matrix branch, every active first-response row has zero unwanted intercept. The genuine protected constant and linear returns remain. The assertion holds for both signs of δ at the actual current and mate gates.*

Proof. The complete uncurled equation at cutoff a, before removing any stationary term, is

$$\nabla S_a + J_{a+1}d_1 + \sum_n \mathcal{L}_{a,n}[b_n] = 0.$$

Here the actual moving potential is $S_a = \sum_{m=0}^{a} J_{a+\tau+1-m}T_m(H_R)$, $\tau = [s < \delta]$; the full baseline is $\sum_{m=2+\tau}^{M-1+\tau} J_{M-m+\tau+1}T_m(H_R) = 0$. These formulas come from the omitted returned columns of the full denominator-return chart. Their index test is $\ell \geq M - k + 1$, and grouping it gives $B_n = -J_{M-n+\tau+1}$; this is the column counterpart of the actual row test in (C.3). At negative δ the same direct test has $\tau = 0$ and $\beta = s - \delta \in (0, r)$; its actual order-zero response is zero. Thus no positive-shift baseline is being transported by notation.

Separate genuine coarse weights, holding the neutral fine field fixed. The part containing b_1 at cutoff two and at the first nonzero J_2 grade gives $\nabla T_{\tau+1} = -b_1'(\vartheta)$. The remaining cutoffs and the whole terminal baseline give

$$\mathcal{L}_a[b_1] = \sum_{b=2}^{a} J_b[z^{a-b}]b_1'(H), \quad 1 \leq a < M, \qquad \sum_{b=2}^{M-1} J_b[z^{M-b}]b_1'(H) = 0,$$

where $\mathcal{L}_a[f] = \sum_D C_{a,D}[z^{D+a-1}]f(H)$. No b_n, $n \geq 2$, has been suppressed; it lies in a different homogeneous component. The actual fine grading makes an active nonzero intercept $c\vartheta^p$. Its weight is $r + s + \delta - j = k - t > \rho = s - t$, so p is an integer at least two. The constant and linear possibilities occur only at the returned locations $r + s + \delta$, $r + t + \delta$; their partners would be the fixed low returns t, s, so they are not active outside pairs.

At the genuine neutral top the entire adjugate is $\kappa G_k C_{M-j}$. Direct finite multiplication gives

$$[z^{M+a-1}]\mathcal{T}_{A,B}(f(H)) = \sum_{i \geq a,\ j < a} (A_iB_j - B_iA_j)f(H)_{M+a-1-i-j}.$$

The actual numerator support gives $\operatorname{ord} D \geq M + 1$, $\deg D \leq 2M - 2$. The preceding cutoffs and terminal equation therefore imply

$$\mathcal{T}_{G,C}(H^p) = p[H^{p-1}D]_{<2M}. \tag{C.10}$$

The coefficient $2M-1$ is part of this equality. Off the neutral top the full adjugate is $\kappa G \otimes C + eK$; its independent e-coefficient gives a second finite defect for the full antisymmetric inverse array. It remains in the original equations. We need only the necessary neutral equation (C.10) to obtain a contradiction.

Here G, C are the opposite-end coframes of the actual nonself board V, because the nonzero matrix branch has already earned $V' = V^*$. The active exponent is the integer $p \geq 2$ obtained above, the span is this board's supplied M, and the actual cutoffs include the terminal coefficient. Thus the opposite-end unshifted exclusion [6] applies to (C.10) and excludes the active exponent. The genuine constant and linear returns remain intact. This argument is carried out separately in both rebuilt fields. □

Appendix D. Cofaces, two-copy identities, and native residues

D.1. **Actual cofaces, signs, and two-copy comparisons.**

Lemma D.1. *Order every positive connected block and every isolated unit of the proper corner of an actual connected J from low to high as $E_1, \dots, E_m$. Let a_d, h_d be their least columns and greatest rows. For a genuine proper column k in block ℓ, put $h = h_\ell$. Then*

$$G_k = \sigma_G(k)\frac{s_{S_k}}{\Pi_G(k)}, \quad \Pi_G(k) = \prod_{d\le\ell} s_{E_d}, \quad \sigma_G(k) = (-1)^{|Y_J\cap(0,k]|},$$

where

$$X_{S_k} = X_J \cap (0,h], \qquad Y_{S_k} = (Y_J \cap [0,h]) \setminus \{k\}.$$

For an available numerator a, retain every block with least column at most a, let H be the maximum of a and all their heads, and let $\Pi_P(a)$ be their full product. Then

$$P_a = \sigma_P(a)\frac{s_{A_a}}{\Pi_P(a)}, \quad \sigma_P(a) = (-1)^{|X_J\cap(0,a)|}, \qquad X_{A_a} = (X_J\cap(0,H])\cup\{a\}, \quad Y_{A_a} = Y_J\cap[0,H].$$

These are actual connected strict cell subdiagrams. Every denominator component has smaller span than its connected numerator.

For an active ordinary PG pair the complete physical identities are

$$i + j = N_J + \delta - \epsilon, \quad \epsilon = v - s - t, \qquad G^{*'}_{j+\epsilon} = \sigma G'_j, \quad P^{*'}_{i+\epsilon} = \sigma P'_i, \quad \sigma \in \{1,-1\}. \tag{D.1}$$

They give both whole connected-child equalities and both complete denominator-product equalities. For a mixed active pair the corresponding identities are (4.17)*; their signs are opposite.*

Proof. The full triangular solve for G_k cancels exactly the blocks above the block containing k. Cramer's rule on its complete lower prefix replaces column k by column zero. Moving that new column to its actual position gives $\sigma_G(k)$, including k itself in the count. The child's heights are

$$h_{S_k}(x) = h_J(x) - [x \ge k], \qquad 0 \le x < h.$$

A positive component has height at least two on the affected interval. For a unit $k = h$, the subtraction is outside the occupied interval. Thus the child is connected with singleton extreme diagonals; its actual endpoint steps delete a northwest strip from the original cells. The full bordered determinant for P_a inserts row a and column zero into precisely the blocks whose least column is at most a. Sorting gives the stated sign. It agrees with J below a and deletes a southeast strip on $a, \dots, H-1$. If a is in a gap the strip is empty and this is the actual first-a prefix. A unit cannot end an insertion at its own row because a is an available numerator. This proves connectedness and strict area also for internal insertions and gaps. The least column of every proper denominator component is positive, so its span is strictly smaller than the child's head.

The full one-period mark solve gives

$$g_t = \Gamma/A_{s,0}(W), \qquad g^*_t = \Gamma/A_{v,t}(W), \qquad R_0 = s_W/\Gamma, \quad \Gamma = A_{s,t}(W),$$

and returned-row coefficient $1/g^*_t$. Consequently the ordinary first-response equations are exactly $F_{1,P_i} = -G^{*'}_{N_J+\delta-i}$ and $F_{1,G_j} = P^{*'}_{N_J+\delta-j}$. Pure PG, actual weights, and the absence of an active returned intercept give (D.1) with an initially nonzero rational constant. Clear the two complete proper denominators. The largest connected numerator cannot divide a mark factor of ceiling at most v or a proper component of smaller span. Connected primality identifies the two actual numerators. Primitivity and positive specialization then make the remaining rational constant a sign. This also proves the complete component-product equality; no ordering of coequal components is inferred from a product.

For the mixed normalization, the actual full proper corner has diagonal blocks

$$N_W = \det \mathrm{JT}(X_W \setminus v, Y_W \setminus t), \quad C_{\mathrm{priv}}, \quad D_W = \det \mathrm{JT}(X_W, (Y_W \cup \{s\}) \setminus 0).$$

Alternating maximal minors of the low rectangular mark matrix and the inverse s-row of the replaced mark give

$$g_t = \chi\widehat{A}/D_W, \qquad \eta = \psi N_W/\widehat{A}, \qquad \eta g_t = \xi N_W/D_W, \quad \xi = \chi\psi.$$

These are full determinants, also at $t = 0$. For a constant mixed entry, before removing its mark-only intercept, the equation

$$(\xi N_W/D_W)Q_a - cG_k = \gamma, \qquad \gamma \in \mathbb{Q}(h_1, \dots, h_v),$$

shows $a > v$: a nonzero G_k, $k > v$, has a genuine fresh head letter beyond v, while a low Q_a is mark-only. Its two connected heads must agree, since a larger fresh letter on either side could not cancel. After Lemma C.4, $\gamma = 0$, so the same prime argument gives the whole child equality and

$$N_W\Pi_G(k) = D_W\Pi_Q(a).$$

The actual sign is

$$c = \xi(-1)^{|Y_J\cap(b,N_J)|+|Y_J\cap(0,k]|}, \qquad b = N_J - a.$$

For the two skew rows $j < k$, their opposite signs say exactly that

$$|Y_J \cap (j, k]| + |Y_J \cap (j - \delta, k - \delta]|$$

is odd. This retains both closed port conventions and all units.

In particular the mixed actual children of common head h, with $L = N_J - h$, are

$$\begin{aligned} X_S &= X_J \cap (0, h], & Y_S &= (Y_J \cap [0, h]) \setminus \{k\}, \\ X_T &= (X_J \cap [L, N_J]) - L, & Y_T &= ((Y_J \cap [L, N_J)) \cup \{j - \delta\}) - L. \end{aligned}$$

For every closed $L \le x \le h$, their row difference is zero and their column difference is $[x = j - \delta] + [x = k]$. The first inserted column is actually absent, the deleted column actually present, so they are distinct. Both endpoint cases $j - \delta = L$ and $k = h$ are retained. The opposite equation supplies its own complete pair and product. □

For a wholly private ordinary pair, the full component and rank data can be written explicitly. Here $\Gamma = A_{s,t}(W)$ is the same signed marked cofactor as in the two own normalizations above. Order the private strict blocks from low to high as $E_1, \dots, E_m$, retaining each unit as a rank-one block. Let a_d, h_d be their least columns and heads. Let k contain the input column j, let l contain the target row $i - \delta$, and set

$$\alpha = \max\{d : a_d \le i\}, \qquad \beta = \min\{d : h_d \ge j - \delta\}.$$

After the low and high marked corners cancel against their respective own normalizers, the four denominators of the two paired identities are

$$\Gamma \prod_{d\le k} s_{E_d}, \quad \Gamma \prod_{d\ge l} s_{E_d}, \quad \Gamma \prod_{d\le \alpha} s_{E_d}, \quad \Gamma \prod_{d\ge \beta} s_{E_d}.$$

The two separate whole identities therefore retain

$$\prod_{d\le k} s_{E_d} = \prod_{d\ge l} s_{E_d}, \qquad \prod_{d\le \alpha} s_{E_d} = \prod_{d\ge \beta} s_{E_d}. \tag{D.2}$$

Equal whole connected numerators also fix their row counts. Writing $q_d = \operatorname{rank} E_d$, this gives

$$\sum_{d\le k} q_d = \sum_{d\ge l} q_d, \qquad \sum_{d\le \alpha} q_d = \sum_{d\ge \beta} q_d.$$

Positive connected factors match with multiplicity; the rank equations also retain the number of units, invisible as factors equal to one. No ordering of coequal components is inferred from these product identities. The denominator-child heads satisfy $h_k = N_J - a_l$, and the numerator-child heads satisfy

$$\max(i, h_\alpha) = \max(i + \epsilon, N_J - a_\beta).$$

These formulas include genuine internal and gap insertions.

Lemma D.2. *The ordinary P-child comparison transports to an actual G-child comparison on the antitransposed whole diagrams, preserving its primitive sign and strict area. Writing $J^{\dagger} = \mathrm{AT}(J)$, with $\mathcal{I}(f)(z) = \iota(f(-z))$,*

$$G^{\dagger} = \mathcal{I}(P), \quad P^{\dagger} = \mathcal{I}(G), \quad C^{\dagger} = \mathcal{I}(Q), \quad Q^{\dagger} = \mathcal{I}(C).$$

Its returned normalization and clock are

$$g^{\dagger} = (-1)^{s}\iota(g_t R_0), \qquad R_0^{\dagger} = (-1)^{t-s}\iota(1/R_0),$$

and the opposite field uses $s^* = v - t$ and its own g_t^*.

Proof. Actual antitranspose has complete flags $X^{\dagger} = (0, N_J] \setminus Y_J$, $Y^{\dagger} = [0, N_J) \setminus X_J$. Since $\mathcal{I}(H) = 1/H$, the full identity $P = HG$ below N_J gives $H\mathcal{I}(P) = \mathcal{I}(G)$. Its support and annihilator are exactly the antitransposed flags, with constant one; uniqueness of the actual proper-corner solve proves the first two equations. The opposite solve proves the other two. The returned coefficient is the actual coefficient at s; reading it and the coefficient at t gives the displayed normalization. Thus

$$G_a^{\dagger\prime} = (-1)^{a-s}\iota(P_a'/R_0), \qquad P_a^{\dagger\prime} = (-1)^{a-s}\iota(G_a'/R_0).$$

Since $s^* - s = \epsilon$, substitution in (D.1) cancels every parity and clock factor and gives the same sign on its antitransposed pair. The inserted-row child's actual complement flags are exactly the deleted-column child on $J^{\dagger}$; if its head is an old positive head it is not a proper column, and if it is a gap insertion the complement deletes that new endpoint. These cover all cases. Antitranspose preserves connectedness, strict occupied area, and the whole Schur equality under ι. Its target finer is ordinary (b); no (a) compiler is reused in that field. □

Lemma D.3. *Let an active ordinary pair in* (D.1) *have all four indices strictly between v and r, with its actual targets $i - \delta, j - \delta$ private. On the actual $Z = \mathrm{Row}_2 \circ_{\mu} J$, of span $2r + v$, its full global records satisfy*

$$C_{Z,r+j+\epsilon}/g_t^* = \sigma G_{Z,r+j}/g_t, \qquad Q_{Z,r+i+\epsilon}/g_t^* = \sigma P_{Z,r+i}/g_t.$$

They give actual equal connected strict children and both complete component products. The statement includes negative δ for an arbitrary nonliteral coequal coarse pair. At an empty finer its two identities have $v = s = t = \epsilon = 0$, the genuine empty chart, and both normalizers one.

Proof. Put $B_+ = X_W \setminus s$, $C_+ = Y_W \setminus t$, $B_0 = B_+ \sqcup B_-$, $C_0 = C_+ \sqcup C_-$. The complete actual flags are

$$B_2 = B_0 \sqcup (r + B_0) \sqcup (2r + B_+), \quad X_Z = B_2 \sqcup \{2r + s\},$$
$$C_2 = C_0 \sqcup (r + C_0) \sqcup (2r + C_+), \quad Y_Z = C_2 \sqcup \{t\}.$$

The fixed matrix in increasing-period order is

$$D_2 = \begin{pmatrix} D_0 & 0 & 0 \\ D_1 & D_0 & 0 \\ E_2 & E_1 & K_0 \end{pmatrix}, \qquad K_0 = \mathrm{JT}(B_+, C_+).$$

All lower crossing blocks remain. Thus $\det D_2 = (\det D_0)^2 \det K_0 \neq 0$. Its inverse row at actual column zero is confined to initial mark rows at most v, and its inverse column at the actual top is confined to final mark columns at least $2r$. A vector is absent if its extreme is returned. In particular its $(0, 2r + v)$ entry is zero.

The dense full chart is

$$P_{\mathrm{full}} = P' + R_1 z^{r+s} + R_2 z^{2r+s} + \sum_i \alpha_{P_i} z^{r+i}, \qquad G_{\mathrm{full}} = G' + \sum_j \alpha_{G_j} z^{r+j}.$$

The actual D_2 solve recovers every source coefficient and is its rational inverse; remove $R_1 z^{r+s}$ before reading the neutral P' if $s < v$. In native series this is the simultaneous substitution $R_0 \mapsto R_0 + R_1 u + R_2 u^2$, $x \mapsto x + u\alpha$. The already supplied coefficient two therefore gives

$$F_{2,P_i} = -\sigma\alpha_{G_j}, \qquad F_{2,G_j} = \sigma\alpha_{P_i}.$$

At positive δ the possible $R_2d_0+R_1d_1$ terms are absent at these private active rows. At negative δ, $F_0=0$ and only R_1d_1 is possible; its putative partner would be the fixed P_s. This excludes it without a returned-row value assumption.

In native-plus-returned order write

$$M_Z=\begin{pmatrix} D_2 & b\\ c & d\end{pmatrix},\quad v_0=(-D_2^{-1}b,1)^T,\quad w_0=(-cD_2^{-1},1),\quad R_2=d-cD_2^{-1}b.$$

If K embeds D_2^{-1} by zero on the returned ports, the *whole* adjugate is $\det D_2(v_0w_0+R_2K)$, with one common actual sorting sign. If p,j_t are the positions of s,t in the decreasing mark lists, that sign is $(-1)^{p+j_t}$. Thus, writing $\Gamma=A_{s,t}(W)$, $\Delta=C_{\mathrm{priv}}$,

$$s_Z=\Gamma^3\Delta^2R_2,\qquad \kappa_Z=\Gamma A_{s,0}(W)A_{v,t}(W)\Delta^2.$$

The true low and high mark solves give $(v_0)_0=1/g_t$, $(w_0)_{2r+v}=1/g_t^*$. Consequently the complete global coframes are

$$G_Z/g_t=v_0+R_2K_{.,2r+v}/(w_0)_{2r+v},\qquad C_Z/g_t^*=w_0+R_2K_{0,.}/(v_0)_0.$$

Both compensators are retained. The proved inverse supports place each, and its H-convolution, in degrees at least $2r$. Every relevant amplified label lies below $2r$ and above $r+v$. Hence

$$G_{Z,r+j}/g_t=\alpha_{G_j},\quad P_{Z,r+i}/g_t=\alpha_{P_i},\quad C_{Z,2r+v-(i-\delta)}/g_t^*=-F_{2,P_i},\quad Q_{Z,2r+v-(j-\delta)}/g_t^*=F_{2,G_j}.$$

The last formula sums the full omitted-column residual; its potential last private omission has zero constant inverse component by the true mark solve. The pair weight identifies the two target labels with $r+j+\epsilon$ and $r+i+\epsilon$, proving the equations.

For example, if the original G-child head is $h<r$, put $L=N_J-h$. The enlarged G-children have flags

$$\begin{aligned} X_S&=X_Z\cap(0,r+h], & Y_S&=(Y_Z\cap[0,r+h])\setminus\{r+j\},\\ X_T&=((X_Z\cap[L,2r+v])\setminus\{i-\delta\})-L, & Y_T&=(Y_Z\cap[L,2r+v))-L.\end{aligned}$$

The inserted-row pair is given by the second displayed equation with the full head rule of Lemma D.1. The same connected-prime argument proves both whole equalities and retains both full denominator products. The children are strict cell subdiagrams of Z; the exact composition area law and the nonline original coarse outer give $|Z|<|E|<n$.

At empty width use $X_Z=I\cup(r+I)\cup\{2r\}$, $Y_Z=C_0\cup(r+C_0)\cup\{0\}$. Both extremes are returned, so K has no corresponding extreme row or column and the two compensators are identically zero. Its corner is $(\det D_0)^2$, its whole determinant is that corner times R_2, and the same full chart has $P_{\mathrm{full}}=P+R_1z^r+R_2z^{2r}+$ its real waves. The active relation is $i+j=r-w$; its returned intercept cannot be active. The preceding calculation proves the two equations and strict-area statement with the actual empty conventions. □

Lemma D.4. *At an active mixed pair satisfying* (4.22)*, put* $a_j=N_J+\delta-j=k+\epsilon$, $b_j=j-\delta$. *The actual* $Z=\mathrm{Row}_2\circ_\mu J$*, of span* $S_2=2r+v$*, satisfies*

$$\eta Q_{Z,r+a_j}=cG_{Z,r+k}/g_t-b_{2,j}(\vartheta),\qquad \eta Q_{Z,r+a_k}=-cG_{Z,r+j}/g_t-b_{2,k}(\vartheta).$$

If the supplied response in a row is zero, that row gives equal whole functions of two actual connected strict children, together with the full component-product equality. Both original own fields are retained.

Proof. The actual flags of Z, including both genuine returned columns, are

$$\begin{aligned} X_Z&=\bigsqcup_{\ell=0}^{2}(\ell r+X_W)\sqcup\bigsqcup_{\ell=0}^{1}(\ell r+B_-),\\ Y_Z&=\bigsqcup_{\ell=0}^{2}(\ell r+(Y_W\setminus t))\sqcup\bigsqcup_{\ell=0}^{1}(\ell r+C_-)\sqcup\{t,s,r+s\}.\end{aligned}$$

Use the full all-s square baseline B_s, whose determinant is $\widehat{A}^3 C_{\mathrm{priv}}^2$, and whose last returned column is $\alpha = 2r + s$. The actual dense chart is

$$P_{\mathrm{full}} = P' + \sum_i \alpha_{P_i} z^{r+i}, \qquad G_{\mathrm{full}} = G' - R_1 z^{r+s} - R_2 z^{2r+s} + \sum_\ell \alpha_{G_\ell} z^{r+\ell}.$$

Its full inverse recovers every raw source coefficient; add back its returned denominator terms before reading the neutral fine chart. Thus its native substitution is $\vartheta \mapsto \vartheta + R_1 u + R_2 u^2$, $x \mapsto x + u\alpha$. At an active private row, the order-zero response and first intercept are zero, and the matrix is constant. Therefore

$$F_{2,G_j} = -c\alpha_{G_k} + b_{2,j}(\vartheta), \qquad F_{2,G_k} = c\alpha_{G_j} + b_{2,k}(\vartheta).$$

Let $r_0 = B_s^{-1} b_t$, $\ell = e_\alpha^T B_s^{-1}$, $f = (r_0)_\alpha = R_2$, $g_\alpha = 1$, $g_\beta = -(r_0)_\beta$ at other actual columns, and $K = (I - e_\alpha e_\alpha^T) B_s^{-1}$. The full one-column replacement identity is

$$\operatorname{adj} M_Z = \det B_s (g\ell + fK).$$

Its true extremes satisfy $g_0 = 1/g_t$, $\ell_{S_2} = \eta$, $K_{0,S_2} = 0$. The final inverse column is supported only on final mark columns at least $2r$, and the initial inverse row only on initial mark rows at most v. Since $g(z) = G_{\mathrm{full}} + f z^{2r+s}$, the complete global records are

$$G_Z/g_t = G_{\mathrm{full}} + f\{z^{2r+s} + K_R/\eta\}, \qquad \eta C_Z = \ell(z) + f K_L/g_0.$$

They retain both compensators for unrestricted f. Their support and the support of their numerator convolutions begin at $2r$. On the genuine private indices this gives

$$G_{Z,r+k}/g_t = \alpha_{G_k}, \qquad -\eta Q_{Z,r+a_j} = F_{2,G_j},$$

and the swapped equation follows identically.

To see the exact cleared identity, let $\Pi_{G,2}$, $\Pi_{Q,2}$ be the complete products of the relevant actual cofaces and use the mark factors of Lemma D.1. It is

$$\xi \sigma_{Q,2} N_W s_{T^*} \Pi_{G,2} - c\sigma_{G,2} D_W s_S \Pi_{Q,2} = -b_{2,j}(\vartheta) g_t D_W \Pi_{G,2} \Pi_{Q,2}.$$

Every enlarged side appends the same full $C_{\mathrm{priv}} \widehat{A}$ to its original component product. The original earned equality $N_W \Pi_G = D_W \Pi_Q$ therefore gives $N_W \Pi_{G,2} = D_W \Pi_{Q,2}$, with multiplicity. When $b_{2,j} = 0$, connected primality and the actual sorting signs give $s_S = s_T$.

For head $h < r$, $L = N_J - h$, the full flags are

$$X_S = X_Z \cap (0, r+h], \qquad Y_S = (Y_Z \cap [0, r+h]) \setminus \{r+k\},$$
$$X_T = (X_Z \cap [L, S_2]) - L, \quad Y_T = ((Y_Z \cap [L, S_2)) \cup \{j - \delta\}) - L.$$

They are the actual cofaces of Lemma D.1. Their closed row defect is zero and their column defect is $[x = j - \delta] + [x = r + k]$. The original end comparison on $[0, v]$ has the two actual column defects $-[x = t] - [x = s]$; both full words and all four cap values are inherited from the displayed lists. The children have smaller occupied area than Z, and the full composition area law gives $|Z| < |E| < n$. Thus lower full recognition is applied only after these actual whole equalities and strict area bounds have been earned. □

D.2. **The complete native words at a zero matrix.**

Lemma D.5. *At every mixed finite comparison with $0 < w < r$, $w \neq v$, the full native row set $\mathcal{B} = X_W \sqcup B_- \subset (0, r)$ is invariant under $b \mapsto b + \delta \pmod r$. Consequently, with $g = \gcd(r, \delta)$, it is a union of complete g-classes, $2 \leq g < r$, and $g \nmid v$. For a nonzero matrix and $\delta < 0$, every active pair satisfies* (4.22)*. For $\delta > 0$, either a genuine connected head exceeds r, or both paired heads are below r and* (4.22) *holds.*

Proof. The full numerator response is zero. Its actual one-period reading is $\eta C_{N_J + \delta - i} = 0$ at each available outside numerator i. The complete support says that this coefficient is nonzero exactly when $i - \delta$ is an actual row.

For $\delta > 0$, fix $b \in \mathcal{B}$. If $b + \delta > v$, the outside input $i = b + \delta$, whenever absent, would have the nonzero response of row b; therefore it is present. An attempted multiple of r is excluded because row phase zero is absent. If $b + \delta \leq v$, use the actual high mark row $r + b$ and input

$r+b+\delta$. At equality with N_J, the target is the genuine top phase v, already present. All other cases are the same available-input contradiction. These choices keep both actual copies of X_W, which is the mixed row word.

For $\delta=-k<0$, $r-k>v$. If $0<b<w$, the actual upper row mate sends b to $(M-1)r+(r+b-k)$. Its complete final-period flag projects to the required phase $b-k \pmod r$. This includes the private $b\in(v,w)$, for which a fictitious row $r+b$ could not be used. At $b=w$ the target is the actual phase v. At $b>w$ an absent outside $i=b-k\in(v,r)$ would have nonzero response from row b, a contradiction. The impossible zero phase is handled by its actual absence. Thus closure holds for either sign. A permutation of a finite cyclic set turns inclusion into equality. Phase zero is absent and v present, giving the stated gcd facts.

At negative δ, the constant native inverse is supported on mark rows at most v; every shifted outside target lies above v or has a positive carry. Hence $F_0=0$. The genuine zero-order aliases kill the entire matrix column at each high mark input $r+c$, including the combined $c=0$ case because the P block is zero. Thus active j,k lie in (v,r). The coface head argument in Lemma D.1 gives $N_J+\delta-j>v$, hence $j-\delta<r$; since $\delta<0$, also $j-\delta>v$. The swapped row gives the other two inequalities. At positive δ, the two genuine heads satisfy

$$h_k\ge\max(k,N_J+\delta-j),\qquad h_j\ge\max(j,N_J+\delta-k).$$

No head equals r, since that row is absent. If neither exceeds r, both are smaller, and these four inequalities give exactly (4.22). □

Lemma D.6. *At the actual mixed finite comparison, retain the full response identities of Propositions 4.3 and 4.4, the native row conclusion of Lemma D.5, and the upper-mate implications* (4.3) *in both own fields. Suppose the entire GG first-response matrix is zero. An arbitrary mark-only intercept is allowed. Set*

$$g=\gcd(r,\delta),\quad m=r/g,\quad v=dg+a,\quad 0<a<g,\qquad \mathcal{C}=(Y_W\setminus t)\sqcup C_-\sqcup\{s\}.$$

The row word is a union of complete g-classes R, and every column class foreign to $t,s \pmod g$ is full or empty. The two port residues are distinct, their occupancies sum to m, and the full foreign column residues T satisfy $|T|=|R|-1$. These are the complete actual signature inputs to the mixed fold. If $d=0$, the original mark remains literally W, and the actual proposed root flags supplied to that constructor are

$$X_K=R\cup(g+(R\cap[0,a])),\qquad Y_K=T\cup(g+(T\cap[0,a]))\cup\{t,s\}.$$

No divisor chart or upper divisibility is assumed in this lemma.

Proof. The entire left coframe and its numerator below N_J are independent of the unique top raw letter. The neutral source solve only solves that fresh letter; the prefix below N_J remains the full free raw field. Thus the zero-matrix equation $-\eta Q_{N_J+\delta-i}=b_i(\vartheta)$ is a whole prefix identity. When $v<A<N_J$, a nonzero Q_A has an actual connected numerator with head at least $A>v$, and every denominator factor has smaller ceiling. It cannot be mark-only. Hence

$$i\in Y_J,\ i>v,\quad v<N_J+\delta-i<N_J\quad\Longrightarrow\quad i-\delta\in Y_J.$$

This uses the full coface, not a selected inverse entry.

For $\delta>0$, take a foreign $c\in\mathcal{C}$. If $v<c<r$, use actual input $i=c$; the head reader gives $c-\delta\in Y_J$, in (w,r), and foreignness excludes both returned exceptions. If $c\in Y_W\setminus t$, $c<\delta$, use the actual high input $i=r+c$, including $c=0$ when $t>0$. Its output is in (v,N_J), so the target $r+c-\delta$ belongs to the native column set. For $c>\delta$, the genuine final source column $Mr+c$ lies strictly above Q; the open upper mate gives $c-\delta$ in the first band $Y_W\cup\{s\}$, and foreignness excludes s,t. At $c=\delta$, the desired target is zero; foreignness means $t\neq0$, so zero is already the genuine native column. No mate at equality or derivative of a fixed unit has been introduced.

For $\delta=-k<0$, split the private columns into $c<r-k$, $c=r-k$, and $c>r-k$. In the first case the head reader gives $c+k\in C_-$. At equality the required zero is a genuine column because its class is foreign to t. In the third case the actual last private column $(M-1)r+c$ lies above $Q=Mr-k$; its mate is $c+k-r$ in the full first band, which may be mark or private.

Foreignness removes only the two genuine exceptions. For a mark column $c \in Y_W \setminus t$, use the actual final column $Mr + c$, including $c = 0$. Its mate is $c + k \in (0, w)$, again in the full first band. These cases exhaust all foreign columns. Closure under $-\delta$ makes each finite foreign g-cycle full or empty.

The native ranks are equal. If the two port residues coincided, that exceptional class would contain s and omit t, with occupancy strictly between zero and m, contradicting the rank equation $m|R| = m|T| +$ exceptional occupancy. Thus they are distinct. Their combined occupancy is positive and strictly below $2m$; the same rank equation forces it to equal m and $|T| = |R| - 1$. If $d = 0$, $a = v < g$, so the actual mark flags are $X_W = R \cap [0, a]$ and $Y_W = (T \cap [0, a]) \cup \{t\}$. The displayed root flags therefore use this same literal mark. Their actual construction and legality, including the short-width case, are supplied in [9, Mixed residue fold]. □

Lemma D.7. *At the actual comparison* (4.1)*, retain the full response identities of Propositions 4.3 and 4.4 and the upper-mate implications* (4.3) *in both own fields. If the whole ordinary first-response matrix is zero, its complete native sets $\mathcal{B} = X_W \sqcup B_-$, $\mathcal{C} = Y_W \sqcup C_-$ satisfy*

$$(\mathcal{B} \setminus s) + \delta \subseteq \mathcal{B} \pmod r, \qquad (\mathcal{C} \setminus t) - \delta \subseteq \mathcal{C} \pmod r.$$

At an empty finer the corresponding conditions are

$$(I - w) \pmod r \subseteq I \cup \{0\}, \qquad (C_0 + w) \pmod r \subseteq C_0 \cup \{0\}.$$

All are actual two-sided conditions, with the indicated returns retained.

Proof. For $\delta > 0$, a nonreturned row b with $b + \delta > v$ uses the available-input test $i = b + \delta$, just as in Lemma D.5. If $b + \delta \le v$, use its actual high mark row and input $r + b + \delta$, treating the true top separately. The absent initial returned row s is precisely the exception. For a private column c, use $j = c$; for a nonreturned mark column, use the actual high $j = r + c$. The zero response forces $j - \delta$ to be an actual column except at the genuine $j = r + \delta + t$ intercept, whose required residue is already t. It does not force the false high column $r + t$.

For $\delta = -k$, a row $b < w$ uses its actual upper mate, $b = w$ has the true target v, and $b > w$ uses the outside input $b - k$. This input is below the protected intercept $r + s - k$. For a mark column use the actual final $Mr + c$. For a private column, use the upper mate if $c + k > r$, the genuine column zero if $c + k = r$, and the available-input coface test if $c + k < r$. In the last case its index is strictly below the protected intercept $r + t - k$. These are the complete negative-shift choices; nonexistent high private rows are never used.

At empty width put $k = r - w$. For a native row $b < w$ the actual mate gives $b + k$; at $b = w$ the target is the real returned zero; at $b > w$ an absent numerator P_{b-w} would have nonzero output C_{r-b}, below the sole intercept at k, a contradiction. For a native column $c > k$ the last-period mate gives $c - k$; at equality the target is zero; at $c < k$ its zero G_c response forces $c + w$ to be an actual column by the exact opposite numerator annihilator. This proves every displayed incidence. □

Appendix E. Endpoint scalars and ordinary reflection

E.1. **The endpoint supply and the mixed sign restriction.**

Lemma E.1. *For an actual positive upper of width $0 < w < v$ on an ordinary* (a) *or mixed* (d) *finer source, the complete source windows give*

$$s > w, \qquad t < \delta, \qquad h_W(w) = h_W(\delta - 1) = 1.$$

The same statements hold after the actual whole transports needed for the other orientations.

Proof. The upper's true private diagonals $w, Q - 1$ are singletons. For a mixed upper this can also be checked directly: its low word has k rows and $k + 1$ columns, its high word k rows and $k - 1$ columns, and the complete private lists have equal rank. Thus each of these heights is one, with endpoint incidences included. For an ordinary (a) finer the first and last windows give

$$h_E(w) = h_W(w) + [s \le w], \qquad h_E(Q - 1) = h_W(\delta - 1) + [t \ge \delta].$$

For a mixed (d) finer the complete supplemented words give the same two formulas; their added low tail begins at s and their high tail ends at t. Since W is connected, both mark heights are positive. The equations force precisely the four assertions. Actual antitranspose preserves content heights, so applying it to a mixed (c) upper proves the same source heights before rereading the original finer windows. Whole antitranspose of an ordinary (b) finer transports both marks and both fields and reduces it to this calculation. □

Lemma E.2. *Suppose the finer is ordinary (a), with $s = v, t = 0$, and the actual upper is ordinary of width $0 \le w < v$, with its genuine gauge and whole current. Put $N = N_J = r + v$, $\delta = v - w$. For $w > 0$ let*

$$B = W|_{[0,\delta-1]}, \qquad D = W|_{[w,v-1]}$$

in their own southwest normalizations; at $w = 0$ set $B = D = W$. Then B, D are actual connected diagrams of span δ. The horizontal splices

$$S = D \triangleleft J, \qquad T = J \triangleleft B$$

are actual connected strict-lower-area diagrams and satisfy

$$s_S = s_T, \qquad \kappa_B = \kappa_D, \qquad |S|, |T| < |E| < n.$$

Their complete shared word is

$$\mathrm{word}_S(x) = \mathrm{word}_T(x + \delta) = \mathrm{word}_J(x), \qquad 0 < x < N.$$

Their first δ cell windows agree literally as B, their last δ windows as D, and their true splice seams are $S(N) = T(\delta) = (0, 0)$. No equality of the two opposite seam values is asserted.

Proof. We derive the needed whole scalar before naming a cap. Write $K = N + \delta$, $s^* = v - t$, $t^* = v - s$ temporarily at general ordinary contacts. The complete first responses are

$$F_{1,P_i} = -G^{*'}_{K-i}, \qquad F_{1,G_j} = P^{*'}_{K-j}.$$

Multiply by the actual outside coordinates and sum. Skewness cancels the matrix and gives

$$\sum_{i>v} P'_i G^{*'}_{K-i} - \sum_{j>v} G'_j P^{*'}_{K-j} = P'_{K-t^*} - R_0 G'_{K-s^*}.$$

Every omitted low P term except $P'_s = R_0$ is zero. Indeed a surviving nonreturned low index i would have $\delta < i \le v$; if it were a genuine missing mark row, the opposite high column would require the actual low row $i - \delta$, and its open upper mate would force i present, a contradiction. The endpoint $i = v$ is an actual row or the returned row. Thus the omitted sum is $R_0 G^{*'}_{K-s}$. For a low nonreturned G'_j, the actual final source column $Mr + j$ and its upper mate force $j - \delta$ to be a low source column; the opposite high numerator is therefore zero. Only the fixed $G'_t = 1$ remains, contributing $P^{*'}_{K-t}$.

The full returned values in (C.1) give $G^{*'}_{K-s} = G'_{K-s^*}$. If $s > \delta$, this is the actual low-row solve at $s - \delta$; if that row is absent both sides are zero. At $s = \delta$, the target is the genuine top and is absent. At $s < \delta$, a putative nonzero high right side would have a mark-only low mate; the returned equation would identify it with an independent opposite outside coframe coordinate, which is impossible. This includes the carry case without an extra observation. Consequently

$$[z^K]\{P'(G^{*'} - z^{t^*}) - (G' - z^t)P^{*'}\} = 0.$$

Using only $P' = [HG']_{<N}$ and $P^{*'} = [HG^{*'}]_{<N}$, expand the exact discarded tails. The identity becomes

$$[z^K]H(G'A^* - G^{*'}A) = 0, \quad A = G'_{\le\delta} + [t > \delta]z^t, \quad A^* = G^{*'}_{\le\delta} + [t^* > \delta]z^{t^*}.$$

A returned monomial above the truncation is visibly retained.

Now $s = v, t = 0$, so $t^* = 0$, both normalizers are one, and the small coframes are the actual G_W, C_W. The returned $n = 0$ value gives $C_{W,\delta} = G_{W,\delta}$; if the row is absent both coefficients

are absent, and at $w = 0$ their degree v slots are absent. The genuine source top coefficients $[HG_J]_N = [HC_J]_N = s_J/\kappa_J$ agree. Thus the degree-δ tail cancels as a whole, and the scalar is

$$[z^{N+\delta}]H\{G_J(C_W)_{<\delta} - C_J(G_W)_{<\delta}\} = 0. \tag{E.1}$$

For $w > 0$, Lemma E.1 gives the two actual singleton cuts $h_W(w) = h_W(\delta - 1) = 1$. The two restrictions B, D therefore inherit positive interval diagonals, legal endpoint steps, and singleton extremes, so are connected. They may overlap in W; no separated-mark hypothesis on W is needed. Their exact flags are

$$X_B = \{\delta\} \cup (X_W \cap (0, \delta)), \quad Y_B = Y_W \cap [0, \delta),$$
$$X_D = (X_W \cap (w, v]) - w, \quad Y_D = \{0\} \cup ((Y_W \cap (w, v)) - w).$$

The low proper equations see no column at least δ, because every such matrix entry has genuinely negative index. Uniqueness of the complete proper-corner solve gives $G_B = (G_W)_{<\delta}$. Applying this argument to the actual rotated suffix gives $C_D = (C_W)_{<\delta}$. At $w = 0$ these statements are immediate.

For any two connected diagrams A, Z of spans a, c, the actual horizontal splice has

$$X_{A \triangleleft Z} = (c + X_A) \cup (X_Z \setminus c), \quad Y_{A \triangleleft Z} = (c + (Y_A \setminus 0)) \cup Y_Z.$$

Its full block determinant, expanded at the merged horizontal seam, is

$$\frac{s_{A \triangleleft Z}}{\kappa_A \kappa_Z} = [z^{a+c}] H C_A G_Z.$$

One way to verify the last equality is to expand the two proper corners first: their complete null vectors are C_A, G_Z; the remaining crossing entry at exponent $a + c$ is their full convolution with H. The two actual extreme coefficients fix the signs, so all units and crossings remain in the formula. Applying it to (E.1) gives $\kappa_B s_S = \kappa_D s_T$. Neither connected whole prime, of span $N + \delta$, divides either small proper corner, whose raw ceiling is below δ. Primality and primitive positive normalization give $s_S = s_T$, followed by $\kappa_B = \kappa_D$. Their equal areas and ranks imply $|B| = |D|$ and equal cap ranks.

Finally the full endpoint area law, with $A = |W|$ and $b = \sum B_- - \sum C_- \geq 0$, gives

$$|J| = r + 2A - v + b, \qquad |E| \geq 3|J| - 2A + v, \qquad |E| - (|J| + A) \geq 2r + A - v + 2b > 0.$$

It uses the actual nonliteral coarse bounds $M \geq 3$, $2 \leq \operatorname{rank} V \leq M - 1$, and $|V| \geq M$. Since the cap areas are at most A, both splices lie strictly below the original occupied-area horizon. Their complete flag formulas prove the shared word, the literal end windows and the two stated 00 seams directly. The other seams are genuine values of J and are not altered. □

Lemma E.3. *At the actual finite comparison* (4.1)*, let the finer be ordinary* (*a*)*, with overlap W of width v and contacts s, t. Suppose the actual mixed upper has width $0 < w < v$, and put $\delta = v - w$. If $s \geq \delta$ and $t \leq w$, its contacts $0 \leq t_\Omega < s_\Omega \leq w$ satisfy*

$$t_\Omega < t < s_\Omega \quad \textit{in type } (d), \qquad t_\Omega < s - \delta < s_\Omega \quad \textit{in type } (c).$$

In particular $s = v, t = 0$ is impossible for either actual mixed upper.

Proof. The actual pinches are supplied by Lemma E.1. Put $\Gamma = A_{s,t}(W)$, $\beta = s - \delta$, and $y = \delta + t$. The $n = 0$ case of (C.1), with $\tau = 0$, gives $L_\beta(0) = -G'_y$. The complete mark solves give

$$L_\beta(0) = -A_{\beta,t}(W)/\Gamma, \qquad G'_y = A_{s,y}(W)/\Gamma,$$

with absent ports read as zero. Hence

$$A_{s-\delta,t}(W) = A_{s,\delta+t}(W).$$

Since $s > w$, the missing finer row s lies outside the initial window $[0, w]$. Since $t < \delta$, the missing final column $Mr + t$ lies outside the terminal window $[Mr + \delta, Mr + v]$. By (4.6), the source flags on these windows read $X_W(x), Y_W(x)$ and $X_W(x + \delta), Y_W(x + \delta)$, respectively. The

actual upper coding table, including the supplemented columns, therefore gives for every closed $0 \le x \le w$

$$\begin{aligned}(d):&\quad X_W(x+\delta) = X_W(x),\ Y_W(x+\delta) = Y_W(x) - [x = t_\Omega] - [x = s_\Omega],\\ (c):&\quad X_W(x+\delta) = X_W(x) + [x = t_\Omega] + [x = s_\Omega],\ Y_W(x+\delta) = Y_W(x).\end{aligned}$$

The closed caps $X_W(0) = 0, X_W(v) = 1, Y_W(0) = 1, Y_W(v) = 0$ are included. These are actual finite recurrences only on this window.

Every existing single row/column deletion from a connected strict Jacobi–Trudi matrix is a nonzero weak skew determinant: its sorted diagonal differences are one of $x_i - y_i > 0$, $x_i - y_{i+1} > 0$, $x_{i+1} - y_i \ge 0$. Actual units factor triangularly, leaving a nonzero Schur product. Thus its only sign is the genuine cofactor parity. In type (d), the row recurrence makes β an actual row, so $\beta > 0$ and the left cofactor is nonzero. Equality forces y to be an actual column; hence t is neither upper contact. In type (c), the column recurrence makes y an actual column, so $t < w$; equality forces β a row and different from both contacts. This first disposes of absent ports and closed endpoints before taking signs.

Write $R = |X_W \cap [0, \delta)|$, $C = |Y_W \cap [0, \delta)|$. The actual singleton at $\delta - 1$ gives $C - R = 1$. The sign ratio of the two cofactors is

$$(-1)^{|X_W \cap [\beta, s)| + |Y_W \cap [t, y)|}.$$

Its exponent must be even, since both underlying weak determinants are positive on the same sufficiently large positive alphabet. In type (d), telescoping the two complete recurrences makes it $2R + 1 - [t_\Omega < t] - [s_\Omega < t]$. Exactly one contact is below t. In type (c), it is $2C - 1 + [t_\Omega < \beta] + [s_\Omega < \beta]$. Exactly one contact is below β. The already excluded equalities prove the two strict-between statements. At $s = v, t = 0$, the first would put a nonnegative contact below zero, and the second would put w strictly between contacts in $[0, w]$. Both are impossible. □

Lemma E.4. *At a nonzero ordinary first matrix, an active pair supplies either the complete terminal signed comparison of Theorem 6.9, in a lawfully transported orientation, or the wholly private input of Lemma D.3 and Proposition 6.10. For an empty finer every active pair is private.*

Proof. In the actual ordinary (a) flags, the full proper corner has three square triangular diagonal pieces

$$r + \mathrm{JT}(X_W \setminus v, Y_W \setminus t), \qquad \mathrm{JT}(B_-, C_-), \qquad \mathrm{JT}(X_W \setminus s, Y_W \setminus 0).$$

Every high-piece block has least column at least r; all other blocks have head below r. For an available P_i, its full threshold product sees a high block if and only if $i \ge r$. For a genuine G_j, its containing block is high if and only if $j \ge r$. Thus, for either kind of actual coface, its head is at least r if and only if its input is at least r. This proves the required head assertion from actual strict blocks, even with units or empty pieces. The other own ordinary field has its separately rebuilt identical three-band property.

The two numerator equalities in (D.1) have equal heads on their respective pairs. If a G-pair has a head at least r, both actual inputs $j, j + \epsilon$ are terminal. If $\epsilon < 0$, swap the two actual own fields: the new ϵ is $-\epsilon$, and its first input is $j + \epsilon$, still terminal. A terminal P-pair is changed into a terminal G-pair by the full physical antitranspose of Lemma D.2. Both true word orders and the actual finer (b) mark are transported.

For completeness, in the direct ordinary (a) terminal orientation write $j = r + c$, $k = j + \epsilon$, h for the common head, $L = N_J - h$, $D = h - v$, and $\ell = N_J - k = s + t - c$. Its four complete cuts are those of Theorem 6.9. The genuine high mark column has $c \in Y_W \setminus t$. If $c \ge t$, then $\ell \le s$ and its strict block starts at $L \le s$. If $c < t$, then $s < \ell < v$; the low source height on $[s, v)$ is $h_W + 1 \ge 2$, so there is no intervening strict cut and that same block still starts at or below s. Thus

$$1 \le L \le s \le v - t, \quad \ell \le v, \quad j \ge r, \quad k \le h, \quad h \ge r > N_J/2.$$

Directly from the full cuts, for every closed $L \le x \le h$,

$$X_T(x - L) - X_S(x) = -[x = \ell], \qquad Y_T(x - L) - Y_S(x) = [x = j].$$

The actual two mark copies give, for every closed $0 \le x \le v$,

$$X_T(x+D) - X_S(x) = [x = s], \qquad Y_T(x+D) - Y_S(x) = -[x = t].$$

The complete sign, products, whole equality and strict areas have already been proved in Lemma D.1; these displayed inequalities and word equations give every remaining terminal input. At the transported (b) orientation the actual mark errors are exchanged and complemented as in Lemma D.2, which is one of the explicit physical orders in the signed theorem.

If neither connected pair has a terminal head, the three-band property in both fields gives $v < i, j, i+\epsilon, j+\epsilon < r$. The exact weight equation then gives

$$i - \delta = N_J - (j+\epsilon) \in (v, r), \qquad j - \delta = N_J - (i+\epsilon) \in (v, r).$$

These are precisely the four genuine private inputs and two private targets of Lemma D.3. Choose the larger of the two connected heads, antitransposing the pair if it is the P-head; this is the maximal-head convention used in Proposition 6.10. It changes neither a scalar gate nor the supplied finite horizon. At an empty finer the active weight is $i + j = r - w < r$, so both inputs are strictly private from the outset. These alternatives exhaust every nonzero skew PG entry. □

Lemma E.5. *For an actual ordinary-marked source E of area a, span S, and upper period Q, put $w = S - Q$. At positive width,*

$$\mathcal{G}_Q(\mathrm{AT}\, E) = (-1)^{w+1} \iota(\mathcal{G}_Q(E)).$$

At the genuine empty upper, the exact identity is instead

$$\mathcal{G}_S(\mathrm{AT}\, E) = \iota(s_E^2 - \mathcal{G}_S(E)).$$

These are whole-source identities in the rebuilt fields.

Proof. Let R, L be the actual normalized right and left coframes, and put $U = HR, V = HL$. Their complete null equations give

$$U_x = 0 \ (x \in X_E \setminus S), \quad V_{S-y} = 0 \ (y \in Y_E \setminus 0), \quad U_S = V_S = s_E/\kappa.$$

Moreover $\mathcal{G}_Q(E) = \kappa^2 [z^{S+Q}] HLR$. In the inverse alphabet $g = H^{-1}$, the actual complementary dual coframes are exactly $P_R = [HR]_{<S}$, $P_L = [HL]_{<S}$: their constant, support and annihilator equations identify them uniquely by the full dual proper-corner solve. The squared dual corner in that alphabet is κ^2; the corner itself has sign $(-1)^{a-1}$, obtained from the unique top coefficient.

For $Q < S$, expand both discarded tails before taking degree $S + Q$. Their product cannot contribute below $2S$, and the exact formula is

$$\mathcal{G}_{\mathrm{dual}}(g) = -\mathcal{G}_Q(E) + \kappa^2 \left\{ \sum_{\substack{x \in X_E \\ 0 < x < w}} L_x U_{Q+x} + \sum_{\substack{y \in Y_E \\ Q < y < S}} R_y V_{S+Q-y} \right\}.$$

Here L_x denotes the keyed left coefficient at row x. The genuine ordinary mark gives both open mate implications. Thus each indicated numerator is an actual source null equation and both complete tails vanish. At $Q = S$, the tail product contributes its one real coefficient $(s_E/\kappa)^2$, giving the second displayed formula. Finally the whole scalar has weight $2a - w$; replacing the inverse alphabet by the Hall involution contributes $(-1)^w$. This gives the positive-width sign and the separate empty correction. The actual (b) case follows by applying the same proved identity to its actual antitransposed (a) mark. No unmarked class-only duality has been used. □

E.2. **Endpoint products and the principal ordinary comparison.**

Lemma E.6. *Let D be connected, with m rows, span N, decreasing flags X, Y, and signed cofactors C_{ij} of its complete Jacobi–Trudi matrix. Put*

$$P_L(D;z)=\sum_i C_{i,m}z^{N-x_i}, \qquad P_R(D;z)=\sum_j C_{1,j}z^{y_j},$$
$$a=C_{1,m}, \qquad L_D=P_L/a, \qquad R_D=P_R/a.$$

For at least two rows, P_L, P_R are irreducible over the rational moment field. The pair (N, L_DR_D) determines the unordered pair $\{D, D^*\}$. A one-row diagram is determined by its span alone.

Proof. Every coefficient $C_{i,m}$ is nonzero. After the deletion, its sorted diagonal comparisons are either an original positive row length or an adjacent connectivity comparison $x_{j+1}-y_j\ge 0$. Thus its minor is a feasible weak skew determinant. The same holds for P_R by rotation. An h-only common divisor of the coefficients would divide both $s_D=\sum_i C_{i,m}h_{x_i}$ and its lower-weight corner a, contradicting connected primality.

Replace the last column by z^{N-x_i}. Its determinant is P_L. The largest remaining moment h_H, $H=x_1-y_{m-1}$, occurs once, at $(1,m-1)$, and its coefficient C is nonzero. For two rows the polynomial has the form $h_Hz^{x_1-x_2}-h_{x_2-y_1}$, with coprime coefficients, so is irreducible. For more rows, a nontrivial factorization has a factor F independent of h_H, since the degree in this letter is one. The preceding primitivity gives $\deg_z F>0$, and $F\mid C$. Desnanot–Jacobi on the first and last rows and last two columns shows that F divides aT, where $T=P_L$ for the connected prefix. Induction makes T irreducible, and $F\nmid a$, so F is associated to T. But T has degree one in $h_{x_1-y_{m-2}}$, while C uses indices at most $x_2-y_{m-2}<x_1-y_{m-2}$. This is impossible. Gauss's lemma proves the assertion over the moment field.

Rotation exchanges the two endpoint polynomials exactly. Unique factorization of L_DR_D, with each factor normalized to have constant term one, therefore recovers L_D, R_D up to exchange. Their nonzero supports are $\{N-x_i\}$ and $\{y_j\}$. They recover both full flag lists, and exchange is precisely whole rotation. A repeated factor causes no exception. □

Lemma E.7. *Use the standing ordinary(a) normalization of* (4.6). *Let $E=V\circ_\mu J$ and $U=V'\circ_\mu J$ be actual connected compositions through the same positive ordinary mark, with finer period r, width v, common coarse function, and common source function. Let $0\le w<v<r$, $M=N_V=N_{V'}$, and $Q=Mr+v-w$. If their whole global contractions $\mathcal{G}_Q$ agree, then $V'=V$ or $V'=V^*$. The hypotheses hold at the matched original upper comparison whenever Lemma 4.1 supplies its current in this rebuilt field. The lemma assumes only the entire raw scalar, not an actual upper board or a source minimum.*

Proof. For one coarse row, or span at most two, the common coarse class gives the literal diagram. Otherwise put $a=|V|$, let $m\ge 2$ be the common coarse row count, and use decreasing coarse flags X_V, Y_V.

The actual source blocks and endpoint records. Write

$$B_+=X_W\setminus\{s\}, \qquad C_+=Y_W\setminus\{t\}, \qquad A=s_W,$$
$$B=\det(h_{b-c})_{b\in B_+,c\in C_+}, \qquad C=\det(h_{b-c})_{b\in B_-,c\in C_-}.$$

These are the complete marked deletion and private determinants from (4.6); an empty block has determinant one and every unit row is retained. Set $n_0=|B_+|+|B_-|=|C_+|+|C_-|$. The native lists at coarse span M are

$$I_M=\bigsqcup_{j=0}^{M}(rj+B_+)\sqcup\bigsqcup_{j=0}^{M-1}(rj+B_-), \qquad C_M=\bigsqcup_{j=0}^{M}(rj+C_+)\sqcup\bigsqcup_{j=0}^{M-1}(rj+C_-).$$

The source adds exactly the rows $rx+s$, $x\in X_V$, and columns $ry+t$, $y\in Y_V$. Let $N=\mathrm{JT}(I_M,C_M)$, in decreasing native order. Its marked blocks B_j lie in $[rj,rj+v]$, and its private blocks C_j in $(rj+v,r(j+1)-1]$. The singleton private ends become empty cell diagonals in the native deletion, giving square triangular cuts, not deletions of unit vertices. If the original

marked interval is $[a_z, b_z]$, the weak marked strip is, up to a common diagonal translation, $[a_z - \mathbf{1}_{z<t}, b_z - \mathbf{1}_{z<s}]$. On a private diagonal $[u_z, v_z]$ the strip is $[u_z, v_z - 1]$. The initial and terminal deletions supply the missing end tails, so there are $M+1$ whole marked blocks and M whole private blocks. Consequently $\det N = B^{M+1}C^M$. Reinserting row $rx+s$ and column $ry+t$, in their sorted positions, leaves the actual row composition of length $x-y$ in the middle and $M-x+y$ of each native strip outside it. Thus its determinant is $G_{x-y}(BC)^{M-x+y}$ when $x \geq y$, where $G_j = s_{\mathrm{Row}_j \circ_\mu J}$ and $G_0 = A$. For $x < y$ a private cut has negative height, so this bordered determinant is zero. This derives the embedding from the complete flags and their actual cuts.

Let i_s, j_t be the zero-based positions of s, t in the decreasing mark lists, and put $\epsilon = (-1)^{i_s+j_t}$. The sorted insertion counts are

$$a_x = \#\{b \in I_M : b > rx+s\} = i_s+(M-x)n_0, \qquad b_y = \#\{c \in C_M : c > ry+t\} = j_t+(M-y)n_0.$$

In native-plus-port order the source and its Schur complement are

$$\mathbf{M} = \begin{pmatrix} N & Q_0 \\ P & H \end{pmatrix}, \qquad Z = H - PN^{-1}Q_0 = \eta \mathsf{R}\, T \mathsf{S}, \qquad \eta = \epsilon A/B, \quad T = \mathrm{JT}_f(V), \quad f_j = G_j/A,$$

with $f_0 = 1$, $f_j = 0$ for $j < 0$, and the genuine diagonal factors

$$\mathsf{R} = \mathrm{diag}_{x \in X_V}(\rho_x), \quad \mathsf{S} = \mathrm{diag}_{y \in Y_V}(\sigma_y), \qquad \rho_x = (-1)^{xn_0}(BC)^{-x}, \quad \sigma_y = (-1)^{yn_0}(BC)^y.$$

Indeed the bordered determinant just computed gives $Z_{xy} = (-1)^{a_x+b_y}(A/B)(BC)^{y-x}f_{x-y}$. These scales are fixed by the sorted insertions, not free gauges. In particular $\det \mathsf{R} \det \mathsf{S} = (-1)^{n_0 a}(BC)^{-a}$, since $\sum X_V - \sum Y_V = a$, so this product is common.

Put $U_0 = (-N^{-1}Q_0, I)^T$, $V_0 = (-PN^{-1}, I)$, and $E_N = \mathrm{diag}(N^{-1}, 0)$. The full inverse identity gives

$$\mathrm{adj}\, \mathbf{M} = \det N \det Z\, E_N + \det N\, U_0 \,\mathrm{adj}\, Z\, V_0. \tag{E.2}$$

Selectors are keyed by actual physical flags, transported by this same permutation. Define both native seeds by

$$p_N = e_0^T E_N, \qquad q_N = E_N e_{Mr+v}.$$

Thus $p_N = 0$ when $t = 0$, and $q_N = 0$ when $s = v$. Otherwise the square cuts make p_N an inverse row supported on the initial B_0 rows, and q_N an inverse column supported on the terminal B_M columns. They use only local marked moments and B^{-1}, not private inverse denominators.

For the local mark, let ℓ_W, u_W be the full one-port null vectors, with value one at its selected row s and column t and native parts

$$-(h_{s-c})_{c \in C_+}\, \mathrm{JT}(B_+, C_+)^{-1}, \qquad -\,\mathrm{JT}(B_+, C_+)^{-1}(h_{b-t})_{b \in B_+},$$

respectively. Put $L_W = \ell_W(v)$, $R_W = u_W(0)$, also when these endpoints are selected. The two square cuts surrounding a marked block show that its complete source one-port vectors have these same extreme coordinates: lower crossings cannot feed back into the left extreme, nor upper crossings into the right extreme. Hence

$$e_0^T U_0 \mathsf{S}^{-1} = R_W \sigma_0^{-1} e_{\mathrm{last}}^T, \qquad \mathsf{R}^{-1} V_0 e_{Mr+v} = L_W \rho_M^{-1} e_{\mathrm{first}}.$$

With $e = \det T$, L, R its complete last-column and first-row cofactor records, set $E_L = \mathsf{R}^{-1}V_0$, $E_R = U_0\mathsf{S}^{-1}$, and $K_0 = \det N\, \eta^{m-1} \det \mathsf{R} \det \mathsf{S}$. Equation (E.2) gives

$$\begin{aligned} L_E &= K_0(R_W \sigma_0^{-1} L E_L + \eta e\, p_N), \\ R_E &= K_0(L_W \rho_M^{-1} E_R R + \eta e\, q_N). \end{aligned} \tag{E.3}$$

Returning to physical order contributes the same permutation sign to both records, which squares to one in their contraction.

The local marked determinant has the exact Wronskian identity

$$\epsilon B^2 L_W R_W = \mathcal{W} := B\partial_{h_v} A - A \partial_{h_v} B. \tag{E.4}$$

The full bordered inverse, differentiated at its unique maximum entry $(v, 0)$, gives

$$\partial_{h_v} A = \epsilon B L_W R_W + A(\mathrm{JT}(B_+, C_+)^{-1})_{0,v}.$$

The last entry is zero if either endpoint is selected; otherwise it equals $(\partial_{h_v} B)/B$. Both A, B are affine in h_v, so $\mathcal{W}$ omits h_v. The nonzero weak minor B has smaller ordinary degree than the connected prime A, as does its nonzero extreme cofactor $\partial_{h_v} A$. Therefore $\mathcal{W} \bmod A = B\partial_{h_v} A \neq 0$. In particular $B, \mathcal{W}, L_W, R_W$ are units at A. No coprimality between A and C is asserted.

The whole negative mask. Subtract the two raw global contractions first. Jacobi's identity expresses their nonnegative-label aggregate in terms of the whole source, its signed proper corner, and their complete positive derivatives. At the unit label the group for a rank-k determinant P is $kP - \sum_{j>0} h_j \partial_{h_j} P$. Those data and ranks are common, so this entire aggregate cancels. This uses no upper board. The remaining matrix is the full mask

$$(K_Q)_{\xi,\zeta} = \mathbf{1}_{\xi-\zeta<0}\, h_{Q+\xi-\zeta} \quad (\xi \in X_E,\ \zeta \in Y_E),$$

with its actual raised units and negative zeros. A tied raw letter occurs at every one of its actual positions. Only the paired contraction may replace the raised matrix by this mask. After dividing by the common nonzero $\Lambda = K_0^2 R_W L_W/(\rho_M \sigma_0)$, its four channels are

$$\begin{aligned} &LE_L K_Q E_R R, \\ &\frac{\eta\sigma_0}{R_W}\, e\, p_N K_Q E_R R, \qquad \frac{\eta\rho_M}{L_W}\, e\, LE_L K_Q q_N, \\ &\frac{\eta^2 \rho_M \sigma_0}{R_W L_W}\, e^2 p_N K_Q q_N. \end{aligned}$$

All four remain in the paired identity.

Moving pivots and their weighted budget. Use the triangular raw pivots $z_k = h_{rk+v}$, $1 \le k \le M$, retaining every other positive raw letter as a spectator. The actual row probes have

$$G_k = a_k z_k + \mathscr{R}_k, \qquad a_k = \epsilon_k B^{k-1} C^k \mathcal{W}, \qquad G_k = A f_k, \qquad \epsilon_k = (-1)^{kn_0}.$$

Here $\mathscr{R}_k$ involves only earlier pivots and spectators. For the coefficient a_k, apply (E.2) to the row-k source. Its native extreme inverse entry is zero by the square cuts, so its corner is its full port product $(-1)^{kn_0} \epsilon B^{k+1} C^k L_W R_W$. Equation (E.4) gives the stated coefficient and its actual sign. The successive affine solves $z_k = (A f_k - \mathscr{R}_k)/a_k$ are a rational coordinate change.

Give f_j weight j and every spectator weight zero. Write $\deg_f$ for this weighted polynomial degree, Top_k for the homogeneous part of weight k, and $|\nu| = \sum_j \nu_j$ for monomial length, distinct from the weight $\sum_j j\nu_j$. Let $\mathcal{O}_A$ be the local ring of the spectator polynomial ring at the prime A; only polynomials not divisible by A are units. The bounds below lie in $C^{-k}\mathcal{O}_A$, not necessarily in $\mathcal{O}_A$.

For a spectator t, let a dot mean its derivative with all f_k and the other spectators fixed. The full compensated chain rule is

$$\begin{aligned} a_k \dot{z}_k &= f_k \partial_t A - z_k \partial_t a_k - \partial_t \mathscr{R}_k|_{z_1,\dots,z_{k-1}} - \sum_{j<k} (\partial_{z_j} \mathscr{R}_k)\dot{z}_j, \\ \dot{F} &= \partial_t F|_{z_1,z_2,\dots} + \sum_j (\partial_{z_j} F)\dot{z}_j, \qquad \dot{N}^{-1} = -N^{-1}\dot{N}N^{-1}. \end{aligned}$$

The finite sum includes every pivot on which the complete expression depends, including every affected later probe and both endpoint records.

Let $-d < 0$ be the least coarse label, taking the larger depth of the two diagrams. Connectedness gives $1 \le d \le M-2$. Put

$$L_0 = M - d, \qquad t_* = rL_0 + 2v - w, \qquad D = 2a - M + d.$$

This tag is a nonpivot, above $r+v$, but $t_* - (Mr+v) = v - w - rd < 0$. It lies inside the source. In its compensated derivative ∂_{t_*}, the local factors A, a_k are fixed, whereas the later pivots and the complete inverse need not be fixed.

Assign potential j to both vertex kinds in B_j, and $j + \frac{1}{2}$ to those in C_j. Every nonzero native entry has nonnegative row-minus-column potential. An entry labeled $rk+v$ costs at least k; in a mixed marked/private sector it costs at least $k + \frac{1}{2}$. Indeed a marked-row/private-column

residual is at most -1, and a private-row/marked-column residual at most $r-1$. These bounds include genuine zero-index entries, of weight zero. The row-k source adds one marked row of potential k and one marked column of potential zero. Every determinant allocation thus has total available potential k. Induction in its affine solve gives $\deg_f z_k \le k$: an earlier pivot of weight at most j occupies an entry of cost at least j, and division is by the f-independent a_k.

At weight exactly k, every used entry must attain its cost. No mixed-sector entry can do so. Within the balanced private sector, a strictly forward block entry cannot close a permutation; each of its k diagonal blocks therefore contributes its entire determinant C. If $Z_j = \operatorname{Top}_j z_j$, this proves

$$\operatorname{Top}_k G_k = C^k \Psi_k(Z_1, \dots, Z_k), \qquad \Psi_k = \epsilon_k B^{k-1} \mathcal{W} Z_k + \Psi_k^{<}(Z_1, \dots, Z_{k-1}).$$

The polynomial $\Psi_k^{<}$ is homogeneous of weighted degree k, with source-local polynomial coefficients. The displayed Z_k-coefficient is an A-unit. Its homogeneous equation is

$$Z_k = \frac{A f_k / C^k - \Psi_k^{<}(Z_1, \dots, Z_{k-1})}{\epsilon_k B^{k-1} \mathcal{W}}.$$

For a weight-k monomial f^ν, each product of earlier Z_j's has total C-denominator exponent k, while its A-exponents add to $|\nu|$. The direct term $A f_k / C^k$ has the same rule. Dividing by the displayed unit proves the first bound in

$$[f^\nu] \operatorname{Top}_k h_{rk+v} \in A^{|\nu|} C^{-k} \mathcal{O}_A, \qquad \deg_f \partial_{t_*} h_{rk+v} \le k - L_0 - 1. \tag{E.5}$$

This coefficientwise induction remains valid when $A \mid C$, with any multiplicity; it never calls an individual pivot A-regular.

For the second bound, a direct source occurrence of $t_* = rL_0 + 2v - w$ costs at least L_0. Equality is possible only in the private/private sector, since $v < 2v - w \le 2v$. Such a forward entry cannot close within the balanced private blocks and connect the two marked source ports without a mixed-sector entry. Its half-unit slack and integral free weight give a loss of at least $L_0 + 1$. Every other direct placement already has that loss. In the compensated chain rule, the cofactor of an earlier pivot z_j has degree at most $k - j$, while induction bounds $\dot{z}_j$ by $j - L_0 - 1$. Every summand has degree at most $k - L_0 - 1$; division by a_k proves the second bound for every later pivot. Zero derivatives have degree minus infinity.

The normalized mask budget. For actual coarse ports x, y, let

$$\ell_x = (-h_{rx+s, C_M} N^{-1}, e_x^T), \qquad u_y = \begin{pmatrix} -N^{-1} h_{I_M, ry+t} \\ e_y \end{pmatrix}, \qquad \epsilon_{xy}^{\mathrm{ins}} = (-1)^{a_x + b_y},$$

in the complete native-plus-port orders. These are the full row of V_0 and column of U_0, not truncated local selectors. Define the normalized restoration kernel by

$$\mathcal{P}_{xy} = \epsilon_{xy}^{\mathrm{ins}} (BC)^{x-y} \ell_x K_Q u_y.$$

The main channel entry $(E_L K_Q E_R)_{xy}$ is $\epsilon \mathcal{P}_{xy}$, by the actual diagonal factors. The mask retains all strictly negative physical positions, including native/port and native/native positions.

Put $q = x - y$. Finite triangular inverse paths, equivalently complete Cramer allocations, bound a left path ending at potential p by $x - p$, and a right path beginning at p' by $p' - y$. A restored entry has available cost $M + p - p'$; its pivot weight does not exceed that cost, with strict mixed-sector slack because $w < v$. Thus the total kernel budget is $M + q$. A differentiated pivot loses $L_0 + 1$ by (E.5). An internal tag occurrence in an inverse has the same loss: its only cost-L_0 placement is private/private, which cannot join the two marked endpoints critically. This accounts for the moving inverse terms as well as pivot terms.

Only direct restoration of t_* can reach weight $q + d = M + q - L_0$. Its original label is $-rd + v$. A critical marked/marked placement is therefore exactly row $rj + v$, column $r(j + d)$; every such actual position is summed. Its critical paths stay in marked blocks, so their local inverse denominators are powers of the A-unit B. The private determinants cancel in these marked-to-marked Cramer ratios. The first bound in (E.5) gives each weight-$(q + d)$ coefficient

the factor $A^{|\nu|}C^{-(q+d)}$. The actual normalization contributes C^q, leaving $C^{-(q+d)}C^q = C^{-d}$. Therefore

$$[f^\nu]\,\mathrm{Top}_{q+d}\,\partial_{t_*}\mathcal{P}_{xy} \in A^{|\nu|}C^{-d}\mathcal{O}_A. \tag{E.6}$$

No lower-weight or per-entry pole assertion is used. At $q = -d$ the supported extreme is unique. Its two local selectors are L_W, R_W, so its weight-zero coefficient is exactly

$$\epsilon_{xy}\frac{\mathcal{W}}{B^2(BC)^d}, \qquad \epsilon_{xy} = \epsilon^{\mathrm{ins}}_{xy}\epsilon.$$

This follows also by taking that extreme cofactor in (E.2); all other occurrences of its raw label remain in the full kernel and have lower free weight.

All four channels and the tail observation. The main pair $q = x - y$ has coarse cofactor-product weight $2a - M - q$. To reach total weight D, a pair $q > -d$ needs positive kernel weight $q + d$. After multiplication by C^d, every such coefficient vanishes modulo A by (E.6). The native-left channel has coarse weight $2a - M + y$, hence needs weight $d - y$; its factor σ_y^{-1} has C-depth y. The native-right channel has coarse weight $2a - x$, needs weight $x - M + d$, and its factor $\rho_M\rho_x^{-1}$ has C-depth $M - x$. A negative required weight contributes nothing. Otherwise the same marked-path argument, with a local native selector in place of one port, gives total depth d in each channel. Both have the additional genuine $\eta = \epsilon A/B$. Their local seeds and L_W^{-1}, R_W^{-1} are A-regular, so both entire cleared coefficients vanish modulo A, even at free weight zero. The native/native restored indices are at most $2v - w < t_*$; its seeds and scales are local, so its compensated derivative is exactly zero.

Apply $C^d\,\mathrm{Top}_D\,\partial_{t_*}$ to the whole paired scalar and only then reduce at A. All coefficients to be reduced have just been proved regular. The surviving extreme multiplier is $\epsilon_{xy}\mathcal{W}/B^{d+2}$, a nonzero A-unit. The coarse alphabet remains independent in the residue field. Unequal depths would leave a nonzero whole tail product on one side and zero on the other. At equal depth the insertion parity is common: $x + y \equiv x - y = -d \pmod 2$, and all ranks and native counts agree. Thus the whole products of the two coarse endpoint cofactors agree.

For completeness, if the coarse least label $q \geq 0$, no negative reader is required. The row count and the all-row intersection $\max(0, q + m - 1)$ are Schur invariants [4], so every mate has the same q. Put $c = [f_M]e$, the common signed proper corner. Its unique minimum cofactor and the corresponding native cofactor are positive derivatives of the common e, c, or their unit Euler groups when $q = 0$. Formula (A.4) supplies the same tail product. The positive all-row intersection also precludes mixing a nonnegative and a negative minimum in one class.

In either sector these factors are the actual first-row deletion $T_+(V)$ and last-row deletion $T_-(V)$. In decreasing coarse order their endpoint entries are $R_{y_{\max}} = \mathrm{cof}_{1,1}\,T = s_{T_+(V)}(f)$ and $L_{x_{\min}} = \mathrm{cof}_{m,m}\,T = s_{T_-(V)}(f)$. The signs introduced in lifting these records are precisely the insertion signs already retained above. Each deletion is nonempty and connected: at least one row remains, and all adjacent overlaps between remaining rows are inherited from V. Connected primality matches their unordered Schur pair, retaining repeated factors. The coefficient-one normalization and positivity fix the scalar signs. Rotate V' if the matching is exchanged; rotation exchanges the two deletion functions. We then have both the common whole function and the common first-row-deletion function. The theorem [5, Reconstruction from two boundary observations] identifies these connected diagrams up to translation. Hence $V' = V$ or $V' = V^*$.

Finally, the actual common finer fixes A, every physical source probe, and all modified-composition normalizers. The common whole source fixes its corner and literal ranks. If the original selector was column, the genuine identity $A_\omega s_{F_R} + B_\omega s_{F_C} = s_E^2$ earns row equality; at an empty mark $A_\omega = B_\omega = 1$. The ordinary activation or the current identity (4.5) gives the required whole global scalar. These operations use the supplied whole observation, not an additional parent or an independently observed cofactor. □

E.3. **Self-reflecting inner factors.**

Lemma E.8. *Retain the landed original comparison through a common full marked ordinary or empty inner J, and the minimum held on its actual source E. Suppose $J = J^*$. At every*

narrow upper width $0 \le w < r$, $w \ne v$, the original whole mate belongs to the prescribed full marked fiber. Alternatively, the construction in the proof produces an actual proper finer chart on E of strictly smaller score, contradicting that minimum.

Proof. We first check the physical geometry needed by the scalar calculation. The first and last r-diagonal windows of E are the respective end windows of its actual copies of J; denote the terminal copy origin by o. Since $w < r$, these contain the entire overlap bands and their neighboring contact diagonals. If the upper displacement is u, its actual lower overlap is

$$W_\omega = J \cap (J - d), \qquad d = u - o, \qquad W_\omega + u = (W_\omega + d) + o.$$

The half-turn fixing J takes W_ω to $W_\omega + d$. It also interchanges the two local complements and their neighbor contacts. Therefore the two attachments are of the same kind: every narrow upper is ordinary. This conclusion uses the actual intersections and contacts, not just a symmetric height sequence. The common coarse function fixes span, row count, and column count, hence the same o on E, U. Locality gives their same actual upper mark; the genuine ordinary gauge and the already decoded domino class give its matching selector.

For a vertical finer, apply Lemma A.1 to the complete pair, including both upper presentations, and rebuild both rational fields. At an empty finer the genuine edge conversion also applies to both coarse outers. Reapply Lemma 4.1 on these actual image charts. A vertical upper uses Lemma E.5 with its actual two mate equations, not class-only antitranspose covariance. Thus the two complete global contractions are earned afresh in the horizontal finer field before the principal reader is used. Whole rotation of the finer mark interchanges its two actual end copies. In the positive case its contacts obey $s + t = v$, and its marked overlap is self-reflecting.

The empty finer. Here $v = 0$, $0 < w < r$. Write $X_J = \{r\} \sqcup I$, $Y_J = C \sqcup \{0\}$, with $I = r - C$, and let $D = \{1, \dots, r-1\} \setminus I$. Set $T = z^r$ and, for $i \in I, c \in C$, use the true native series

$$K_{ic} = \sum_{j \ge 0} h_{jr+i-c} T^j, \quad Z_i = \sum_{j \ge 0} h_{jr+i} T^j, \quad U_c = \sum_{j \ge 0} h_{jr+r-c} T^j, \quad g = K^{-1} Z.$$

After indexing row i by $c = r - i$, the matrix is symmetric and U is the corresponding Z. Thus the full left and right transfers coincide. Put

$$P(z) = 1 - \sum_{c \in C} g_c(T) z^c, \qquad H(z) = \sum_{j \ge 0} h_j z^j.$$

The native row equations and the actual returned-zero equation are

$$HP = \Phi(T) + \sum_{d \in D} q_d(T) z^d =: Q(z), \quad \phi_0 = 1, \quad \phi_j = h_{jr} - [T^{j-1}] U K^{-1} Z. \tag{E.7}$$

There is no numerical specialization of the clock or the inverse. At a complete finite raw cutoff $r(L+1) - 1$, the coefficients $\phi_1, \dots, \phi_L$, $g_{c,j}$, and $q_{d,j}$, $0 \le j \le L$, are rational coordinates: their number is $r(L+1) - 1$, and the inverse formula $H = Q/P$ reconstructs all retained raw moments. They are therefore algebraically independent. One may take $L = 2M - 1$. Completing its last block adds only unused raw spectators, and adds neither an equation nor a higher response.

Let $A(T) = \sum a_i T^i$, $B(T) = \sum b_j T^j$ be the complete normalized coarse endpoint polynomials in the alphabet ϕ. Their degrees are at most $M - 1$. If P_n cuts each g_c before level n, the whole global comparison is

$$\sum_{i,j=0}^{M-1} \Delta(a_i b_j) C_{M-i, M-j} = 0, \qquad C_{nm} = [z^{(n+m)r-w}] H P_n P_m. \tag{E.8}$$

The common complete source corner has been divided out; no entry of this kernel is separately observed. Put $k = r - w$, $s = n + m - 1$, and $G = 1 - P$. The product of the two omitted tails begins strictly above the required degree, so exactly

$$C_{nm} = [z^{rs+k}] Q(1 + G - G_{<n} - G_{<m}). \tag{E.9}$$

The direct term is $q_{k,s}$ when $k \in D$. Every other term is

$$q_{d, s-\epsilon-j} g_{c,j} (1 - [j < n] - [j < m]), \qquad d + c = k + \epsilon r, \quad \epsilon \in \{0, 1\},$$

where $d \in D \cup \{0\}$ and $q_0 = \Phi$. Thus both residue carries, the returned zero, and both finite tails are retained.

If $k \in D$, taking the coefficient of $q_{k,\ell}$ in the one whole equation (E.8) gives its antidiagonal sum with $n + m - 1 = \ell$; that variable occurs in no product term. If instead $d \in D, c \in C$ and $d + c = k + \epsilon r$, the monomial $q_{d,\ell} g_{c,0}$ gives minus the antidiagonal with $n + m - 1 = \ell + \epsilon$. The coefficient is -1 since $n, m \geq 1$. Taking all actual indices gives $AB = A'B'$, the entire endpoint product.

If both readers fail, $k \in I$ and $i + k \bmod r \in I \cup \{0\}$ for every $i \in I$. The reflected column condition holds too. Put $g = \gcd(r, k)$. Every nonzero g-class is full or empty in I; the zero class contains every nonzero multiple of g, by iteration from k. For $g = 1$, J is a literal column, covered by Lemma A.2; equivalently, after antitransposition the row has a direct reader. For $g \geq 2$, take the first g diagonals as K. Both words, singleton ends, and positive actual heights give

$$J = \mathrm{Col}_{r/g} \circ_{\mathrm{EMPTY}_H} K.$$

This is a literal cell union: its within-block flags are the full nonzero residue classes, and its zero-residue flags are those of $\mathrm{Col}_{r/g}$. Hence the empty edge is legal, including its two hatted conditions. Both factors are proper and strict. Aligned gluing through the actual finer mark gives a period-g empty chart on E, contradicting its held score $(0, r)$.

The positive finer, intermediate width. The principal range $w < v$ is covered by Lemma E.7; consider $v < w < r$. Let

$$\mathcal{B} = X_W \sqcup B_-, \quad \mathcal{C} = Y_W \sqcup C_-, \quad B_0 = \mathcal{B} \setminus \{s\}, \quad C_0 = \mathcal{C} \setminus \{t\}.$$

The complete self-reflecting flags give $\mathcal{B} = v - \mathcal{C} \pmod r$, including every private phase. Use the actual returned-port coordinate chart

$$\begin{aligned} K_{bc}(T) &= \sum_{j \geq 0} h_{jr+b-c} T^j, & z_b(T) &= \sum_{j \geq 0} h_{jr+b-t} T^j, \\ p &= -K^{-1} z, & P(z) &= z^t + \sum_{c \in C_0} p_c(T) z^c, \\ HP = Q &= \sum_{d \notin B_0} q_d(T) z^d, & \Phi &= q_s, \quad \phi_j = \Phi_j / \Phi_0. \end{aligned} \tag{E.10}$$

Here $b \in B_0, c \in C_0$ in the matrix, the sum for Q runs over $d \in \{0, \dots, r-1\} \setminus B_0$, and the fixed returned coefficient is $p_t(T) = 1$. The low matrix $K(0)$ is a feasible weak Jacobi–Trudi matrix: its height is the actual first-period height minus the returned column step. Hence it is nonzero even with its genuine units. If C_{ab} denotes the signed cofactor of the actual mark matrix, then Cramer's rule gives

$$\Phi_0 = A / C_{st}, \qquad p_{c,0} = C_{sc} / C_{st} \quad (c \in Y_W). \tag{E.11}$$

In particular $p_{0,0} \neq 0$. At a finite complete raw cutoff, (E.10) has the inverse $H = Q/P$. The sole low relation is $q_{0,0} = p_{0,0}$, with both equal to one when $t = 0$. Equality of the coordinate count and the raw count therefore proves independence of all remaining parameters. Replacing Φ_j by $\Phi_0 \phi_j$ is invertible. The coarse alphabet ϕ is independent of the entire original low field.

We must use the global endpoint records, not just the returned ones. Let $P_n = [P]_{<nr+v}$, let A_V, B_V be the two coarse normalized endpoint polynomials. For $m_V = \operatorname{rows} V$, define the signed proper corner

$$\begin{aligned} \alpha_V(\phi) = [\phi_M] s_V(\phi) &= (-1)^{m_V - 1} \det(\phi_{x-y})_{x \in X_V \setminus \{M\},\, y \in Y_V \setminus \{0\}}, \\ f_V = \frac{s_V(\phi)}{\alpha_V(\phi)}, &\qquad c_V = \Phi_0 f_V. \end{aligned}$$

The sign is the extreme cofactor sign in (2.4), not an unsigned proper determinant. The whole coarse equality fixes the row count and this signed coefficient, so both quotients are common

on the two coarse boards. If $s = v$, put $Z = 0$. Otherwise Z is the inverse column of the actual cap $\mathrm{JT}(X_W \setminus \{v\}, Y_W \setminus \{t\})$ specified by

$$[HZ]_b = [b = s] \quad (b \in X_W \setminus \{v\}).$$

Its coefficients use only h_j, $j < v$. The complete normalized global records are

$$R_E = \frac{[B_V P]_{<S} - c_V T^M Z}{p_{0,0}}, \qquad L_E = \frac{[A_V P]_{<S} - c_V T^M Z}{p_{0,0}}. \tag{E.12}$$

Indeed $HP = Q$ annihilates the fixed rows, and the coarse null equations annihilate all returned rows except $rM + s$. Its response is c_V. If $s < v$, this is a nonglobal row, and the terminal seed subtracts exactly that response. It affects no earlier row by the actual negative slots and no constant coefficient. If $s = v$ the returned row already is global. Uniqueness of the normalized source null record proves the right formula. Full marked rotation proves the left one, with the same seed. This proves both cap corrections rather than dropping them.

Put $\tau = 2v - w < v$. Exact finite degree bounds give

$$C_{nm} := [z^{(n+m)r+\tau}]HP_nP_m = [z^{(n+m)r+\tau}]Q(P_n + P_m - P), \qquad [z^{nr+\tau}]HZP_n = [z^{nr+\tau}]QZ.$$

The first discarded product starts above its target; in the second the omitted tail starts at $nr + v > nr + \tau$. The common seed-square cancels in the paired scalar. The whole comparison is consequently

$$\sum_{i,j} \Delta(a_i b_j) C_{M-i,M-j} - c_V \sum_i \Delta(a_i + b_i)[z^{(M-i)r+\tau}]QZ = 0. \tag{E.13}$$

All coefficients a_i, b_i belong to the independent coarse field.

Let $d = (\tau - t) \bmod r$, which differs from s. If $d \notin B_0$, its independent coefficient $q_{d,\ell}$, with $\ell = 2M + e - K$, $e = (\tau - d - t)/r \in \{0, -1\}$, reads exactly $\Delta[A_V B_V]_K$, for $0 \le K \le 2M - 2$. The only matching native phase is the fixed t-coefficient one, and the cap seed has no t-coordinate. If instead $c \in C_0$, $d \notin B_0$, $d \ne s$, and $d + c = \tau - er$, take the coefficient of $q_{d,\ell}$ and then its constant term in all positive-level p's. With the same $K = 2M + e - \ell$, the result is

$$p_{c,0}\, \Delta[A_V B_V]_K = \Phi_0 f_V Z_c\, \Delta[A_V + B_V]_{K-M}. \tag{E.14}$$

For a private $c > v$, $Z_c = 0$, and the independent nonzero coefficient $p_{c,0}$ can be canceled. The possible carry $e = -2$ then uses independent low private data at $\ell = 0$, never the returned-zero relation. For a marked $c < v$, $e \ge -1$, so $\ell \ge 1$ at every required K, and the low cap is fixed during extraction. Divide by the true Φ_0 and use (E.11). The left coefficient becomes C_{sc}/A, with a genuine pole at the connected prime A; its nonzero cofactor has smaller ordinary degree. The right coefficient $f_V Z_c$ is regular at A, since the cap uses only $h_{<v}$ and f_V belongs to the independent coarse field. Therefore $\Delta[A_V B_V]_K = 0$. This proves the entire product equality whenever either reader exists, with all carries and cap terms retained.

If no reader exists, the complete residue conditions give

$$b + v - w \pmod r \in \mathcal{B} \qquad (b \in \mathcal{B}).$$

Reflection gives the full dual column condition. Thus $\mathcal{B}$ is a union of complete cycles of this shift. Since $v \in \mathcal{B}$ and $0 \notin \mathcal{B}$, the number $g = \gcd(r, |v - w|)$ satisfies $2 \le g < r$ and $g \nmid v$. The full ordinary fold in [9, Ordinary residue fold] constructs on J a proper period-g, width-v mod g chart, with both words, all four legality conditions, and strict factors. One aligned lift through the supplied finer mark contradicts its minimum held on E.

In all remaining cases the complete endpoint product and Lemma E.6 give $V' = V$ or $V' = V^*$; the same conclusion in the principal range came from Lemma E.7. The actual self-reflection of J and its finer mark makes the latter case $U = E^*$. If R_E denotes this source half-turn, the actual intersection equations at the start give $R_E(W_\omega + u) = W_\omega$ and $R_E(W_\omega) = W_\omega + u$. Hence the entire upper marked presentation is transported. Its only other reading is the genuine ordinary gauge with the corresponding outer involution. The decoded domino selector therefore gives the original full marked fiber. □

Appendix F. An empty upper over a mixed inner

This case admits a direct finite reader. Its useful feature is that the last missing cofactor can be recovered from the actual row count; no extra response coefficient is needed.

Lemma F.1. *Let μ be a full legal mixed mark on a connected diagram J, of period r and width $0 < v < r$. Let*

$$E = V \circ_\mu J, \qquad U = V' \circ_\mu J$$

be actual connected proper charts with the same literal full marked inner, $s_V = s_{V'}$, and $s_E = s_U$. Suppose an EMPTY domino composition on E and an actual proper composition on U have equal whole Schur functions. Then, after the genuine EMPTY edge gauge, the upper marks and domino selectors agree, and $V' = V$ literally. Consequently $E = U$ and the original whole mate belongs to the prescribed full marked fiber. No minimum on J or additional lower-area call is used in this lemma.

Proof. An actual whole rotation converts mixed type (c) to type (d), carrying both original charts, all supplemental copies, and both upper presentations. We work in that frame and undo the rotation at the end. Put $M = N_V = N_{V'} \geq 2$ and $N = Mr + v$.

First align the actual EMPTY upper. If the competing upper has width u, period $p' = N - u$, and outer span k, whole span equality says

$$N + (k-1)p' = 2N.$$

Hypothesis II gives $p' > N/2$. Thus $k = 2$ and $u = 0$; a connected diagram of span two is a domino. Convert any vertical EMPTY reading to the horizontal reading by its genuine edge gauge, which applies the outer involution and leaves the whole diagram unchanged. The extreme finer copies of both sources have the same actual origins: in the mixed case their displacement is $(M-1)v_\mu$. Hence E, U have the same actual row count, say t. The horizontal EMPTY row-domino parent has $2t-1$ rows and the column-domino parent has $2t$ rows. Since whole equality preserves row count, their domino selectors agree.

If the common selector is the column domino, the two-junction identity

$$s_E^2 = s_{\mathrm{Row}_2 \circ_\varnothing E} + s_{\mathrm{Col}_2 \circ_\varnothing E} \tag{F.1}$$

and its identical formula for U give equality of the row-domino parents. Identity (F.1) follows by the two Laplace expansions at the single horizontal or vertical junction; the EMPTY overlap and complementary overlap both have function one. Equivalently, a pair of tableaux is assigned to the horizontal or vertical junction according to the weak or strict inequality of its two joining entries. Thus it suffices to recover V from s_E and $P_E = s_{\mathrm{Row}_2 \circ_\varnothing E}$.

The actual pencil. Let $B \subseteq \{1, \ldots, r-1\}$ be the occupied row residues of the mixed finer; $v \in B$. Its two returned column phases satisfy $0 \leq t_0 < s_0 \leq v$. To avoid conflict with the source row count, we use t_0, s_0 for these phases. The complete mixed coding of [9, Boundary coding lemma] gives

$$\begin{aligned} X_E &= \{rj + b : 0 \leq j < M,\ b \in B\} \cup \{rM + b : b \in B,\ b \leq v\}, \\ Y_E &= \{rj + c : 0 \leq j < M,\ c \in Y_J \setminus \{t_0, s_0\}\} \\ &\qquad \cup \{ry + t_0 : y \in Y_V\} \cup \{rj + s_0 : j \in L_V\}, \\ L_V &= \{0, \ldots, M-1\} \setminus X_V. \end{aligned} \tag{F.2}$$

These are set unions. Two representatives $c, c+r$ in the first column set extend one fixed string; they are not duplicate columns. In particular X_E is closed under subtraction of r while positive, and has no zero residue.

Let ℓ_x be the signed cofactor at source row x, column zero, and R_y the signed cofactor at source row N, column y. Set $\alpha = \ell_N = R_0 = [h_N]s_E$ and $\Phi(a) = \sum_y R_y h_{a-y}$, the determinant with its top row replaced in its original position by a. The actual EMPTY row parent has flags

$$(X_E + N) \cup (X_E \setminus \{N\}), \qquad ((Y_E \setminus \{0\}) + N) \cup Y_E.$$

The lower-left block is zero, because every proper row is at most $N-1$ and every positive column is at least one. Laplace expansion along its first block of columns therefore gives

$$P_E = \sum_{x\in X_E}\sum_{y\in Y_E} \ell_x R_y h_{x+N-y}. \tag{F.3}$$

Apply the one-parameter substitution $h_j \mapsto h_j + \zeta h_{j-r}$, including its actual negative-index zeros. Every R_y is fixed. In a last-column cofactor, a shifted row either duplicates its predecessor or is zero, unless the shifted rows form a consecutive chain from the omitted row. Sorting that chain with the cofactor sign gives exactly

$$\ell_x \longmapsto \sum_{j\ge 0}(-\zeta)^j \ell_{x+jr}, \qquad \ell_a = 0 \quad (a \notin X_E).$$

Substitution into (F.3), followed by reindexing, proves, for every $L \ge 1$,

$$[\zeta^L]P_E(h+\zeta h_{\cdot -r}) = (-1)^{L-1}\sum_{b=1}^{r} \ell_{b+(L-1)r}\, Q_b, \qquad Q_b = \Phi(N+b-r). \tag{F.4}$$

The two consecutive chain terms cancel except at the first positive representative of each residue. Thus this formula does not assume separate access to tied cofactors.

Let

$$S = \{b \in B : (b+v) \bmod r \notin B\}, \qquad S_+ = S \cap [1, v].$$

The set S is nonempty. Otherwise translation by v would permute B, and the predecessor of $v \in B$ would force $0 \in B$. If $S_+ \neq \varnothing$, put $I = S_+$ and $L_* = M+1$; otherwise put $I = S$ and $L_* = M$. A term with $b \in B \setminus S$ has $Q_b = 0$ because its replacement row is an existing proper row.

The last r content diagonals contain $|B|$ rows and $|B|-1$ fixed columns. This follows from the actual last-window copy and the singleton private diagonal of J at v. No returned t_0- or s_0-column lies in this window. Its matrix Win has rank $|B|-1$: deleting any one row leaves a nonzero weak Jacobi–Trudi determinant, since its sorted row entries are at least the next original row, hence at least the corresponding column. Consequently, at each row x of that window,

$$\ell_x = \kappa_x \alpha, \qquad \kappa_x = (-1)^{i-1}\frac{\det \mathrm{Win}_{-i}}{\det \mathrm{Win}_{-1}} \neq 0, \tag{F.5}$$

where the top row has position one. Every κ_x depends only on $h_0, \dots, h_{r-1}$ and on the same marked J. At L_* all surviving rows lie in this window. Thus the common whole data determine

$$O = \frac{(-1)^{L_*-1}}{\alpha}[\zeta^{L_*}]P_E(h+\zeta h_{\cdot -r}) = \sum_{b\in I}\kappa_b Q_b. \tag{F.6}$$

Here $\kappa_b = \kappa_{b+(L_*-1)r}$.

Complete fixed-block elimination. We give the finite elimination behind the terminal reader, including its constant correction. Introduce residue series $H_d(u) = \sum_{n\ge 0} h_{rn+d}u^n$, $0 \le d < r$, and the causal symbols

$$L_{bc}(u) = u^{\mathbf{1}_{b<c}} H_{(b-c) \bmod r}(u).$$

Choose carrier row phase v. Let C_0 be the fixed column residue set; its size is $|B|-1$. The fixed zero-order matrix D_0 has rows $B \setminus \{v\}$ and columns C_0. Its prefix height is, with $H_J(a) = |\{y \in Y_J : y \le a\}| - |\{x \in X_J : x \le a\}|$,

$$H_J(a) - \mathbf{1}_{a\ge t_0} - \mathbf{1}_{a\ge s_0} + \mathbf{1}_{a\ge v}.$$

This is nonnegative: on $s_0 \le a < v$ the mixed middle tail gives $H_J(a) \ge 2$, and elsewhere connectedness suffices. Thus $\det D_0 \neq 0$ by weak sorted feasibility. The last partial fixed block D_{cap} has rows $B \cap [1, v-1]$ and columns $C_0 \cap [0, v-1]$. Its prefix height is $H_J(r+a) - 1 \ge 0$, so its determinant is also nonzero. The entire finite fixed determinant is

$$K_J = (\det D_0)^M \det D_{\mathrm{cap}}.$$

Empty determinants are one.

Eliminate this complete fixed block. Put $B_0 = B \setminus \{v\}$ and $D(u) = (L_{bc}(u))_{b\in B_0, c\in C_0}$, so $D(0) = D_0$. The ordinary rows have two remaining symbols

$$G = L_{v,s_0} - L_{v,C_0} D^{-1} L_{B_0,s_0}, \qquad A = L_{v,t_0} - L_{v,C_0} D^{-1} L_{B_0,t_0},$$

for the missing-row and original outer-column strings. The inverse is causal, and its finite truncations equal the inverses of the corresponding finite lower triangular blocks. Their constants are nonzero: their bordered determinants have prefix heights $H_J(a) - \mathbf{1}_{a\ge t_0}$ and $H_J(a) - \mathbf{1}_{a\ge s_0}$. Put

$$\tau = A_0/G_0, \qquad T = A/(\tau G), \qquad T_0 = 1.$$

The ratio τ has positive raw weight $s_0 - t_0$, so some derivative in h_d, $1 \le d < r$, is nonzero. Holding all residue constants fixed, the coefficient change from the positive $H_{d,n}$ to T_n is triangular with nonzero diagonal $(\partial\tau/\partial h_d)/\tau$. Thus the positive T_n are legitimate independent coordinates after localization.

For a terminal row of phase β put

$$Z^\beta = L_{\beta,s_0} - L_{\beta,C_0} D^{-1} L_{B_0,s_0}, \qquad W^\beta = L_{\beta,t_0} - L_{\beta,C_0} D^{-1} L_{B_0,t_0},$$

and put $\Delta_\beta = W^\beta/\tau - Z^\beta T$. A row at copy M with $\beta \le v$ first eliminates the fixed cap columns. Its constant fixed input is supported on the low columns of D_0. The low-row/high-column block is zero, and its low diagonal block is exactly D_{cap}. More explicitly, let $C_{\mathrm{low}} = C_0 \cap [0, v-1]$ and $B_{\mathrm{low}} = B \cap [1, v-1]$. If $a = L_{\beta,C_{\mathrm{low}}}(0) D_{\mathrm{cap}}^{-1}$ and D_{low} denotes the low rows of D, cap subtraction changes the fixed input to $L_{\beta,C_0} - aD_{\mathrm{low}}$, which has zero constant coefficient on *all* fixed columns. It changes the direct variable entries to $L_{\beta,s_0} - aL_{B_{\mathrm{low}},s_0}$ and $L_{\beta,t_0} - aL_{B_{\mathrm{low}},t_0}$. Since $D_{\mathrm{low}} D^{-1}$ is the low-row selector, the subsequent bulk elimination cancels the terms involving a exactly and gives Z^β, W^β. Thus the complete elimination equals the bulk causal inverse, without a virtual copy-M column. A row at copy $M-1$ already has zero entries at every copy-M fixed column and introduces a factor u.

Normalizing the ordinary rows by G^{-1} and clearing their identity columns leaves terminal series

$$W/\tau - (Z - Z_0)T.$$

The coefficient Z_0 is not used in clearing any column and must be retained. For the original carrier $Z = G, W = A$, this gives $G_0 T$. In particular,

$$s_E = \epsilon_V K_J G_0^{M+1} \tau^{\mathrm{rows}\, V} s_V(T). \tag{F.7}$$

For a readable replacement $rj' + \beta$, where $j' = M-1$, or $j' = M$ with $\beta \le v$, the same elimination gives

$$\Phi(rj' + \beta) = c_V \sum_{y\in Y_V} C_y(T) \left(u^{M-j'} \Delta_\beta + \mathbf{1}_{j'=M} Z_0^\beta T \right)_{M-y}, \tag{F.8}$$

where $C_y(T)$ is the actual signed first-row cofactor of V and $c_V = \epsilon_V K_J G_0^M \tau^{\mathrm{rows}\, V}$. The sign and all fixed factors are the same as for the source, because both determinants replace the same row of the same finite matrix.

For the rows in (F.6), write

$$\beta_b = (b+v) \bmod r, \qquad \nu_b = 1 - \left\lfloor \frac{b+v}{r} \right\rfloor \in \{0,1\}.$$

Then the replacement row is $r(M - \nu_b) + \beta_b$. When $\nu_b = 0$ one has $\beta_b < v$, so every row used is covered by (F.8). We obtain

$$O = c_V \sum_{y\in Y_V} C_y(T) R^O_{M-y}, \qquad R^O = \sum_{b\in I} \kappa_b \left(u^{\nu_b} \Delta_{\beta_b} + \mathbf{1}_{\nu_b=0} Z_0^{\beta_b} T \right). \tag{F.9}$$

A cyclic translation with delay at most two. Work in the cyclic series ring with $z^r = u$. Put $D_{\mathrm{res}} = C_0 \cup \{t_0, s_0\} = Y_J \cap [0, r-1]$. There is a unique annihilator

$$g(z) = \sum_{c \in D_{\mathrm{res}}} g_c(u) z^c, \qquad g_0 = 1, \qquad F = Hg, \quad F_b = 0 \quad (b \in B).$$

Its constant pivot has prefix height $H_J(a) - 1 \geq 0$ and is a nonzero weak determinant. The Cramer pivots for the two returned coordinates have prefix heights $H_J(a) - \mathbf{1}_{a \geq t_0}$ and $H_J(a) - \mathbf{1}_{a \geq s_0}$; both are nonnegative. They give $g_{t_0}(0), g_{s_0}(0) \neq 0$. Comparing its one-dimensional kernel with the Schur-complement kernel gives

$$T = -\frac{g_{s_0}}{\tau g_{t_0}}, \qquad \Delta_\beta = \frac{F_\beta}{\tau g_{t_0}} \quad (\beta \notin B). \tag{F.10}$$

Fix one $b \in I$ and put $\beta = \beta_b$. For any series $Z(u)$, make the lawful substitution

$$H'(z) = H(z) + \frac{z^{r+\beta} Z(z^r)}{g(z)}.$$

It leaves all short data and all κ_b, Z_0^β unchanged. Moreover $H'g = F + uz^\beta Z$, so uniqueness keeps g, and hence T, fixed. Only Δ_β changes, by $uZ/(\tau g_{t_0})$. Formula (F.7) keeps c_V fixed as well. Since $b \mapsto \beta_b$ is injective, the complete observable changes by

$$O(H') - O(H) = c_V \sum_{y \in Y_V} C_y(T) \left(\frac{\kappa_b u^d Z}{\tau g_{t_0}} \right)_{M-y}, \qquad d = \nu_b + 1 \in \{1, 2\}.$$

Choose $Z = \tau g_{t_0} u^k / \kappa_b$. The resulting actual common substitution reads $c_V C_{M-d-k}(T)$, including its zero entries, for $0 \leq k \leq M - d$.

Apply the same substitution to the common whole observable for E and U. Their fixed factors and row counts agree. Thus their outer first-row cofactors agree up to a common nonzero sign for every $y \leq M - d$. If $d = 1$, this is the whole vector. If $d = 2$, it is the vector through $M - 2$, and that is enough. Indeed $C_y \neq 0$ precisely at actual column positions y, while for $1 \leq q < M$,

$$\sum_{y \leq q} C_y(T) T_{q-y} = 0 \quad \Longleftrightarrow \quad q \in X_V \setminus \{M\}.$$

For an absent row, sorting the inserted row gives a nonzero feasible weak determinant, so this is an exact zero-set test. The read coefficients determine $Y_V \cap [0, M-2]$ and $X_V \cap [1, M-2]$. If $n = \operatorname{rows} V$, then

$$\begin{aligned} M - 1 \in X_V &\iff |X_V \cap [1, M-2]| = n - 2, \\ M - 1 \in Y_V &\iff |Y_V \cap [0, M-2]| = n - 1. \end{aligned}$$

The known row count therefore recovers the remaining incidences. Both full outer lists agree, proving $V' = V$. The already aligned EMPTY selectors finish the original marked-parent conclusion. □

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Computational verification. The proofs do not depend on numerical computation. Scripts used for verification and figures are available at `https://github.com/johnnyshields/papers`.

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